THE INTERSECTION ALGEBRA OF SAFETY, COSAFETY, LIVENESS, AND COLIVENESS OVER LINEAR TEMPORAL LOGIC

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Abstract. Runtime verification of linear-temporal-logic (LTL) properties needs a precise classification of which omega-regular properties can be monitored, in what verdict alphabet, and with what residual evidence on the unverified portion. Alpern and Schneider partitioned properties into safety and liveness; Kupferman and Vardi added cosafety; coliveness completes the symmetric structure. We integrate these four families into a single intersection algebra over omega-regular properties, develop its operational counterpart as a verdict-alphabet ladder spanning binary, ternary, four-valued, and quantitative monitor systems, and connect the two layers through a universal-attribution theorem.

On the classification side, the four families generate a sixteen-cell exclusive partition of the omega-regular properties under combined topological and prefix-quantifier membership. The partition is equivalent, modulo two degenerate singletons, to the nine-cell verdict-axis partition of Peled and Havelund. We give independent topological and prefix-quantifier derivations of the empty cells, a De Morgan duality structure on the partition induced by the LTL temporal dualities, and a literature-anchored worked-example catalog with provenance per cell.

On the operational side, a verdict-alphabet ladder spans verdict cardinalities one, two, three, four, and infinity (with the trivial baseline at one, the qualitative rungs at two, three, and four, and the quantitative limit at infinity). Definitive verdicts on every trace characterise the clopen cell, and on every violation or every satisfaction the safety or cosafety properties, which the three-valued rung covers in a single monitor; there definitive verdicts saturate: the four-valued and quantitative rungs add revocable and quantitative information but resolve no further cell, and monitorability in the sense of Pnueli and Zaks is not a function of the cell. A universal-attribution theorem shows that every non-trivial verdict monitor canonically yields, for each verdict, a persistence pullback property on its input, the pre-image of a Liveness-and-Coliveness property of its verdict trace; this pullback is itself Liveness-and-Coliveness exactly when, after every input prefix, the verdict trace can still both settle on that verdict and fail to settle on it. The framework supplies a precise vocabulary for the calibration analysis of monitor-based epistemic guarantees on omega-regular properties.

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1. Introduction

Runtime verification asks whether an ω-regular property holds on an infinite execution as the execution is consumed prefix by prefix. Two questions immediately follow. *First*, into which canonical families do ω-regular properties decompose, and how do these families intersect? *Second*, what verdict alphabet does a runtime monitor need in order to resolve membership in each family from finite prefixes? The first question has a classical topological answer in Alpern–Schneider [2, 3] and an automata-theoretic one in Kupferman–Vardi [38], but the intersection structure of the four primitive families — Safety, Cosafety, Liveness, Coliveness — has not been enumerated in a single exhaustive classification. The second question has been answered rung by rung across multiple-valued LTL variants ([6, 7, 22]) but without a unified organising principle mapping verdict cardinality to resolvable property classes.

This paper develops a single classification algebra over the four primitive families, an operational verdict-alphabet ladder spanning binary, ternary, four-valued, and quantitative monitor systems, and a universal-attribution theorem connecting the two layers — showing that every non-trivial verdict monitor canonically yields a Persistence/Recurrence pullback property of its input, regardless of which cell its target property occupies.

Classical foundations. The safety/liveness dichotomy originates in Lamport [41] and is formalized topologically by Alpern and Schneider [2, 3]: a property is safety iff it is closed in the Cantor topology on Σ^ω, and liveness iff it is dense. Manna and Pnueli [48] refine this into a temporal-property hierarchy, while Sistla and Clarke [57] determine the complexity of satisfiability and model checking for LTL fragments defined by their temporal operators; Sistla [56] gives syntactic characterisations of the LTL formulas expressing safety, liveness, and fairness properties. Vardi and Wolper [61] formalise the automata-theoretic translation of LTL to Büchi automata, providing the operational substrate for the present work. The topological perspective on ω-languages is developed systematically in Staiger [58] and Chang–Manna–Pnueli [15]; Thomas [60] is the standard reference for ω-automata.

Cosafety ("guarantee") properties - those recognizable by a finite good prefix - are treated automata-theoretically by Kupferman and Vardi [38] through their good prefixes. Runtime monitoring introduces a three-valued semantics (accept/reject/inconclusive) for prefix-based verdicts [7, 45, 5]; Falcone, Havelund, and Reger [25] give a tutorial introduction, and Ferrando–Cardoso [27, 26] explore partial monitoring. Cimatti et al. [16] extend runtime verification to assumption-based settings.

Relation to the Manna–Pnueli hierarchy. Manna and Pnueli [48] classify temporal properties into six levels ordered by Borel complexity: Safety (Π_1^0), Guarantee (Σ_1^0), Obligation (Δ_2^0), Recurrence/Response (Π_2^0), Persistence (Σ_2^0), and Reactivity (Δ_3^0). Our four-class intersection algebra is orthogonal: it partitions properties by topological character (open/closed $\times$ dense/co-dense) rather than by Borel level. The difference is visible at $\mathsf{L} \cap \mathsf{K}$, which contains both Recurrence formulas (GFp) and Persistence formulas (FGp). The two classifications complement each other: the hierarchy orders properties by complexity; the intersection algebra partitions them by prefix-determinability.

Adjacent hierarchies. The CLASP partition—our four-class partition of the ω-regular properties by Safety, Cosafety, Liveness, and Coliveness ($\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$), read as a *CL*assification by liveness *A*nd *S*afety *P*artition (Safety and Liveness generate the lattice; Cosafety and Coliveness are their De Morgan duals)—coexists with several adjacent refinements of ω-language classification that are not directly competitive but enrich the positioning. *Wadge degrees* (Wadge [62]) impose a continuous topological hierarchy on ω-languages refining the Borel levels; within the ω-regular fragment, each CLASP cell corresponds to a finite union of Wadge classes. The exact topological classification of the ω-regular sets is the *Wagner hierarchy* (Wagner [63]), the effective restriction of the Wadge hierarchy to Büchi-recognisable languages; the CLASP cells are a coarsening of the whole hierarchy: Safety and Cosafety pin down the low $\mathbf{\Sigma}^0_1/\mathbf{\Pi}^0_1$ strata, while Liveness and Coliveness each fold together ω-regular sets of arbitrarily high Wagner degree (every dense, respectively empty-interior, ω-regular language), so the four Safety/Cosafety/Liveness/Coliveness blocks group rather than resolve the finer Wagner degrees. The standard reference for this topological and Borel theory of ω-languages is Perrin–Pin [51] (alongside Staiger [58] and Thomas [60]). *Hyperproperties* (Clarkson–Schneider [17]), monitored at runtime for HyperLTL by Bonakdarpour–Finkbeiner [9], extend the setting to properties of sets of traces and are orthogonal to our single-trace partition; the CLASP construction lifts to hyperproperties via tuple-trace observations but the lift is not developed here. The alternation hierarchy of the modal μ-calculus (introduced by Kozen [36]) refines temporal classification at the modal level; the LTL fragment we classify sits at low alternation depths within that hierarchy. Finally, our entire setting lives in the regular layer of the Chomsky hierarchy (ω-regular $\equiv$ Büchi-recognisable, cf. Thomas [60]); the CLASP partition is a refinement *inside* that single Chomsky-level, with no escape upward.

Prior intersection enumeration: Peled–Havelund 2019. Peled and Havelund [50] introduced a four-class refinement of Lamport's safety/liveness dichotomy using *guarantee* (= cosafety) and *morbidity* (= coliveness), and added a fifth class *quaestio* to cover the residual properties. Their classification axis is verdict-availability: for each property they ask whether a finite *negative* verdict is Always (AFR), Sometimes (SFR), or Never (NFR) available; and symmetrically for the positive verdict (AFS / SFS / NFS). This yields a 3×3 partition of all temporal properties into nine cells, with the dictionary

$$AFR \equiv \mathsf{S}, \qquad AFS \equiv \mathsf{C}, \qquad NFR \equiv \mathsf{L}, \qquad NFS \equiv \mathsf{K},$$

and *quaestio* occupies the central cell $SFR \cap SFS = \overline{\mathsf{S} \cup \mathsf{L}} \cap \overline{\mathsf{C} \cup \mathsf{K}}$. Havelund and Peled revisit monitorability for runtime verification in [28]. The exact correspondence with our membership-axis lexicon is recorded in Remark 3.6; the polarity schemas of the four classes are catalogued in Remark 3.5. Broy [11] refines the safety–liveness classification according to *realizability*.

The missing membership-axis lattice. The topological characterization recognizes four class families (Safety S, Cosafety C, Liveness L, Coliveness K). Their full *membership-axis* algebra–determining which of the $2^4 - 1 = 15$ subset-lattice cells over $\{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$ are nonempty, including the four triple cells and the quadruple–is not present in the Peled–Havelund decomposition, which terminates at the nine $XFR \times XFS$ cells (each a two-class intersection in our terminology). The four *corner* Peled–Havelund cells coincide with our pair-intersections (Theorem 8.2), but the five remaining Peled–Havelund cells

are not intersections of the four classes: the four edge cells also exclude classes (e.g. $AFR \cap SFS = \mathsf{S} \cap \overline{\mathsf{C}} \cap \overline{\mathsf{K}}$), and the central cell lies outside their union.

Research questions. Given the state of the art above, this paper addresses the following master research question:

> *How does the topological CLASP classification of ω-regular properties — Safety, Cosafety, Liveness, and Coliveness — extend operationally to runtime verification, and what semantic content does the coliveness class carry in this extended framework?*

The question decomposes into six sub-questions, each addressed by a dedicated portion of the paper:

- SQ1**:** (section 10) How does verdict-alphabet cardinality $n \in \{1, 2, 3, 4, \infty\}$ determine which intersection cells a monitor system can resolve? Answered by the verdict-alphabet ladder (degenerate baseline at n=1, section 11.1; definitive-verdict resolution at $n \in \{2, 3\}$, revocable refinement at n=4; quantitative at n=∞) and the per-rung coverage matrix.
- SQ2**:** (section 10.6) What is the canonical class attribution of an arbitrary verdict monitor to the input alphabet? Answered by the universal attribution theorem (Theorem 10.23).
- SQ3**:** (section 10.6) When does monitor composition lift a base monitor's cell to higher cells of the intersection algebra? Answered by the composition class-promotion theorem (Theorem 10.20).
- SQ4**:** (section 9) What is the semantic content of the coliveness class K on ω-regular properties? Answered by five readings of the K-class (Remarks 9.6 and 9.10 to 9.13), all operating at Layer 2 (verdict-trace properties produced by monitors of the input, per Theorem 10.23).
- SQ5**:** (Remark 9.6) Does the intersection $\mathsf{L} \cap \mathsf{K}$ admit a further compositional refinement? Answered by the LK/KL bipartition into Persistence (Σ_2^0) and Recurrence (Π_2^0) polarities.
- SQ6**:** (section 10.7) What structure on a verdict trace lifts a refutation-only monitorable property to finite-time confirmation? Answered by the refinement-based confirmation framework, with intrinsic (V, A) regularity and extrinsic regularity as the two in-framework sources; a third source (active-tuning, parametric-monitor architecture) is left to future work.

Contributions. This paper presents the membership-axis formulation of the Peled–Havelund classification and proves its equivalence to the verdict-axis original; the classification itself is not new (Remark 8.17). The contributions are organised in two blocks — *foundational* (F1–F8, combinatorial-topological framework) and *operational* (O1–O6, verdict-monitor ladder and confirmation framework). To keep the novelty surface explicit: the genuinely new results are the 16-cell $\leftrightarrow$ PH 9-cell bijection up to the two degenerate singletons (**F3**), the parallel topological/prefix-quantifier dual derivation of the empty-cell results (**F4**), the universal-attribution theorem (**O2**), and the composition class-promotion theorem (**O3**); the remaining items are explicit repackaging or exposition — the four primitive classes themselves, the De Morgan lifting (**F5**), the worked-example catalogue (**F6**), and the

variable-space taxonomy (**F8**) — retained for completeness and reproducibility rather than claimed as research novelty.

Foundational layer.

F1**:** *Cell enumeration.* A subset-monotone enumeration of all 15 intersection cells of $\{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$ with 12 non-empty cells and 3 provably empty cells (Theorems 5.4 and 5.5).

F2**:** *Totality.* A totality theorem (Theorem 8.4) showing that the four primitive classes together with the Peled–Havelund quaestio cell partition $\mathcal{P}(\Sigma^\omega)$ exactly into 16 exclusive cells (11 non-empty, 5 empty); the two count formulations are reconciled in Remark 8.16.

F3**:** *Bijection 16-cell ↔ PH 9-cell.* A bijection (Theorem 8.2 and Remark 8.16) between our 16-cell membership-axis partition and the 9-cell verdict-axis partition of [50]: nine non-degenerate non-empty cells coincide bijectively with the $XFR \times XFS$ cells; the two extra non-empty cells of our partition are the degenerate singletons $\{\Sigma^\omega\}$ and $\{\varnothing\}$, which Peled–Havelund aggregate inside their $AFR \cap AFS$ cell.

F4**:** *Dual derivations.* Two independent derivations of the empty-cell results: a topological derivation (Theorem 5.4) and a prefix-quantifier logical derivation (Proposition 8.10). The second transfers to settings (e.g. finite-trace prefixes) where the Cantor topology is unavailable.

F5**:** *De Morgan duality on the partition.* A De Morgan duality structure (Theorem 8.13 and Corollary 8.14) on the 16-cell partition, induced by the LTL temporal dualities (Theorem 2.4): four swap-pairs of non-empty cells, three self-dual non-empty cells, and a matching structure on the empty cells.

F6**:** *Expository: literature-anchored example catalog.* A worked-example catalog (section 9 and table 8) attributing each cell to its earliest LTL witness, predominantly from Peled–Havelund Fig. 1.

F7**:** *Horizon refinement of the enumeration.* An explicit horizon-projection operator (Definition 6.1) and a clopen-collapse proposition (Proposition 6.2) showing that on any finite horizon the 11-cell exclusive partition reduces to $\mathsf{S} \cap \mathsf{C}$; a default-expression table (table 2) recording the canonical unbounded-LTL witness per cell, and a bounded-operator migration table (table 3, Proposition 6.4) showing how each cell collapses to $\mathsf{S} \cap \mathsf{C}$ once G, F, U, R are replaced with bounded counterparts. The horizon-sensitivity gradient (section 6.4) labels each cell with the horizon discipline that the implicit-ω convention of section 3 silently fixes; reconciles with Remark 5.8 via Remark 6.11.

F8**:** *Expository: variable-space catalogue of traces and monitors.* An explicit catalogue (section 7) of the 23 axes parameterising a trace (T_1–T_{12}, table 15) and a monitor (D_1–D_{11}, table 14), together with the unique mandatory cross-axis dependency between them (Proposition 7.1: meta-trace alphabet at layer $k{+}1$ equals verdict alphabet at layer k) and the horizon-independence remark (Remark 7.2) for asymmetric horizons across layers. A defaults table (table 16) declares the slice of the 23-axis space that this paper inhabits and identifies eight previously-implicit trace axes. An effective-verdict cardinality operator (Definition 7.4, Proposition 7.5) identifies the cells $\{\Sigma^\omega, \varnothing, \mathsf{L} \cap \mathsf{K}\}$ as the effective-1 class of LTL_3, an information-theoretic grouping orthogonal to the operational monitorability gradient of Remark 5.8. Three layer regimes (stable bootstrap, collapsing, explosive) classify how the iteration layer-k → layer-$(k{+}1)$ behaves.

Operational layer.

- O1**:** *Verdict-alphabet ladder.* A ladder (section 10) organising verdict-monitor systems by alphabet cardinality $n \in \{1, 2, 3, 4, \infty\}$, with the trivial n=1 rung (section 11.1) as degenerate baseline, the qualitative rungs $n \in \{2, 3, 4\}$ as the discriminating core, and n=∞ as the quantitative limit, together with a coverage matrix recording per rung the newly resolved intersection cell, the witness mechanism, and the canonical LTL example.
- O2**:** *Universal attribution.* A universal-attribution theorem (Theorem 10.23): every verdict monitor with $|V| \geq 2$ canonically attributes to the input alphabet persistence and response pullback properties, the pre-images of $\mathsf{L} \cap \mathsf{K}$ properties of its verdict trace, via the continuous monitor-extension map, with an explicit characterisation of when a pullback is itself in $\mathsf{L} \cap \mathsf{K}$ (non-absorbing verdict) and when it is not (absorbing verdict).
- O3**:** *Composition class-promotion.* A class-promotion theorem (Theorem 10.20) lifting a base monitor's intersection cell to higher cells of the algebra under Layer-2 composition with a verdict-trace property (regularity condition: open / prefix-surjective ω-extension).
- O4**:** *Five semantic readings of* K. Five remarks on the coliveness class (Remarks 9.6 and 9.10 to 9.13), including the framework-relativity diagnosis (K-membership is a trace-only verdict; model checking and refinement-based confirmation can lift it) and the LK/KL bipartition of $\mathsf{L} \cap \mathsf{K}$ into Persistence (FGp, Σ_2^0) and Recurrence (GFp, Π_2^0) polarities, verified to be structurally unique to $\mathsf{L} \cap \mathsf{K}$ within the 16-cell partition.
- O5**:** *Refinement-based confirmation framework.* A framework (section 10.7) lifting refutation-only verdict traces to finite-time confirmation via the (D, A_0, U, E) refinement tuple, with intrinsic (V, A) regularity and extrinsic regularity (inherited from a function behind the monitor) as the two in-framework source types. A third source — active-tuning regularity internal to a parametric monitor architecture — is identified as a natural further direction beyond the verdict-only scope of the present paper.
- O6**:** *Extended verdict-alphabet ladder and meta-classification.* An extension of the ladder of **O1** to include the trivial rung n=1 (table 11) and the split sub-ladder at n=2 (the safety/cosafety regions $\mathsf{S} \setminus \mathsf{C}$, $\mathsf{C} \setminus \mathsf{S}$ and clopen $\mathsf{S} \cap \mathsf{C}$, section 11.2); and the meta-classification conjecture (Conjecture 11.1) stating that the four Level-1 CLASP classes themselves classify under CLASP applied one level up: the limit-classes L, K sit at meta-level Morbidity ($\mathsf{S} \cap \mathsf{K}$), the fixed-witness classes S, C sit at meta-level Mixed ($\mathsf{L} \cap \mathsf{K}$), with no Level-1 class admitting a finite positive verdict at the meta-level. Related in spirit to the descriptive-set-theoretic study of monitorability by Camerlo–Dagnino [12], whose $\mathbf{\Pi}_3^0$ bound concerns countable spaces (Conjecture 11.1).

Novelty discipline. The classification is due to Peled–Havelund ([50, 28]). The prefix definitions and topological correspondences are classical [2, 3, 38, 7, 45, 19]. Novelty claims are restricted to the equivalence bijection (**F3**), the parallel logical/topological dual derivations (**F4**), the universal-attribution theorem (**O2**), and the composition class-promotion theorem (**O3**); the totality theorem, the De Morgan duality structure, and the literature-anchored worked-example catalog are structural or expository, not claimed as research novelty.

Related work. The classical foundations summarised above (Alpern–Schneider [2, 3], Manna–Pnueli [48], Kupferman–Vardi [38], Peled–Havelund [50, 28]) together with the runtime-verification tutorial of Falcone, Havelund, and Reger [25] fix the substrate against which this paper places its contributions. We position the work along four axes that are not subsumed by the foundations themselves.

Topology of monitorability. Diekert and Leucker [19] characterise monitorability of ω-regular properties via the Borel hierarchy and identify the threshold at which a property becomes non-monitorable; Aceto, Achilleos, Francalanza, Ingólfsdóttir, and Lehtinen [1] give an operational reformulation that aligns finite-trace verdict structure with regular-property classes. Camerlo and Dagnino [12] measure the Borel complexity of the family of monitorable subsets of a countable space. These works fix the *monitorability boundary* at the topological layer; our verdict-alphabet ladder (section 10) crosses that boundary in the opposite direction by asking which intersection cells become *resolvable* as the verdict alphabet grows from $n = 2$ to $n = \infty$, and which remain unresolved at every n (all cells outside $\mathsf{S} \cup \mathsf{C}$, of which only $\mathsf{L} \cap \mathsf{K}$ is non-monitorable throughout).

Verdict-alphabet variants. The three-valued LTL_3 semantics of Bauer, Leucker, and Schallhart [7] uses the $\{\top, \bot, ?\}$ verdict alphabet; RV-LTL [6] refines the inconclusive verdict into presumably-true and presumably-false, giving four verdicts; Jaǩsić et al. [34] give a semiring-based framework that unifies qualitative and quantitative runtime verification. The survey by Sánchez et al. [54] reviews challenges for runtime verification in advanced application domains; Henzinger and collaborators study quantitative and approximate monitoring [31] and quantitative safety and liveness [30]. Each of these works fixes a single n and develops it in depth; our contribution is orthogonal: we present a single ladder $n \in \{1, 2, 3, 4, \infty\}$ (with the trivial n=1 rung as degenerate baseline, section 11.1) with a uniform coverage matrix mapping each rung's verdict cardinality to the intersection cells it resolves, and show that the definitive-verdict resolving power saturates at $n = 3$: no richer alphabet issues a sound definitive verdict beyond the good and bad prefixes already reported by LTL_3 (Proposition 10.17); the $n = 4$ and $n = \infty$ rungs add revocable and quantitative information only.

Quantitative refinements. Maler and Nickovic [47] introduce Signal Temporal Logic (STL) for continuous-signal monitoring; Donzé and Maler [22] develop its robust quantitative semantics; Tabuada and Neider [59] give a robust semantics for LTL itself (rLTL). These works supply real-valued or partially ordered verdict domains for finer discrimination of property violations. We position them as the $n \to \infty$ rung of the ladder (section 10.5): quantitatively informative but adding no new qualitative cell beyond those resolved by $n = 3$.

Enforcement. Falcone, Fernandez, and Mounier [24] characterise what can be *enforced* at runtime by enforcement monitors that edit the execution sequence (suppressing or inserting events), extending Schneider's EM-enforceability [55]. This frames enforcement as a transducer applied to the trace. A structurally distinct enforcement mechanism — via parametric monitor adaptation rather than verdict-edit — is a natural further direction whose refinement layer is sketched in section 10.7; the two mechanisms are not substitutes: edit-based enforcement modifies the monitored execution; adaptation modifies the monitor's parameters and hence the alphabet V itself.

Paper structure. Sections 2, 4, 5 and 8 develop the foundational layer (**F1**–**F6**): prefix-quantifier and topological definitions of the four classes, the 15-cell subset-monotone enumeration, the 16-cell exclusive partition with totality theorem, the bijection to the Peled–Havelund verdict-axis partition, the dual derivations of empty cells, and the De Morgan duality structure. Section 6 adds the horizon refinement (**F7**): explicit horizon-projection, clopen collapse at finite horizon, default-expression and bounded-operator migration tables, and the horizon-sensitivity gradient. Section 7 catalogues the full 23-axis variable space (**F8**): twelve trace axes T_1–T_{12}, eleven monitor axes D_1–D_{11}, the meta-layer alphabet constraint that ties their composition, the slice declared by this paper, and the effective-verdict cardinality grouping that unifies the cells from which an LTL_3 monitor extracts no variation. Section 9 catalogues the literature-anchored worked examples (**F6**). Section 10 develops the operational layer (**O1**–**O5**): the verdict-alphabet ladder $n \in \{1, 2, 3, 4, \infty\}$ (with the trivial baseline at n=1 recorded in section 11.1), the universal-attribution theorem, the composition class-promotion theorem, five semantic readings of the coliveness class, and the refinement-confirmation framework. Section 11 extends the ladder to the trivial rung n=1, the split sub-ladder at n=2, and states the meta-classification conjecture (Conjecture 11.1) (**O6**). Section 12 closes with implications and open directions.

A note on length. This paper exceeds the customary LMCS length guideline because it presents a *complete* classification: the sixteen-cell totality, the dual empty-cell derivations, the verdict-alphabet ladder, and the per-cell worked examples are individually compact but jointly exhaustive, and omitting any part would leave the classification incomplete.

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2. Preliminaries: Syntax and Semantics

Let AP be a finite set of atomic propositions and let $\Sigma = 2^{AP}$. Traces are elements of Σ^ω; properties are subsets of Σ^ω.

2.1. **Trace Structures.**

Definition 2.1 (Prefix Relation)**.** For $\pi \in \Sigma^*$ and $\tau \in \Sigma^\omega$, define

$$\pi \sqsubseteq \tau \iff \exists n \in \mathbb{N}\,\big(|\pi| = n \wedge \forall i\,(0 \le i < n \Rightarrow \pi_i = \tau_i)\big).$$

2.2. **LTL Syntax.**

Definition 2.2 (LTL Syntax)**.** The set $\mathcal{L}_{\mathrm{LTL}}(AP)$ is the least set such that:

(1) If $p \in AP$, then $p \in \mathcal{L}_{\mathrm{LTL}}(AP)$.
(2) If $\varphi, \psi \in \mathcal{L}_{\mathrm{LTL}}(AP)$, then $\neg\varphi$, $\varphi \wedge \psi$, $X\varphi$, $\varphi U \psi$, and $\varphi R \psi$ are in $\mathcal{L}_{\mathrm{LTL}}(AP)$.
(3) $F\varphi$ abbreviates $\top U \varphi$, and $G\varphi$ abbreviates $\neg F \neg \varphi$.

2.3. **LTL Semantics.**

Definition 2.3 (LTL Semantics)**.** For each trace $\sigma \in \Sigma^\omega$ and each position $i \in \mathbb{N}$:

$$\begin{aligned}
\sigma, i \models p &\iff p \in \sigma_i, \\
\sigma, i \models \neg\varphi &\iff \neg(\sigma, i \models \varphi), \\
\sigma, i \models \varphi \wedge \psi &\iff (\sigma, i \models \varphi) \wedge (\sigma, i \models \psi), \\
\sigma, i \models X\varphi &\iff \sigma, i+1 \models \varphi, \\
\sigma, i \models \varphi U \psi &\iff \exists j \ge i\,\big(\sigma, j \models \psi \wedge \forall k\,(i \le k < j \Rightarrow \sigma, k \models \varphi)\big), \\
\sigma, i \models \varphi R \psi &\iff \forall j \ge i\,\Big((\forall k\,(i \le k < j \Rightarrow \neg(\sigma, k \models \varphi))) \Rightarrow \sigma, j \models \psi\Big).
\end{aligned}$$

Define denotation:

$$[\![\varphi]\!] := \{\sigma \in \Sigma^\omega \mid \sigma, 0 \models \varphi\}.$$

Theorem 2.4 (Temporal Dualities)**.** *For every formula φ and every formulas α, β,*

$$\begin{aligned}
\neg G\varphi \equiv F\neg\varphi, \qquad & \neg F\varphi \equiv G\neg\varphi, \\
\neg(\alpha U \beta) \equiv (\neg\alpha) R (\neg\beta), \qquad & \neg(\alpha R \beta) \equiv (\neg\alpha) U (\neg\beta).
\end{aligned}$$

Proof. Each equivalence is obtained by replacing each abbreviation using Definition 2.2, then negating the quantified semantic clause in Definition 2.3 and applying propositional equivalence pointwise. □

3. Foundational Class Definitions

This section fixes the four classes whose intersections generate the entire partition—Safety, Cosafety, Liveness, and Coliveness. The following definitions are adopted exactly as project primitives.

Definition 3.1 (Safety)**.** A property $\Phi \subseteq \Sigma^\omega$ is Safety iff

$$\forall \sigma \notin \Phi,\ \exists \pi \sqsubseteq \sigma \text{ s.t. } \forall \tau \in \Sigma^\omega,\ (\pi \sqsubseteq \tau \Rightarrow \tau \notin \Phi).$$

The quantifier-order convention used here (universal over non-satisfying traces, existential over a finite refutation prefix) is discussed in Remark 3.7.

Definition 3.2 (Cosafety)**.** A property $\Phi \subseteq \Sigma^\omega$ is Cosafety iff

$$\forall \sigma \in \Phi,\ \exists \pi \sqsubseteq \sigma \text{ s.t. } \forall \tau \in \Sigma^\omega,\ (\pi \sqsubseteq \tau \Rightarrow \tau \in \Phi).$$

Definition 3.3 (Liveness)**.** A property Φ is Liveness iff

$$\forall \pi \in \Sigma^*,\ \exists \tau \in \Sigma^\omega \text{ s.t. } \pi \sqsubseteq \tau \wedge \tau \in \Phi.$$

Definition 3.4 (Coliveness)**.** A property Φ is Coliveness iff

$$\forall \pi \in \Sigma^*,\ \exists \tau \in \Sigma^\omega \text{ s.t. } \pi \sqsubseteq \tau \wedge \tau \notin \Phi.$$

Remark 3.5 (Polarity Schemas)**.** Each class definition admits an equivalent "polarity" formulation via prefix quantifiers: FIN-INV(Φ) $\Leftrightarrow$ Safety, FIN-CONF(Φ) $\Leftrightarrow$ Cosafety, NO-FIN-INV(Φ) $\Leftrightarrow$ Liveness, NO-FIN-CONF(Φ) $\Leftrightarrow$ Coliveness. Equivalence follows by literal comparison of Definitions 3.1 to 3.4.

Remark 3.6 (Equivalence with the Peled–Havelund Verdict-Axis Classes)**.** Peled and Havelund [50] classify temporal properties by the availability of finite verdicts. Their AFR (*always-finite-refutation*) coincides with our S by Definition 3.1, since the existence of an invalidating prefix for every non-satisfying trace is exactly "a finite refutation is always available". Symmetrically, AFS coincides with C, NFR with L, and NFS with K. Their intermediate classes $SFR = \overline{\mathsf{S}} \cap \overline{\mathsf{L}}$ and $SFS = \overline{\mathsf{C}} \cap \overline{\mathsf{K}}$ are definable from our four classes but are not themselves among our primitive families; they emerge as derived constructs in section 8.

Remark 3.7 (The "Always" Quantifier in Classical Safety)**.** The outer universal quantifier $\forall \sigma \notin \Phi$ in Definition 3.1 carries the "*always*" reading: *every* non-satisfying trace admits a finite invalidating prefix. This is the classical Lamport–Alpern–Schneider [41, 2, 3] formulation of safety. Its Boolean negation,

$$\neg\mathsf{S} \;=\; \big\{\Phi : \exists \sigma \notin \Phi,\ \forall \pi \sqsubseteq \sigma,\ \exists \tau \sqsupseteq \pi,\ \tau \in \Phi\big\},$$

aggregates two distinct cases that the classical dichotomy "safety vs. non-safety" leaves implicit:

(1) Some violations admit finite invalidating prefixes and some do not ($SFR = \overline{\mathsf{S}} \cap \overline{\mathsf{L}}$, the residual case in Peled–Havelund's tripartition).

(2) No violation admits a finite invalidating prefix ($NFR \cap \overline{\mathsf{S}}$: every prefix extends to satisfaction, which is classical liveness L by Theorem 4.6, here minus Σ^ω, the one property that is both L and S).

Definitions 3.1 and 3.3 commit to the classical quantifier convention: S is the *always* class, L is the *never* class, and the "sometimes" residual SFR is definable but not primitive. The corresponding observation for the satisfaction axis is symmetric: $\mathsf{C} = AFS$ (*always*) and $\mathsf{K} = NFS$ (*never*), with $SFS = \overline{\mathsf{C}} \cap \overline{\mathsf{K}}$ (*sometimes*) emerging by Boolean complementation.

Remark 3.8 (The Tripartite (Always/Sometimes/Never) Refinement as Cross-Classification)**.** The tripartition of each axis into *always*, *sometimes*, and *never* categories is an *introduction* of Peled–Havelund [50] relative to the classical safety/liveness dichotomy. In particular it is *not* a promotion of safety: the class S itself remains the classical safety class (= AFR), and the "sometimes" modifier on either axis is an *orthogonal* sub-indexing. When we write a cell as $AFR \cap SFS$ (equivalently $\{\mathsf{S}\}_{\text{excl}}$ in the membership-tuple notation of section 8.1), we are reading off two independent facts about Φ:

- "Φ is safety" on the refutation axis (AFR);
- "Φ is neither cosafety nor coliveness" on the satisfaction axis (SFS).

The conjunction $AFR \cap SFS$ refines the classical statement "Φ is safety" by adding an independent satisfaction-axis constraint. The tripartite refinement is therefore a *cross-classification* rather than a redefinition of any of the four primitive classes. This explains why all nine $XFR \times XFS$ cells of Peled–Havelund [50, Fig. 1] are realised by LTL witnesses without invoking any new quantifier primitive beyond the four of Definitions 3.1 to 3.4.

4. Topological Layer

We now read the four classes through the Cantor topology on Σ^ω, where Safety and Cosafety appear as closed and open sets and Liveness and Coliveness as dense and empty-interior sets.

4.1. Cantor Basis and Operators.

Definition 4.1 (Cylinder Sets)**.** For each finite word $\pi \in \Sigma^*$,

$$[\pi] := \{\tau \in \Sigma^\omega \mid \pi \sqsubseteq \tau\}.$$

Definition 4.2 (Cantor Topology)**.** Let $\mathcal{B} := \{[\pi] \mid \pi \in \Sigma^*\}$. Define

$$\tau_C := \left\{ \bigcup \mathcal{U} \mid \mathcal{U} \subseteq \mathcal{B} \right\}.$$

Lemma 4.3 (Basis Property)**.** *$\mathcal{B}$ is a basis of a topology on Σ^ω.*

Proof. **Coverage.** $[\varepsilon] = \Sigma^\omega$.

Refinement. Assume $\tau \in [\pi_1] \cap [\pi_2]$. Then $\pi_1 \sqsubseteq \tau$ and $\pi_2 \sqsubseteq \tau$. Thus one of π_1, π_2 extends the other. Let π be the longer one. Then $\tau \in [\pi]$ and $[\pi] \subseteq [\pi_1] \cap [\pi_2]$.

Both basis axioms hold; therefore $\mathcal{B}$ generates a topology. □

Definition 4.4 (Topological Operators)**.** For each $X \subseteq \Sigma^\omega$:

$$\mathrm{cl}(X) := \bigcap \{F \subseteq \Sigma^\omega \mid X \subseteq F \wedge F \text{ closed}\},$$

$$\mathrm{int}(X) := \bigcup \{O \subseteq \Sigma^\omega \mid O \subseteq X \wedge O \text{ open}\},$$

$$\mathrm{bd}(X) := \mathrm{cl}(X) \setminus \mathrm{int}(X).$$

Lemma 4.5 (Cylinder Characterizations)**.** *For every $\sigma \in \Sigma^\omega$ and $X \subseteq \Sigma^\omega$:*

$$\sigma \in \mathrm{cl}(X) \iff \forall \pi \sqsubseteq \sigma\ ([\pi] \cap X \neq \varnothing),$$
$$\sigma \in \mathrm{int}(X) \iff \exists \pi \sqsubseteq \sigma\ ([\pi] \subseteq X).$$

Proof. By Lemma 4.3, neighborhoods are unions of cylinders. The closure clause is equivalent to nonempty intersection of every basis neighborhood with X. The interior clause is equivalent to existence of a basis neighborhood contained in X. □

4.2. **Class–Topology Correspondence.**

Theorem 4.6 (Topological Characterization)**.** *For each property $\Phi \subseteq \Sigma^\omega$:*

$$\Phi \in \mathsf{S} \iff \Phi \textit{ is closed},$$
$$\Phi \in \mathsf{C} \iff \Phi \textit{ is open},$$
$$\Phi \in \mathsf{L} \iff \mathrm{cl}(\Phi) = \Sigma^\omega,$$
$$\Phi \in \mathsf{K} \iff \mathrm{int}(\Phi) = \varnothing.$$

Proof. The Safety ⇔ closed and Liveness ⇔ dense equivalences are the classical topological characterisation of Alpern and Schneider [2] (in automata-theoretic form [3, Thms. 1–2]; see also [19, §3.2]). The Cosafety ⇔ open and Coliveness ⇔ empty-interior equivalences follow by complementation: Φ is open iff its complement is closed, and $\mathrm{int}(\Phi) = \varnothing$ iff $\Sigma^\omega \setminus \Phi$ is dense, i.e. Φ is codense in the sense of [12, §2]. In each direction, the cylinder characterisation of Lemma 4.5 furnishes the prefix-quantifier translation: an invalidating cylinder $[\pi] \subseteq \Sigma^\omega \setminus \Phi$ in the safety direction; a confirming cylinder $[\pi] \subseteq \Phi$ in the cosafety direction; $[\pi] \cap \Phi \neq \varnothing$ for every π in the liveness direction; and absence of $[\pi] \subseteq \Phi$ in the coliveness direction. □

Theorem 4.7 (Boundary Characterization)**.** *For each $\Phi \subseteq \Sigma^\omega$:*

(1) Φ *is clopen* $\iff \mathrm{bd}(\Phi) = \varnothing$.
(2) $\Phi \in \mathsf{L} \cap \mathsf{K} \iff \mathrm{bd}(\Phi) = \Sigma^\omega$.

Proof. (1) If Φ is clopen, $\mathrm{bd}(\Phi) = \mathrm{cl}(\Phi) \setminus \mathrm{int}(\Phi) = \Phi \setminus \Phi = \varnothing$. Conversely, $\mathrm{bd}(\Phi) = \varnothing$ gives $\mathrm{cl}(\Phi) \subseteq \mathrm{int}(\Phi)$; combined with $\mathrm{int}(\Phi) \subseteq \Phi \subseteq \mathrm{cl}(\Phi)$, we get $\mathrm{int}(\Phi) = \Phi = \mathrm{cl}(\Phi)$.

(2) If $\Phi \in \mathsf{L} \cap \mathsf{K}$, Theorem 4.6 gives $\mathrm{cl}(\Phi) = \Sigma^\omega$ and $\mathrm{int}(\Phi) = \varnothing$, so $\mathrm{bd}(\Phi) = \Sigma^\omega$. Conversely, $\mathrm{bd}(\Phi) = \Sigma^\omega$ forces $\mathrm{cl}(\Phi) = \Sigma^\omega$ (since $\mathrm{bd}(\Phi) \subseteq \mathrm{cl}(\Phi)$) and $\mathrm{int}(\Phi) = \varnothing$ (since $\mathrm{cl}(\Phi) \setminus \mathrm{int}(\Phi) = \Sigma^\omega$). □

5. Algebra of Class Combinations

Definition 5.1 (Class Families)**.**

$$\mathsf{S} := \{\Phi \subseteq \Sigma^\omega \mid \Phi \text{ is Safety}\}, \quad \mathsf{C} := \{\Phi \subseteq \Sigma^\omega \mid \Phi \text{ is Cosafety}\},$$
$$\mathsf{L} := \{\Phi \subseteq \Sigma^\omega \mid \Phi \text{ is Liveness}\}, \quad \mathsf{K} := \{\Phi \subseteq \Sigma^\omega \mid \Phi \text{ is Coliveness}\}.$$

Remark 5.2 (Empty Property and Singleton-Class Notation)**.** Note: $\varnothing$ denotes the empty property; $\{\varnothing\}$ denotes the singleton class containing the empty property.

5.1. **Constructive Witness Properties.** Fix $p \in AP$ and define:

$$\Phi_{Gp} = [\![Gp]\!], \qquad \Phi_{Fp} = [\![Fp]\!], \quad \Phi_{GFp} = [\![GFp]\!],$$
$$\Phi_{FGp} = [\![FGp]\!], \qquad \Phi_{p0} = [\![p]\!], \qquad \Phi_{\top} = [\![\top]\!] = \Sigma^{\omega},$$
$$\Phi_{\bot} = [\![\bot]\!] = \varnothing.$$

Lemma 5.3 (Class Membership of Canonical Witnesses)**.** *The following hold:*

$$\Phi_{Gp} \in \mathsf{S} \cap \mathsf{K} \setminus (\mathsf{C} \cup \mathsf{L}),$$
$$\Phi_{Fp} \in \mathsf{C} \cap \mathsf{L} \setminus (\mathsf{S} \cup \mathsf{K}),$$
$$\Phi_{GFp} \in \mathsf{L} \cap \mathsf{K} \setminus (\mathsf{S} \cup \mathsf{C}),$$
$$\Phi_{FGp} \in \mathsf{L} \cap \mathsf{K} \setminus (\mathsf{S} \cup \mathsf{C}),$$
$$\Phi_{p0} \in \mathsf{S} \cap \mathsf{C} \setminus (\mathsf{L} \cup \mathsf{K}),$$
$$\Phi_{\top} \in \mathsf{S} \cap \mathsf{C} \cap \mathsf{L} \setminus \mathsf{K},$$
$$\Phi_{\bot} \in \mathsf{S} \cap \mathsf{C} \cap \mathsf{K} \setminus \mathsf{L}.$$

Proof. Each claim follows from the prefix quantifiers of Definitions 3.1 to 3.4. Φ_{Gp}: any violation has an invalidating prefix (first $\neg p$ position), so S; every prefix extends to a violation (set p false once), so K; not C or L (prefixes starting with $\neg p$ have no satisfying extension). Φ_{Fp}: the first p-position provides the confirming prefix (C); every prefix extends to a satisfaction (L); not S or K (two-way continuations from any prefix). Φ_{GFp}: every prefix extends to satisfaction (infinitely alternate p) and to violation (continue $\neg p$ forever), giving $\mathsf{L} \cap \mathsf{K}$; not S or C (no finite confirmation or invalidation). Φ_{FGp}: same argument gives $\mathsf{L} \cap \mathsf{K}$. Φ_{p0}: length-one prefix decides membership, giving $\mathsf{S} \cap \mathsf{C}$; not L (prefix $\neg p$ blocks) or K (prefix p locks in). $\Phi_{\top}, \Phi_{\bot}$: immediate. □

5.2. **Subset-Monotone Enumeration of 15 Intersection Cells.**

Theorem 5.4 (Enumeration Theorem for All Nonempty Index Families)**.** *Let* $\mathcal{F} = \{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$. *For each nonempty subfamily* $\Gamma \subseteq \mathcal{F}$, *define* $I(\Gamma) := \bigcap_{X \in \Gamma} X$. *The complete classification of the 15 intersections is given in table 1.*

Index family Γ	Classification
$\{\mathsf{S}\}$	$I(\Gamma) = \mathsf{S}$ (nontrivial; witness Φ_{Gp})
$\{\mathsf{C}\}$	$I(\Gamma) = \mathsf{C}$ (nontrivial; witness Φ_{Fp})
$\{\mathsf{L}\}$	$I(\Gamma) = \mathsf{L}$ (nontrivial; witness Φ_{Fp})
$\{\mathsf{K}\}$	$I(\Gamma) = \mathsf{K}$ (nontrivial; witness Φ_{Gp})
$\{\mathsf{S},\mathsf{C}\}$	clopen nontrivial class; witness Φ_{p0}
$\{\mathsf{S},\mathsf{L}\}$	singleton class $\{\Sigma^\omega\}$
$\{\mathsf{S},\mathsf{K}\}$	nontrivial; witnesses $\Phi_{Gp}, \Phi_\bot$
$\{\mathsf{C},\mathsf{L}\}$	nontrivial; witnesses $\Phi_{Fp}, \Phi_\top$
$\{\mathsf{C},\mathsf{K}\}$	singleton class $\{\varnothing\}$
$\{\mathsf{L},\mathsf{K}\}$	boundary-dense nontrivial class; witness Φ_{GFp}
$\{\mathsf{S},\mathsf{C},\mathsf{L}\}$	singleton class $\{\Sigma^\omega\}$
$\{\mathsf{S},\mathsf{C},\mathsf{K}\}$	singleton class $\{\varnothing\}$
$\{\mathsf{S},\mathsf{L},\mathsf{K}\}$	empty class
$\{\mathsf{C},\mathsf{L},\mathsf{K}\}$	empty class
$\{\mathsf{S},\mathsf{C},\mathsf{L},\mathsf{K}\}$	empty class

Table 1: Complete enumeration of all 15 finite intersections in the *subset-monotone* indexing scheme. Note that this is *not* a partition of $\mathcal{P}(\Sigma^\omega)$: cells share the singletons $\{\Sigma^\omega\}$ and $\{\varnothing\}$ (rows $\{\mathsf{S},\mathsf{L}\}$, $\{\mathsf{S},\mathsf{C},\mathsf{L}\}$ share $\{\Sigma^\omega\}$; rows $\{\mathsf{C},\mathsf{K}\}$, $\{\mathsf{S},\mathsf{C},\mathsf{K}\}$ share $\{\varnothing\}$); the exclusive 16-cell partition of Theorem 8.4 resolves this overlap explicitly. See Remark 8.16 for the bijective reconciliation of the 15 subset-monotone cells with the 16-cell partitional view.

Proof. For $\Gamma \in \big\{\{\mathsf{S}\},\{\mathsf{C}\},\{\mathsf{L}\},\{\mathsf{K}\}\big\}$, nontriviality and witnesses follow from Lemma 5.3.

Pairs. $\{\mathsf{S},\mathsf{C}\}$: clopen; witness Φ_{p0}. $\{\mathsf{S},\mathsf{L}\}$: closed + dense $\Rightarrow \Phi = \mathrm{cl}(\Phi) = \Sigma^\omega$; singleton $\{\Sigma^\omega\}$. $\{\mathsf{S},\mathsf{K}\}$: closed + empty interior; witness Φ_{Gp}. $\{\mathsf{C},\mathsf{L}\}$: open + dense; witness Φ_{Fp}. $\{\mathsf{C},\mathsf{K}\}$: open + empty interior $\Rightarrow \Phi = \varnothing$ (any nonempty open set has nonempty interior); singleton $\{\varnothing\}$. $\{\mathsf{L},\mathsf{K}\}$: dense + empty interior $\Leftrightarrow$ full boundary (Theorem 4.7); witnesses Φ_{GFp}, Φ_{FGp}.

Triples and quadruple. $\{\mathsf{S},\mathsf{C},\mathsf{L}\}$: clopen + dense $\Rightarrow \Phi = \Sigma^\omega$. $\{\mathsf{S},\mathsf{C},\mathsf{K}\}$: clopen + empty interior $\Rightarrow \Phi = \varnothing$. $\{\mathsf{S},\mathsf{L},\mathsf{K}\}$: $\mathsf{S}\cap\mathsf{L} = \{\Sigma^\omega\}$ but $\Sigma^\omega \notin \mathsf{K}$; empty. $\{\mathsf{C},\mathsf{L},\mathsf{K}\}$: $\mathsf{C}\cap\mathsf{K} = \{\varnothing\}$ but $\varnothing \notin \mathsf{L}$; empty. $\{\mathsf{S},\mathsf{C},\mathsf{L},\mathsf{K}\} \subseteq \mathsf{S}\cap\mathsf{L}\cap\mathsf{K} = \varnothing$. □

Theorem 5.5 (Exhaustive Classification Theorem)**.** *Every finite intersection over the class families* $\mathsf{S},\mathsf{C},\mathsf{L},\mathsf{K}$ *is equal to one of the 15 cases in table 1; no additional finite-intersection class exists.*

Proof. There are exactly $2^4 - 1 = 15$ nonempty index sets $\Gamma \subseteq \mathcal{F}$. Theorem 5.4 proves each corresponding $I(\Gamma)$ and table 1 records all such cases. Therefore classification is exhaustive. □

Remark 5.6 (Relation to Alpern–Schneider Decomposition)**.** The classical decomposition result of Alpern–Schneider [2] is not reproved here. In the present four-class algebra, the identity $\mathsf{S}\cap\mathsf{L} = \{\Sigma^\omega\}$ isolates the trivial overlap between closed and dense classes (already noted in [2]).

Remark 5.7 (Nontriviality of the Boundary-Dense Class)**.** The class $\mathsf{L}\cap\mathsf{K}$ is nonempty and nontrivial: Lemma 5.3 gives Φ_{GFp} and Φ_{FGp} as witnesses, while $\Phi_\top \notin \mathsf{K}$ and $\Phi_\bot \notin \mathsf{L}$.

Remark 5.8 (Monitorability Gradient)**.** The intersection algebra relates to the three-valued verdicts of [7] as follows (monitorability itself is not a function of the cell [7, Lemma 3.8], [50, Sec. 4.3], Lemma 5.9 (3)):

(1) $\mathsf{S} \cap \mathsf{C}$ (clopen): every property is finitely determined by a prefix - fully monitorable with definitive accept/reject verdicts.
(2) $\mathsf{S} \setminus \mathsf{C}$ (closed, not open): every violation is finitely detectable, but not every satisfaction is - safety-monitorable (every violation is eventually rejected; some satisfactions stay inconclusive forever, while others may still be accepted, e.g. $p \vee Gq$ with good prefix $\{p\}$).
(3) $\mathsf{C} \setminus \mathsf{S}$ (open, not closed): every satisfaction is finitely detectable, but not every violation is - cosafety-monitorable (dually).
(4) $\mathsf{L} \cap \mathsf{K}$ (full boundary): no finite prefix determines membership - inherently non-monitorable (permanently inconclusive) [50, Sec. 4.3].

This refines the three-valued monitoring semantics by grounding verdict completeness in cell membership; which verdicts are reachable, and whether one remains reachable after every prefix, is recorded in Lemma 5.9.

Lemma 5.9 (Verdict availability of the membership tuple)**.** *For $\Phi \subseteq \Sigma^\omega$ call $u \in \Sigma^*$ a* good *(*bad*) prefix of Φ if $[u] \subseteq \Phi$ ($[u] \cap \Phi = \varnothing$), and* ugly *if no extension uv is good or bad* [7, Defs. 3.1, 3.3]*; Φ is* monitorable *if it has no ugly prefix, the notion of Pnueli–Zaks* [52] *in the form of* [7, Def. 3.4]*, and $\mathcal{M}(\Sigma^\omega)$ denotes the family of monitorable properties (as in Remark 11.2).*

(1) *Φ has a bad prefix iff $\Phi \notin \mathsf{L}$, and a good prefix iff $\Phi \notin \mathsf{K}$.*
(2) *$\Phi \in \mathsf{S}$ iff every $\sigma \notin \Phi$ has a bad prefix, and $\Phi \in \mathsf{C}$ iff every $\sigma \in \Phi$ has a good prefix.*
(3) *u is ugly for Φ iff $u^{-1}\Phi := \{\tau : u\tau \in \Phi\} \in \mathsf{L} \cap \mathsf{K}$. Consequently $\mathsf{S} \cup \mathsf{C} \subseteq \mathcal{M}(\Sigma^\omega)$ and $(\mathsf{L} \cap \mathsf{K}) \cap \mathcal{M}(\Sigma^\omega) = \varnothing$, while each of the cells $\{\mathsf{L}\}$, $\{\mathsf{K}\}$ and quaestio contains both monitorable and non-monitorable properties: $Gp \vee Fq \in \{\mathsf{L}\}$, $Fp \wedge Gq \in \{\mathsf{K}\}$ and $(a \wedge Gc) \vee (\neg a \wedge Fb) \in$ quaestio are monitorable, whereas $\neg a \vee GFp \in \{\mathsf{L}\}$, $a \wedge GFp \in \{\mathsf{K}\}$ and $(p \vee GFp) \wedge q \in$ quaestio are not. Hence $\mathcal{M}(\Sigma^\omega)$ is not a union of cells of the 16-cell partition.*

Proof. (1) and (2) restate Definitions 3.3 and 3.4 (negated) and Definitions 3.1 and 3.2. (3) uv is bad (good) for Φ iff v is bad (good) for $u^{-1}\Phi$; so u is ugly iff every v extends both to a member and to a non-member of $u^{-1}\Phi$, i.e. $u^{-1}\Phi \in \mathsf{L} \cap \mathsf{K}$. If $\Phi \in \mathsf{S}$ and u is neither bad nor good, some $u\tau \notin \Phi$ exists; its bad prefix is longer than u (as u is not bad), hence a bad extension uv of u; so no prefix is ugly (dually for C; cf. [7, Remark 3.7]). If $\Phi \in \mathsf{L} \cap \mathsf{K}$ the empty prefix is ugly. For the witnesses (writing $\{\}$ for the letter in which no proposition holds): every $u\{q\}$ is good for $Gp \vee Fq$, every $u\{\}$ is bad for $Fp \wedge Gq$, and every prefix of $(a \wedge Gc) \vee (\neg a \wedge Fb)$ extends to a decisive one (the first letter fixes the branch, then $\{\}$ refutes the a-branch and $\{b\}$ confirms the $\neg a$-branch); after the prefix $\{a\}$ (resp. $\{q\}$) the residual of $\neg a \vee GFp$ and of $a \wedge GFp$ (resp. of $(p \vee GFp) \wedge q$) is $GFp \in \mathsf{L} \cap \mathsf{K}$. The cells of $Gp \vee Fq$, $Fp \wedge Gq$ and $(p \vee GFp) \wedge q$ are those of table 7 and Examples 9.14 and A.4. $\neg a \vee GFp$ is dense, has the good prefix $\{\}$, and $\{a\}\{p\}^\omega$ satisfies it with no good prefix: tuple $(0, 0, 1, 0)$. $a \wedge GFp$ has the bad prefix $\{\}$, every prefix extends to a violation, and $\{a\}\{\}^\omega$ violates it with no bad prefix: tuple $(0, 0, 0, 1)$. For $(a \wedge Gc) \vee (\neg a \wedge Fb)$, $\{a\}$ is bad and $\{b\}$ is good, while $\{\}^\omega$ violates it with no bad prefix and $\{a, c\}^\omega$ satisfies it with no good prefix: tuple $(0, 0, 0, 0)$. The mixed cells are those of [50, Sec. 4.3], whose table

gives $Gp \vee Fq$ and $Fp \wedge Gq$; the non-monitorable witnesses above are variants of theirs; the characterisation is equivalent to $\mathrm{int}(\mathrm{bd}(\Phi)) = \varnothing$ (Theorem 4.7 (2) applied to residuals; [19, Cor. 2], [12, Prop. 2.2(4)]). □

6. Horizon refinement: explicit horizons and clopen collapse

The enumeration of section 5 classifies ω-regular properties under an implicit convention: the observation horizon is ω, and every verdict is taken on the totality of the trace. Two operational settings depart from this convention – bounded model checking [8] truncates the trace at a finite bound N, and finite-trace LTL (LTLf, [18]) restricts the semantic domain to Σ^* – and both flatten the membership-axis structure to a single clopen cell. This section makes the horizon-implicit convention explicit, isolates the topological invariant responsible for the flattening (Proposition 6.2), and tabulates how the introduction of bounded temporal operators migrates a property between cells (table 3). The horizon discipline emerges as a meta-axis: each cell of the 11-cell exclusive partition carries a horizon-sensitivity label that the membership tuple (s, c, ℓ, k) does not record directly but that the truncated-path views of Eisner et al. [23] and the monitorability gradient of Remark 5.8 reflect at the verdict-discipline boundary.

6.1. **Default expressions per cell.** Under the ω-horizon convention, each non-empty cell of the exclusive 11-cell partition is witnessed by a canonical LTL formula whose temporal operators carry no explicit bound. We catalogue the default expressions in table 2, juxtaposing each cell with its topological character. The formulas are exactly those of Lemma 5.3 and of the worked-example catalog of section 9; the table is not a new enumeration but an explicit acknowledgement that the *unbounded* character of G, F, U, R is the witness of cell membership outside the clopen sub-cell.

Cell (s, c, ℓ, k)	Default expression	Topological character
$(1,1,0,0)$ $\mathsf{S} \cap \mathsf{C}$	p_0	clopen
$(1,0,0,1)$ $\mathsf{S} \cap \mathsf{K}$	Gp	closed, empty interior
$(0,1,1,0)$ $\mathsf{C} \cap \mathsf{L}$	Fp	open, dense
$(0,0,1,1)$ $\mathsf{L} \cap \mathsf{K}$	GFp, FGp	boundary-dense
$(1,0,0,0)$ S-only	$p \vee Gq$	closed, non-empty interior
$(0,1,0,0)$ C-only	$p \wedge Fq$	open, has some bad prefix
$(0,0,1,0)$ L-only	$Gp \vee Fq$	dense, has good prefix
$(0,0,0,1)$ K-only	$Fp \wedge Gq$	empty interior, has some bad prefix
$(0,0,0,0)$ quaestio	$(p \vee GFp) \wedge q$	none of the four
$(1,1,1,0)$ Σ^ω	$\top$	full
$(1,1,0,1)$ $\varnothing$	$\bot$	empty

Table 2: Default LTL expression per non-empty cell of the 11-cell exclusive partition, under the implicit ω-horizon convention. The temporal operators are unbounded; the topological character is exactly the structure that the unbounded form preserves.

6.2. **Horizon-projection and clopen collapse.** We make the horizon explicit through two projections of an ω-trace property onto a finite length-N word.

Definition 6.1 (Horizon-N projection)**.** Let $\Phi \subseteq \Sigma^\omega$ and $N \in \mathbb{N}$. The *strong* and *weak* horizon-N projections are

$$\Phi|_N^{\text{strong}} := \{\, u \in \Sigma^N \mid \forall v \in \Sigma^\omega.\, u \cdot v \in \Phi \,\},$$
$$\Phi|_N^{\text{weak}} := \{\, u \in \Sigma^N \mid \exists v \in \Sigma^\omega.\, u \cdot v \in \Phi \,\}.$$

The strong projection collects length-N words whose every infinite extension satisfies Φ – the good prefixes of Definition 3.2 truncated to length N. The weak projection collects length-N words admitting at least one satisfying extension – the non-bad prefixes of Definition 3.1 truncated to length N. Eisner et al.'s strong/weak truncated-path views [23] are the formula-level instances of these projections.

Proposition 6.2 (Clopen collapse at finite horizon)**.** *For every $\Phi \subseteq \Sigma^\omega$ and every finite $N \in \mathbb{N}$, both $\Phi|_N^{\text{strong}}$ and $\Phi|_N^{\text{weak}}$ are clopen in the discrete topology on Σ^N. Consequently, on the finite-horizon trace domain the membership-axis partition of Theorem 8.4 reduces to a single non-trivial cell, $\mathsf{S} \cap \mathsf{C}$.*

Proof. The discrete topology on the finite set Σ^N makes every subset clopen; both projections are subsets and are therefore clopen. Clopen sets on Σ^N are simultaneously safety (every violation has a bad prefix – the violating word itself) and cosafety (every satisfaction has a good prefix – the satisfying word itself); they thus inhabit $\mathsf{S} \cap \mathsf{C}$. □

Corollary 6.3 (Horizon residue outside $\mathsf{S} \cap \mathsf{C}$)**.** *A property $\Phi \subseteq \Sigma^\omega$ admits a* uniform *finite horizon – an N such that $\Phi|_N^{\text{strong}} = \Phi|_N^{\text{weak}}$ – if and only if $\Phi \in \mathsf{S} \cap \mathsf{C}$ (modulo the two trivial singletons). Properties in $\mathsf{L} \cap \mathsf{K}$ admit no such N uniformly, and properties in $\mathsf{S} \setminus \mathsf{C}$, $\mathsf{C} \setminus \mathsf{S}$ admit no uniform horizon on either verdict side: every violation (resp. satisfaction) has a bad (resp. good) prefix, but these prefixes have unbounded length.*

Proof. If $\Phi|_N^{\text{strong}} = \Phi|_N^{\text{weak}}$, every word of length N is a good or a bad prefix, so Φ is a finite union of length-N cylinders and hence clopen; conversely a clopen Φ is, by compactness of Σ^ω, a finite union of cylinders, which may be taken of one common length N. If every violation of Φ had a bad prefix of length at most N, the complement of Φ would be a finite union of cylinders, so Φ would be clopen; dually for satisfactions. For $\Phi \in \mathsf{S} \setminus \mathsf{C}$ the bad prefixes still exist trace by trace by Definition 3.1 (e.g. *Gp*: a violation beginning with $\{p\}^N\{\}$, where $\{\}$ is the letter with no proposition true, has no bad prefix shorter than $N{+}1$); dually for $\mathsf{C} \setminus \mathsf{S}$. □

The corollary makes precise the sense in which everything outside the clopen cell is the *horizon residue* of the CLASP classification, and $\mathsf{L} \cap \mathsf{K}$ its extreme case, where no finite horizon decides any trace. On any finite-horizon trace domain (Σ^N or Σ^*), the residue vanishes and the four-class taxonomy collapses to $\mathsf{S} \cap \mathsf{C}$.

6.3. **Explicit-horizon migration.** Replacing an unbounded temporal operator with its bounded counterpart – G with $G_{[0,N]}$, F with $F_{[0,N]}$, U with $U_{\leq N}$, R with $R_{\leq N}$ – substitutes the implicit ω-quantifier with a finite-window quantifier over $N{+}1$ positions. The resulting formula describes a finite Boolean combination of atomic propositions on those positions; its denotation is a finite union of length-$(N{+}1)$ cylinders in Σ^ω, hence clopen. We tabulate the effect on the cells of table 2:

Default expression	ω-cell	$\to$	With explicit bound N	Bounded cell
p_0	$\mathsf{S} \cap \mathsf{C}$	$\to$	p_0 (unchanged)	$\mathsf{S} \cap \mathsf{C}$
Gp	$\mathsf{S} \cap \mathsf{K}$	$\to$	$G_{[0,N]}p$	$\mathsf{S} \cap \mathsf{C}$
Fp	$\mathsf{C} \cap \mathsf{L}$	$\to$	$F_{[0,N]}p$	$\mathsf{S} \cap \mathsf{C}$
GFp	$\mathsf{L} \cap \mathsf{K}$	$\to$	$G_{[0,N]}F_{[0,N]}p$	$\mathsf{S} \cap \mathsf{C}$
FGp	$\mathsf{L} \cap \mathsf{K}$	$\to$	$F_{[0,N]}G_{[0,N]}p$	$\mathsf{S} \cap \mathsf{C}$
$p \vee Gq$	S-only	$\to$	$p \vee G_{[0,N]}q$	$\mathsf{S} \cap \mathsf{C}$
$p \wedge Fq$	C-only	$\to$	$p \wedge F_{[0,N]}q$	$\mathsf{S} \cap \mathsf{C}$
$Gp \vee Fq$	L-only	$\to$	$G_{[0,N]}p \vee F_{[0,N]}q$	$\mathsf{S} \cap \mathsf{C}$
$Fp \wedge Gq$	K-only	$\to$	$F_{[0,N]}p \wedge G_{[0,N]}q$	$\mathsf{S} \cap \mathsf{C}$
$(p \vee GFp) \wedge q$	quaestio	$\to$	$(p \vee G_{[0,N]}F_{[0,N]}p) \wedge q$	$\mathsf{S} \cap \mathsf{C}$

Table 3: Explicit-horizon migration. For every default expression of table 2, replacing each unbounded G, F, U, R with the bounded counterpart $G_{[0,N]}, F_{[0,N]}, U_{\leq N}, R_{\leq N}$ collapses the cell to $\mathsf{S} \cap \mathsf{C}$.

Proposition 6.4 (Bounded-operator migration)**.** *Let Φ be an LTL property over Σ^ω and let Φ' be obtained from Φ by replacing every unbounded G, F, U, R with $G_{[0,N]}, F_{[0,N]}, U_{\leq N}, R_{\leq N}$ for a fixed N. Then $\Phi' \in \mathsf{S} \cap \mathsf{C}$, modulo the two trivial singletons $\{\Sigma^\omega, \varnothing\}$.*

Proof. Every bounded operator over $N{+}1$ positions is a finite Boolean combination of atomic propositions, so the denotation of Φ' is a finite Boolean combination of length-$(N{+}1)$ cylinders. Cylinders are clopen and the clopen sets form a Boolean sub-algebra of $\mathcal{P}(\Sigma^\omega)$; hence Φ' is clopen, i.e. $\Phi' \in \mathsf{S} \cap \mathsf{C}$. □

Proposition 6.4 establishes one direction of a duality between bounded LTL and the clopen cell. The full duality has three parts.

Theorem 6.5 (Clopen Attractor under Bounding)**.** *Let $\varphi \mapsto \varphi^{[N]}$ denote the syntactic bounding operation on LTL formulas that replaces each unbounded G, F, U, R with $G_{[0,N]}, F_{[0,N]}, U_{\leq N}, R_{\leq N}$ for a fixed $N \in \mathbb{N}$.*

(1) Image into clopen. *For every φ in LTL and every $N \geq 0$, $[\![\varphi^{[N]}]\!] \in \mathsf{S} \cap \mathsf{C}$, modulo the two trivial singletons $\{\Sigma^\omega, \varnothing\}$.*

(2) Bounded LTL captures clopen ω-regular. *A property $\Phi \subseteq \Sigma^\omega$ is clopen ω-regular iff $\Phi = [\![\varphi^b]\!]$ for some bounded LTL formula φ^b.*

(3) Unique stable cell. *Among the eleven non-empty cells of the 16-cell partition, $\mathsf{S} \cap \mathsf{C}$ is the unique cell whose default witnesses are bounding-stable under the family $\{\varphi \mapsto \varphi^{[N]}\}_{N \geq 0}$: for every other cell C, the default witness φ_C of Lemma 5.3 satisfies $\left[\!\left[\varphi_C^{[N]}\right]\!\right] \in \mathsf{S} \cap \mathsf{C} \neq C$ for every N.*

Proof. Part (1) is Proposition 6.4. Part (2): clopen sets in the Cantor topology on Σ^ω are exactly finite Boolean combinations of basic cylinders $[u]$, and each cylinder is the denotation of the bounded LTL formula $\bigwedge_{i<|u|} X^i p_{u(i)}$ (with p_a the atomic proposition selecting the symbol $a \in \Sigma$). Conversely every bounded LTL formula denotes a finite Boolean combination of length-$\leq N{+}1$ cylinders, hence clopen. Part (3): for each cell $C \neq \mathsf{S} \cap \mathsf{C}$, the default witness φ_C in Lemma 5.3 contains at least one unbounded G, F, U, or R operator; the

substitution $\varphi \mapsto \varphi^{[N]}$ replaces every such operator with its bounded counterpart, producing a bounded formula, hence by (1) a clopen property; clopen $\neq C$ for any non-clopen cell. □

The dual reading of Theorem 6.5 is the *construction principle*: every non-clopen cell is obtainable from the clopen base by injecting unbounded temporal operators in a specific pattern, and the bounding operation β_N is the inverse that collapses any such construction back to $\mathsf{S} \cap \mathsf{C}$. Figure 1 renders the construction direction as a DAG, with each edge labelled by the syntactic injection move that produces the target cell; the global bounding arrow on the left shows that every cell collapses to the clopen attractor under β_N.

Examples 6.6 to 6.9 illustrate the construction automaton of figure 1 in operation, walking through four representative derivations. Each example explicitly tracks (i) the starting cell, (ii) the sequence of transitions fired in the automaton, (iii) the resulting witness formula together with its cell membership, and (iv) the bounded counterpart obtained by β_N (which always lands in $\mathsf{S} \cap \mathsf{C}$ by Theorem 6.5).

Example 6.6 (Constructing $\mathsf{S} \cap \mathsf{K}$)**.** Start at the clopen root with $\varphi = p_0$. Fire the transition $G(\varphi)$ from clopen to S∩K with a fresh atomic proposition p:

$$G(p) \;=\; Gp \;\in\; \mathsf{S} \cap \mathsf{K}.$$

Cell membership is by Lemma 5.3. Bounding via β_N: $G_{[0,N]}p \in \mathsf{S} \cap \mathsf{C}$ (Proposition 6.4).

Example 6.7 (Constructing S-only via a two-step path)**.** Two transitions in figure 1:

(1) clopen $\xrightarrow{G(\varphi)}$ $\mathsf{S} \cap \mathsf{K}$ with fresh atom q: $\varphi_1 = Gq$.

(2) $\mathsf{S} \cap \mathsf{K}$ $\xrightarrow{\varphi \vee p}$ S-only with fresh clopen p: $\varphi_2 = p \vee Gq$.

The result $p \vee Gq$ is the canonical S-only witness of table 2. Bounding: $p \vee G_{[0,N]}q \in \mathsf{S} \cap \mathsf{C}$.

Example 6.8 (Constructing $\mathsf{L} \cap \mathsf{K}$ recurrence subwitness)**.** Direct one-step transition $GF(\varphi)$ from clopen with fresh atom p:

$$GF(p) \;=\; GFp \;\in\; \mathsf{L} \cap \mathsf{K},$$

at Borel level Π^0_2 (recurrence half of Remark 9.6). The sibling transition $FG(\varphi)$ produces $FGp \in \mathsf{L} \cap \mathsf{K}$ at Borel level Σ^0_2 (persistence half). Bounding: $G_{[0,N]}F_{[0,N]}p \in \mathsf{S} \cap \mathsf{C}$.

Example 6.9 (Constructing the Quaestio cell)**.** Two-step composite path through L∩K:

(1) clopen $\xrightarrow{GF(\varphi)}$ L∩K with fresh atom p: $\varphi_1 = GFp$.

(2) L∩K $\xrightarrow{(\varphi \vee p') \wedge q}$ Quaestio with two fresh clopen atoms p', q. Renaming $p' \to p$ (no clash since the inner p inside GFp is a different bound variable in the construction tree):

$$\varphi_2 \;=\; (p \vee GFp) \wedge q \;\in\; \text{Quaestio}.$$

The result is the canonical Quaestio witness of Remark 8.3 and table 2. Bounding: $(p \vee G_{[0,N]}F_{[0,N]}p) \wedge q \in \mathsf{S} \cap \mathsf{C}$.

The four examples cover the three structural patterns of the automaton: a one-step path to a pair cell (Ex. 6.6, 6.8), a two-step path with a clopen shortcut to an atomic single cell (Ex. 6.7), and a composite path with combined $\vee$ and $\wedge$ shortcuts to the Quaestio (Ex. 6.9). Mirror paths produce $\mathsf{C} \cap \mathsf{L}$, C-only, L-only, K-only by symmetric operator choices.

The construction graph admits a second reading as a *2-axis factorisation* of the 11-cell partition. Figure 2 renders it: rows are values of Axis A (the ω-character of the formula

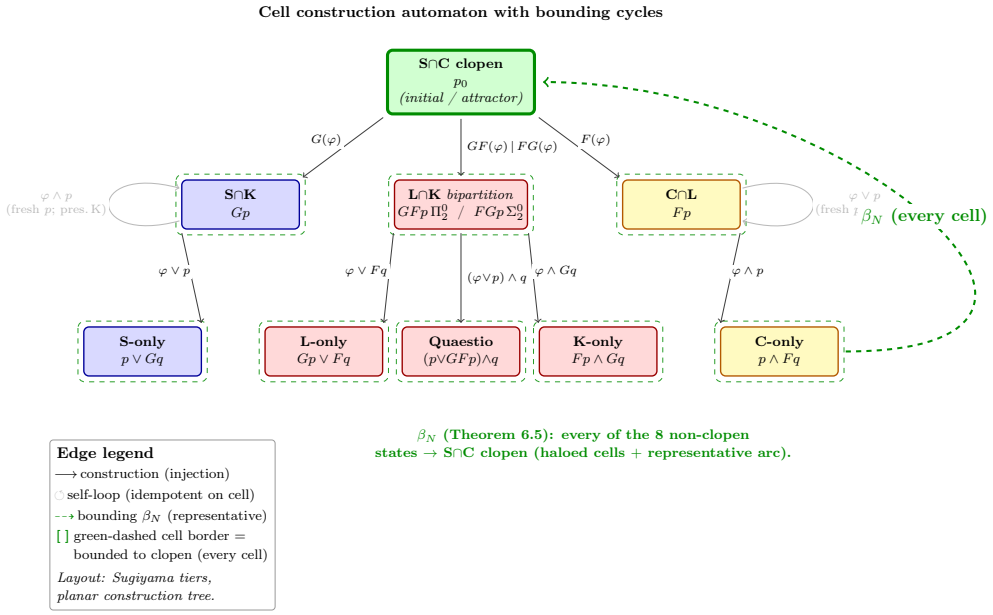


Figure 1: **Cell construction automaton with bounding cycles.** The eleven non-empty cells of the 16-cell partition are arranged as states of a transition system rooted at the clopen attractor $\mathsf{S} \cap \mathsf{C}$, organised in Sugiyama tiers (tier-0: clopen; tier-1: pair cells; tier-2: atomic singles and Quaestio). *Black solid arrows* are operator-injection transitions (construction); each label specifies the syntactic operation applied to the source-cell formula φ. A single G/F injection or nesting GF, FG produces the three pair cells; further $\vee/\wedge$ injections with fresh-atom clopen or with other operators produce the four atomic singles and the Quaestio. The resulting formula is a witness of the target cell but is not necessarily the cell's canonical default expression of table 2: e.g. the $\mathsf{L} \cap \mathsf{K} \xrightarrow{\varphi \vee Fq}$ L-only path produces $GFp \vee Fq \in$ L-only, while the canonical witness shown in the L-only node, $Gp \vee Fq$, is obtainable via the alternative path $\mathsf{S} \cap \mathsf{K} \xrightarrow{\varphi \vee Fq}$ L-only. *Green-dashed cell borders* mark every cell in the bounded set (universal statement of Theorem 6.5); the prominent green-dashed arc from C-only to clopen, labelled "β_N (every cell)", is the family's representative arrow. *Grey self-loops* indicate monotonicity-preserving transitions (taken an arbitrary number of times with fresh atoms without leaving the source state): $\wedge$ preserves K on $\mathsf{S} \cap \mathsf{K}$; $\vee$ preserves L on $\mathsf{C} \cap \mathsf{L}$. The $\mathsf{L} \cap \mathsf{K}$ state is bipartitioned per Remark 9.6: recurrence ($GFp \in \Pi^0_2$) and persistence ($FGp \in \Sigma^0_2$). The two trivial singletons Σ^ω and $\varnothing$ are inhabitants of $\mathsf{S} \cap \mathsf{C}$ and are not shown as distinct states. This automaton operates at Stage 4 of the semantic pipeline (figure 8): each cell is a target CLASP class for a candidate trace property Φ.

nucleus: None, G, F, or GF/FG), columns are values of Axis B (the clopen shortcut added: none, good-prefix via $\vee$, or bad-prefix via $\wedge$). Each non-empty cell of the 16-cell partition lives at exactly one (A, B) coordinate; the remaining grid positions either preserve the source cell (because $\vee$ preserves L and $\wedge$ preserves K, by topological monotonicity) or

trivially collapse back to clopen. The Quaestio cell sits outside the 4×3 factorisation: it requires the combined $\vee$-$\wedge$ shortcut pattern applied to the GF/FG nucleus.

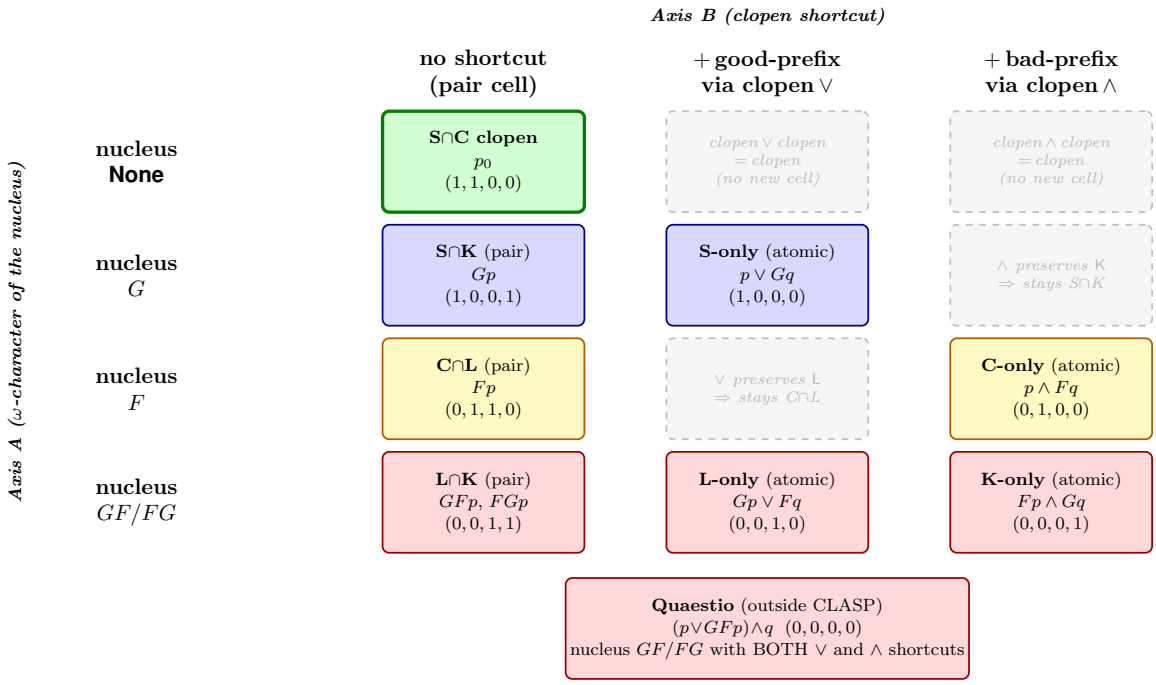


Figure 2: **The 11-cell partition as a 2-axis factorisation.** *Axis A* (rows) is the ω-character of the formula nucleus: None (no unbounded temporal operator), G alone, F alone, or GF/FG nesting. *Axis B* (columns) is the clopen shortcut added on a fresh atomic proposition: none (pair cell), good-prefix via $\vee$, or bad-prefix via $\wedge$. Coloured cells are the nine non-trivial CLASP cells; grey-dashed entries are positions in which the shortcut either preserves the source cell (by topological monotonicity: $\vee$ preserves L, $\wedge$ preserves K) or trivially returns clopen. The Quaestio cell sits outside the 4×3 grid: it requires the combined $\vee$-$\wedge$ shortcut pattern applied to the GF/FG nucleus. The pair-versus-atomic distinction (cf. figure 1) is now visible as the second column movement on each row: $\mathsf{S} \cap \mathsf{K} \to$ S-only on row G (gain $\neg\mathsf{K}$ via non-empty interior), $\mathsf{C} \cap \mathsf{L} \to$ C-only on row F (gain $\neg\mathsf{L}$ via bad prefix), $\mathsf{L} \cap \mathsf{K} \to$ L-only / K-only on row GF/FG (each shortcut breaks one of the two boundary axes). Cell colours match figure 3.

The migration is already captured graphically in figure 1 by the β_N representative arc and the green-dashed cell-borders (universal halo signifier): every non-trivial cell of the 11-cell exclusive partition flows into the central clopen $\mathsf{S} \cap \mathsf{C}$ attractor under the bounded-operator substitution of Proposition 6.4, with the operator whose bounding triggers each cell's migration tabulated in table 4.

To make the migration concrete, table 4 fixes a worked RV-style scenario and exhibits, for each of the eleven non-empty cells of Theorem 8.4, an applied property and the bounded counterpart it migrates to. The domain is a streaming web service with atomic propositions $p =$ "service responds at this tick" and $q =$ "user is authenticated at this tick".

Cell (s, c, ℓ, k)	Default Φ	Applied reading on $\{p, q\}$	After horizon bound N
$\mathsf{S}\cap\mathsf{C}$ $(1,1,0,0)$	p_0	"service responds at tick 0"	p_0 (identity)
$\mathsf{S}\cap\mathsf{K}$ $(1,0,0,1)$	Gp	"service responds at every tick *(forever)*"	$G_{[0,N]}p$
$\mathsf{C}\cap\mathsf{L}$ $(0,1,1,0)$	Fp	"service responds at some tick *(no deadline)*"	$F_{[0,N]}p$
$\mathsf{L}\cap\mathsf{K}$ $(0,0,1,1)$	GFp	"service responds at infinitely many ticks"	$G_{[0,N]}F_{[0,N]}p$
S-only $(1,0,0,0)$	$p \vee Gq$	"service responds at tick 0, or user is authenticated at every tick *(forever)*"	$p \vee G_{[0,N]}q$
C-only $(0,1,0,0)$	$p \wedge Fq$	"service responds at tick 0, and user is authenticated at some tick *(no deadline)*"	$p \wedge F_{[0,N]}q$
L-only $(0,0,1,0)$	$Gp \vee Fq$	"service responds at every tick *(forever)*, or user is authenticated at some tick *(no deadline)*"	$G_{[0,N]}p \vee F_{[0,N]}q$
K-only $(0,0,0,1)$	$Fp \wedge Gq$	"service responds at some tick *(no deadline)*, and user is authenticated at every tick *(forever)*"	$F_{[0,N]}p \wedge G_{[0,N]}q$
Quaestio $(0,0,0,0)$	$(p \vee GFp) \wedge q$	"user is authenticated at tick 0, and service responds at tick 0 or at infinitely many ticks"	$(p \vee G_{[0,N]}F_{[0,N]}p) \wedge q$
Σ^ω $(1,1,1,0)$	$\top$	"no constraint (every ω-trace satisfies)"	$\top$ (identity)
$\varnothing$ $(1,1,0,1)$	$\bot$	"unsatisfiable (no ω-trace satisfies; e.g. $p \wedge \neg p$)"	$\bot$ (identity)

Table 4: Applied readings of the eleven non-empty cells under the streaming-web-service scenario, with the bounded counterpart that realises the migration to the $\mathsf{S} \cap \mathsf{C}$ attractor of figure 1 (via β_N representative arc). The first column lists the cell and its exclusive tuple; the second column gives the canonical default Φ from table 2; the third gives a natural-language reading on $p =$ "service responds" and $q =$ "user authenticated"; the fourth gives the bounded counterpart Φ' resulting from Proposition 6.4. We use the position-quantifier convention "at every tick *(forever)*" for G, "at some tick *(no deadline)*" for F, "at infinitely many ticks" for GF, and avoid the operator-name "eventually", which carries a spurious liveness flavour in informal English. The parenthetical annotations *(forever)* and *(no deadline)* make the ω-bound side of the verdict explicit and prevent a reading from being mistaken for clopen $\mathsf{S}\cap\mathsf{C}$: any annotation of either kind signals a side of the verdict that requires ω-observation. An LTL formula containing F can still land in any CLASP cell depending on its combination with the other operators (e.g. $p \wedge Fq$ is cosafety-only, $Fp \wedge Gq$ is coliveness-only); the topological character is fixed by the cell label in the first column. All bounded counterparts sit in the $\mathsf{S} \cap \mathsf{C}$ attractor regardless of the source cell.

6.4. **The horizon-sensitivity gradient.** The cells of the 11-cell partition group into four classes by horizon-sensitivity:

Horizon-insensitive.: $\mathsf{S} \cap \mathsf{C}$ (clopen). Some finite horizon N (Corollary 6.3) strongly decides both verdict sides. Bounded LTL operates entirely within this class.

Horizon-one-sided safety.: $(1, 0, 0, 1)$ $\mathsf{S} \cap \mathsf{K}$ and the S-only cell. The refutation side is decidable at a finite, trace-dependent horizon (Corollary 6.3); the confirmation side requires the ω-horizon (no good prefix exists, or the confirmation set has only ω-traces in its difference from the clopen shortcut).

Horizon-one-sided cosafety.: $(0, 1, 1, 0)$ $\mathsf{C} \cap \mathsf{L}$ and the C-only cell. Dual: the confirmation side is decidable at a finite, trace-dependent horizon; the refutation side requires the ω-horizon.

Horizon-two-sided.: $(0, 0, 1, 1)$ $\mathsf{L} \cap \mathsf{K}$ and the L-only and K-only cells (the latter two with a clopen shortcut on one side and an ω-horizon on the other). On $\mathsf{L} \cap \mathsf{K}$ no finite horizon decides any single trace (the non-monitorable cell of Remark 5.8 item 4); the L-only and K-only cells decide some traces on their shortcut side and contain both monitorable and non-monitorable properties (Lemma 5.9).

The horizon-sensitivity gradient is not orthogonal to the membership tuple; it is its *projection onto a horizon-axis* collecting cells that share the same horizon discipline. As such it does not enlarge the 11-cell enumeration but labels each cell with the horizon parameter that the implicit-ω convention of section 3 silently fixes.

Remark 6.10 (BMC and LTLf as horizon disciplines)**.** Two operational settings select horizon disciplines explicitly. *Bounded model checking* [8] verifies an LTL property Φ up to a fixed depth k; by Proposition 6.4 every BMC instance encodes the clopen sub-property $\Phi'_k \in \mathsf{S} \cap \mathsf{C}$, and the gap $\Phi \setminus \Phi'_k$ is the $\mathsf{L} \cup \mathsf{K}$ residue. *LTLf* [18] restricts LTL to finite traces; by Proposition 6.2 every LTLf formula is clopen at the trace terminus, recovering the horizon-collapse in the trace-restriction direction. In both settings the $\mathsf{L} \cup \mathsf{K}$ residue is the part of the property that the bounded prefix does not determine: claiming finite-horizon (bounded-prefix) evidence for an ω-horizon property is unjustified without an explicit account of that unverified suffix.

Remark 6.11 (Relation to Remark 5.8)**.** The horizon-sensitivity gradient (section 6.4) and the monitorability gradient (Remark 5.8) read the same partition from two sides: the latter classifies cells by which finite-prefix verdicts are available, the former by which horizon discipline is required to obtain those verdicts. The two gradients coincide on the four-cell coarsening $\{\mathsf{S} \cap \mathsf{C}, \mathsf{S} \setminus \mathsf{C}, \mathsf{C} \setminus \mathsf{S}, \mathsf{L} \cap \mathsf{K}\}$ and refine differently on the atomic cells: the monitorability gradient sees the atomic cells as clopen-shortcut variants of their dominant pair-cell partner, the horizon-sensitivity gradient sees them as one-sided horizon regimes. Neither gradient decides monitorability on the L-only, K-only and quaestio cells, which are mixed (Lemma 5.9).

7. The variable space of traces and monitors

The classifications of sections 5 and 6 operate under a set of conventions that the foregoing prose fixes implicitly: the trace is an element of Σ^ω with $\Sigma = 2^{AP}$, discrete time $\mathbb{N}$, total per-tick observation, single deterministic trajectory, and Σ raw rather than derived from a lower-layer monitor. The horizon discipline of section 6 makes the first conventional choice – observation horizon – explicit as a meta-axis. The present section completes that move: it enumerates the full set of variables along which a trace and a monitor can be parameterised, declares which defaults this paper adopts, and identifies the unique cross-axis constraint that ties a meta-trace to the verdict alphabet of the layer below. The result is a 23-axis

design space; this paper inhabits a single slice, and the classification we develop is the slice's diagonal.

7.1. **The signal–monitor–events–trace pipeline.** The variables of the next subsections inhabit a semantic pipeline that connects a generating process to the trace properties classified by the rest of the paper. Making this pipeline explicit clarifies where each object lives, which property class lives at which stage, and how meta-properties sit *logically prior* to monitor setup. The pipeline has six stages; table 5 summarises them.

Stage 0 — Process and signal. At the root sits a *process* $\mathcal{P}$ that we do not necessarily observe directly. $\mathcal{P}$ produces a *signal* $s\colon T \to D$ from a time domain T (continuous $\mathbb{R}_{\geq 0}$ or discrete $\mathbb{N}$) to a data domain D (categorical, real-valued, structured). The signal is the empirical interface to the process and is the object that meta-properties (Stage 5 below) speak about. The ultimate goal of the framework is to specify properties of $\mathcal{P}$ via specifications on signals derived from s.

Stage 1 — Monitor configuration. A *monitor configuration* μ is a parameter package fixing how the monitor decides events from the signal. It includes the verdict-alphabet cardinality $n \in \{1, 2, 3, 4, \infty\}$ of section 10 and an event-decision mechanism that abstracts the signal into discrete observations. The specific event-decision implementation (e.g. threshold-based discretisation for real-valued signals, predicate evaluation for symbolic signals) is an implementation-level concern beyond the trace-property classification scope of this paper; the present framework treats μ as a parameter, with the constraint that the event-decision mechanism operates on a bounded prefix of s.

Stage 2 — Event generation. The configured monitor M_μ discretises the signal into a sequence of *events*. An event e is a discrete observation from a finite event alphabet $\mathcal{E}_\mu$ determined by μ. Each event is decidable in *finite* signal-time: the event-decision mechanism answers "has this event occurred?" on a bounded prefix of s. Therefore events live as *finitely-witnessable* properties of the signal: a property decided on a bounded prefix is a union of cylinders of one length, as is its complement, hence clopen, so its cell is $\mathsf{S} \cap \mathsf{C}$ (or a trivial singleton), also when the trigger combines a finite shortcut with a bounded recurrence (Proposition 6.4). Events cannot lie strictly in L or K: infinite-trace character cannot be witnessed by a finite signal prefix.

Atoms versus events — notational note. Throughout this paper, propositional-LTL formulae use atoms $p, q, \ldots$ (e.g. in figure 1's node labels and in the CLASP-cell witnesses of table 2), whereas Stage 2 events of the pipeline use letters $e_i \in \mathcal{E}_\mu$ from a finite event alphabet (e.g. the ternary $\{\mathsf{V}, \mathsf{I}, \varepsilon\}$ of [10], where a worked end-to-end pipeline instance is developed for the EXTREMUM encoder applied to $s(t) = \sin t$). The two notations cohabit coherently: each atom p is realised at Stage 2 by an event $e_p \in \mathcal{E}_\mu$ that decides "p holds at this tick" on a bounded signal prefix; conversely each event indicator behaves as an atomic proposition under the pipeline's Stage 4 classification. Whenever a figure shifts between atom and event notation, the underlying CLASP semantics is preserved.

Stage 3 — Trace aggregation. Collecting the events emitted by M_μ over time yields the *trace* $\tau = (e_0, e_1, e_2, \ldots) \in \mathcal{E}_\mu^\omega$. The trace is the discrete abstraction of the original signal under μ. The trace alphabet $\mathcal{E}_\mu$ is the per-tick alphabet of axis T3 at the current layer; cf. Proposition 7.1.

Stage 4 — Properties on the trace. Properties of the form $\Phi \subseteq \mathcal{E}_\mu^\omega$ are trace properties. Their topological character depends on the operator structure:

- *Trace properties finitely decided on at least one side* ($\mathsf{S}, \mathsf{C}, \mathsf{S} \cap \mathsf{C}$, S-only, C-only) add no monitoring power beyond the events already generated by M_μ; they are *inheritable* from the event layer.
- *Infinite trace properties* ($\mathsf{L} \cap \mathsf{K}$, L-only, K-only, Quaestio) have traces whose membership is witnessed by no finite prefix: on $\mathsf{L} \cap \mathsf{K}$ every trace, on the other three cells some traces only (Lemma 5.9). Two complementary responses are possible:
  (1) *Enforcement by an advanced monitor*: a monitor with full ω-trace observation power (e.g. a Büchi automaton on $\mathcal{E}_\mu^\omega$ or the robust signal monitor of section 10.5 for $n = \infty$). This is the operational response to infinite-trace properties.
  (2) *Treat the trace as a new signal*: τ becomes the input to a meta-pipeline. A new monitor $M_{\mu'}$ is configured on τ, emitting events over the verdict alphabet V_μ of M_μ (Proposition 7.1), and the pipeline iterates. This is the recursive response and the source of the layer structure of section 11.

Stage 5 — Meta-properties about the signal. A meta-property is, precisely, the *logically-prior signal → monitor selection problem*: given a signal s (from process $\mathcal{P}$) and a target trace-property Φ, decide *which monitor to deploy before the configuration μ is fixed*. This selection prerequisite decomposes into three sub-questions: (Q1) *verdict availability* — the CLASP cell of Φ, fixing which finite-prefix verdicts exist and whether every violation or satisfaction receives one (Lemma 5.9), but not monitorability in the Pnueli–Zaks sense, which is not a function of the cell (cf. Remark 5.8); (Q2) the *verdict-alphabet cardinality* $n \in \{1, 2, 3, 4, \infty\}$, the rung of the $|V|$ ladder; and (Q3) the *event-generation modality* (touch, crossing, or a combination). This paper answers Q1 and Q2; Q3 is answered in Bragetti [10]. In the CLASP framework, the meta-property of *being-in-cell X* is the canonical such prerequisite: it determines the monitor architecture and the verdict alphabet feasible for properties whose cell is X. We conjecture that meta-properties are themselves classifiable under the same CLASP partition (Conjecture 11.1); if so, this is the bootstrap that closes the framework.

Epistemic demarcation of the three "meta" senses. These three senses of "meta" must not be conflated: the meta-layer recursion (Proposition 7.1, $\Sigma_{k+1} = V_k$) is a theorem, established by construction; the meta-property selection prerequisite of this stage is a definition (an object, not a claim); and the meta-classification (Conjecture 11.1) is a conjecture. Soundness at one of these senses is therefore *not* inherited by the others.

Stage	Object	Action / role	Property class hosted
0	signal $s\colon T \to D$	emitted by process $\mathcal{P}$	domain of meta-properties
1	configuration μ	monitor parameters (n, event-decision)	—
2	event $e \in \mathcal{E}_\mu$	discretisation of s	$\mathsf{S} \cap \mathsf{C}$ (clopen)
3	trace $\tau \in \mathcal{E}_\mu^\omega$	aggregation of events	—
4	trace prop. Φ	verification on τ	all 11 cells; cells outside $\mathsf{S} \cup \mathsf{C}$ need (i) advanced mon. or (ii) recursion
5	meta-property	prerequisite about $s/\mathcal{P}$	CLASP at meta-level (Conjecture 11.1)

Table 5: The signal–monitor–events–trace pipeline. Each stage has a primary object, a primary action, and the property class hosted there. Events at Stage 2 are constrained to the clopen cell $\mathsf{S} \cap \mathsf{C}$ because the event-decision mechanism operates on a bounded prefix of the signal; infinite-trace character emerges only at Stage 4. Meta-properties at Stage 5 are logically prior to Stage 1. The specific event-decision implementation is beyond the trace-classification scope of this paper.

The classifications and figures of the preceding sections inhabit specific stages of this pipeline. The cell-construction automaton (figure 1) operates at Stage 4 (trace properties). The LTL_3 monitor gallery (figure 6) operates at the event boundary, mapping Stage 2 events to verdict-trace properties at Stage 3. The horizon migration captured in figure 1 (via β_N representative arc and halo) operates at Stage 4 also, projecting infinite-trace properties to finitely-decidable ones via bounded operators. The meta-classification (section 11) operates at Stage 5. The trace axes T1–T12 of table 15 parameterise the trace at Stage 3, and the monitor axes D1–D11 of table 14 parameterise the monitor at Stage 1.

A worked end-to-end pipeline instance threading the recurrence formula $\Phi = \mathbf{GF}\, p$ through signal, event stream, trace and cell is developed in [10] as the Extremum-encoder applied to $s(t) = \sin t$; it lands the verdict in the $\mathsf{L} \cap \mathsf{K}$ cell of Theorem 5.4 (recurrence side of the Π_2^0 bipartition) and illustrates parametric adaptation as the canonical alternative to a Büchi-style ω-monitor on the verdict trace.

The full 23-axis design space — the eleven monitor axes (D1–D11) and twelve trace axes (T1–T12), each with this paper's adopted default value — is catalogued in appendix B; in the body we use only the one mandatory cross-axis dependency (Proposition 7.1) and its effective-cardinality consequences (section 7.3).

7.2. **Cross-axis structure.** The twenty-three axes are largely independent. Two relationships are worth isolating because they recur across classification, monitoring, and composition arguments.

Proposition 7.1 (Meta-layer alphabet constraint)**.** *For every $k \geq 1$, the per-tick alphabet of a trace at layer $k{+}1$ equals the verdict alphabet of the monitor at layer k:*

$$\Sigma_{T_6=k+1} \;=\; V_{D_1 \text{ at layer } k}.$$

This is the unique mandatory cross-axis dependency in the 23*-axis design space.*

Proof. By construction of meta-traces: a layer-$(k{+}1)$ trace is the verdict-trace produced by a monitor running on a layer-k trace, whose per-tick symbols are exactly the verdicts the monitor emits. □

Remark 7.2 (Horizon independence across layers)**.** The horizon discipline D2 at layer $k{+}1$ is logically independent of D2 at layer k: a layer-1 trace under ω-horizon may be observed by a layer-2 monitor under finite horizon N, capturing the first N verdicts emitted by the layer-1 monitor. Symmetrically, a finite-horizon layer-1 trace may be summarised by a layer-2 monitor at ω-horizon in degenerate fashion (the layer-2 trace is itself finite, so the layer-2 ω-horizon collapses to finite-horizon). Asymmetric horizons across layers are *not* discussed further in this paper, but they admit a direct extension of the clopen-collapse result of Proposition 6.2.

Remark 7.3 (Trace–monitor cross-axis independence)**.** The trace axes T1–T12 are otherwise independent of the monitor axes D1–D11: any combination of (T, D) values is admissible up to the meta-layer alphabet constraint of Proposition 7.1. In particular, a deterministic trace (T5) may be monitored by a non-deterministic monitor (D6); a stationary trace (T9) may be monitored by a time-varying monitor; a single trace (T10) may be observed through a multi-monitor composition (D10).

7.3. **Layer regimes.** The meta-classification of section 11 treats layer 2. Iteration to layer $k \geq 3$ defines a *layer regime*; three possibilities arise.

Stable bootstrap.: The CLASP partition reproduces with the same structure at every layer: at every k, the layer-k verdict alphabet is partitioned into the same $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$ classes. This is the case conjectured in Conjecture 11.1 (section 11) for layer 2; whether it extends verbatim to layer $k \geq 3$ remains an open direction.

Collapsing.: The effective verdict cardinality (see Definition 7.4 below) drops at some layer. The canonical instance is the $\mathsf{L}\cap\mathsf{K}$ cell observed by an LTL_3 monitor: every trace of the verdict-trace stays in the inconclusive state, so the layer-2 effective cardinality is 1 even though the nominal cardinality is 3 (cf. Remark 5.8 item 4).

Explosive.: The verdict alphabet grows at each layer – the canonical instance is robust monitoring (section 10.5, $n = \infty$): the layer-2 alphabet is a real-valued robustness signal of higher topological complexity than the layer-1 alphabet.

Definition 7.4 (Effective verdict cardinality)**.** For a monitor M with nominal verdict alphabet V of cardinality $n = |V|$ and a property $\Phi \subseteq \Sigma^\omega$, the *effective verdict cardinality* of M on Φ is

$$n_{\text{eff}}(M, \Phi) := \big|\, \{\, v \in V \,:\, \exists w \in \Sigma^\omega,\ \exists k \geq 0.\ M(w[0..k]) = v \,\} \,\big|,$$

the number of verdicts emitted on some non-empty prefix. It satisfies $n_{\text{eff}}(M, \Phi) \leq n$ for every (M, Φ).

Proposition 7.5 (Effective cardinality of LTL_3 per cell)**.** *For LTL_3 (nominal $n = 3$, verdicts $\{\top, \bot, ?\}$) and a property $\Phi \subseteq \Sigma^\omega$:*

(1) $\Phi = \Sigma^\omega$*:* $n_{\text{eff}} = 1$ *(only $\top$).*

(2) $\Phi = \varnothing$*:* $n_{\text{eff}} = 1$ *(only $\bot$).*

(3) $\Phi \in \mathsf{S}\cap\mathsf{C}$ *non-trivial:* $n_{\text{eff}} \in \{2, 3\}$ *($\top$ and $\bot$, plus ? iff some one-letter prefix is neither good nor bad; 2 for p_0).*

(4) $\Phi \in \mathsf{S} \cap \mathsf{K}$ *non-trivial or* K*-only:* $n_{\text{eff}} = 2$ *(? and* $\bot$*).*
(5) $\Phi \in \mathsf{C} \cap \mathsf{L}$ *non-trivial or* L*-only:* $n_{\text{eff}} = 2$ *(? and* $\top$*).*
(6) Φ S*-only,* C*-only, or quaestio:* $n_{\text{eff}} = 3$*.*
(7) $\Phi \in \mathsf{L} \cap \mathsf{K}$*:* $n_{\text{eff}} = 1$ *(only ?).*

Proof. LTL_3 emits $\top$ on good prefixes, $\bot$ on bad prefixes and ? otherwise [7, Remark 3.2], and extensions of good (bad) prefixes are good (bad). By Lemma 5.9 (1), $\bot$ occurs iff $\Phi \notin \mathsf{L}$ and $\top$ occurs iff $\Phi \notin \mathsf{K}$. If $\Phi \notin \mathsf{C}$, the prefixes of a satisfying trace without good prefix are neither good nor bad, so ? occurs; dually if $\Phi \notin \mathsf{S}$; on $\mathsf{L} \cap \mathsf{K}$ only ? occurs. A prefix of a prefix that is neither good nor bad is itself neither, so on $\mathsf{S} \cap \mathsf{C}$ the value ? occurs on a non-empty prefix iff it occurs on a one-letter prefix (3 for $\mathbf{X}p$). Reading off the membership tuple of each cell gives (1)–(7). □

The proposition makes a structurally useful identification: the three cells

$$\{\Sigma^\omega, \varnothing, \mathsf{L} \cap \mathsf{K}\}$$

are unified under LTL_3 as the *effective*-1 class – the cells from which the LTL_3 monitor extracts no verdict variation. This is a different unification from the monitorability gradient of Remark 5.8, which separates the $\mathsf{L} \cap \mathsf{K}$ permanently-inconclusive cell from the two trivial singletons, decided at every prefix; the effective cardinality is an *information-theoretic* grouping rather than an operational one.

Figure 3 consolidates the classifiers introduced by sections 5 to 7 into a single landscape map of the 11 non-empty cells: each cell carries simultaneously its membership tuple, its default LTL witness, its verdict-completeness/horizon class (via colour), and its effective verdict cardinality under LTL_3 (via border thickness).

Beyond the per-cell static view of figure 3, the iterated-monitor structure of Proposition 7.1 and the three layer regimes of this section admit a cross-layer visualisation (figure 9): a tower of trace types tied by the meta-alphabet constraint on the left, and the evolution of verdict cardinality $|V_k|$ along the tower on the right.

The tower of figure 9 is read in trace space: each layer is a different trace alphabet, linked by monitor arrows. A complementary reading in *property space* distinguishes three kinds of property at every layer: *meta-properties* (properties about properties, classifying which cell a given property inhabits), *monitorable properties at layer k* (properties on the trace alphabet at that layer), and *properties on the verdict trace produced by the layer-k monitor* (which are themselves layer-$(k{+}1)$ monitorable properties). The meta-alphabet constraint of Proposition 7.1 ties the layer cascade: the Layer-$(k{+}1)$ trace alphabet is exactly the Layer-k verdict alphabet V_k.

The word "layer" carries three distinct meanings across the preceding discussion that should be disentangled to avoid ambiguity: the *syntactic* layer (operator nesting depth), the *architectural* layer (monitor composition depth), and the *topological* layer (Borel/Manna–Pnueli level). These three axes are not co-extensive: a property has a coordinate on each, and the coordinates are correlated but independent. Table 17 (Rosetta table) pins the three axes against four representative properties so that their coordinates are read at a glance.

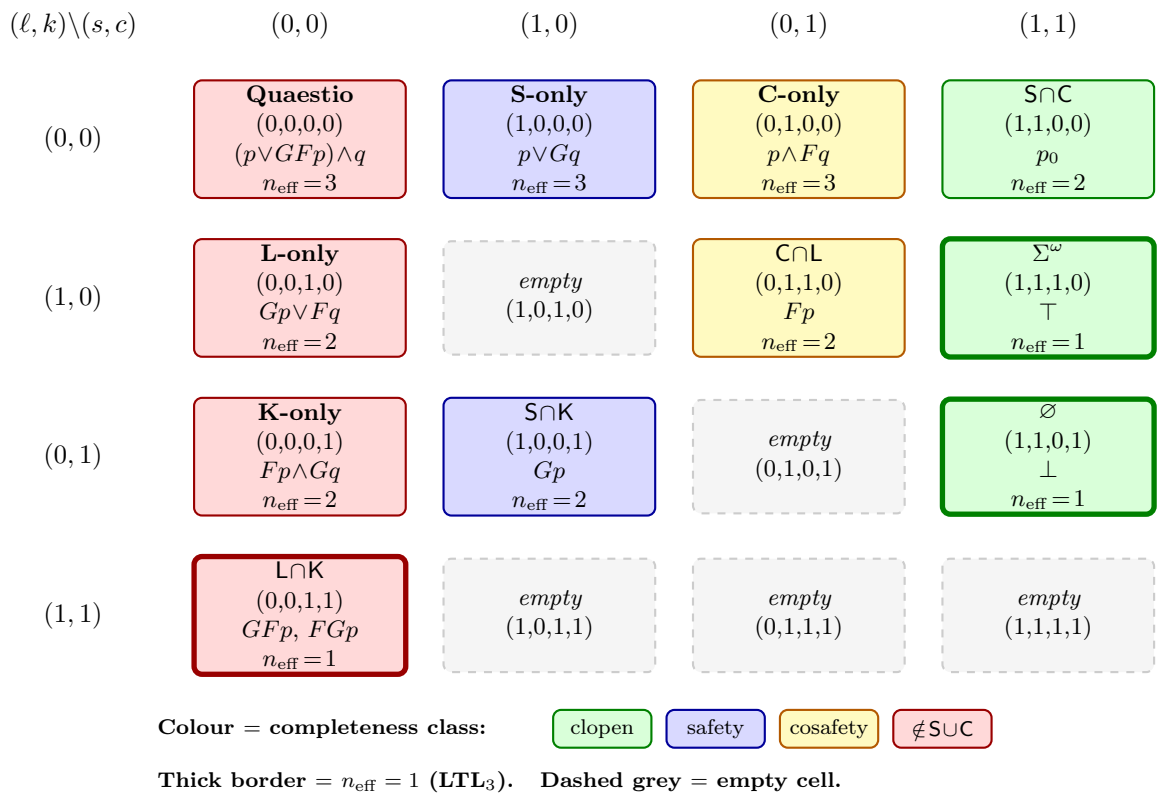


Figure 3: **Cell landscape map.** The 11 non-empty cells of the 16-cell exclusive partition (Theorem 8.4) with three classifiers overlaid simultaneously. *Colour* encodes the verdict-completeness classes of the monitorability gradient (Remark 5.8; red = outside $\mathsf{S} \cup \mathsf{C}$), which by Remark 6.11 coincides with the horizon-sensitivity class of section 6.4. Colour does not classify monitorability: of the red cells only $\mathsf{L} \cap \mathsf{K}$ is non-monitorable throughout, while L-only, K-only and quaestio are mixed (Lemma 5.9). *Thick border* marks effective-$n = 1$ under LTL$_3$ (Proposition 7.5); thin border marks effective-$n \in \{2, 3\}$, printed for the displayed default witness. Cell labels show the membership tuple (s, c, ℓ, k) and the default expression from table 2. In a single view the figure answers four classification questions per cell: membership tuple, default LTL witness, verdict-completeness class, effective verdict cardinality.

8. Joint Classification: Membership Axis versus Verdict Axis

This section relates our 15-cell membership-axis lattice to the 9-cell verdict-axis partition of Peled and Havelund [50]. The relation is a refinement: each verdict-axis cell is a union of one or more membership-axis cells. The Peled–Havelund *quaestio* cell is identified as the unique cell that lies outside the union $\mathsf{S} \cup \mathsf{C} \cup \mathsf{L} \cup \mathsf{K}$.

Definition 8.1 (Verdict-Axis Classes)**.** For each property $\Phi \subseteq \Sigma^\omega$, define the *Peled–Havelund verdict-availability classes* as

$$AFR(\Phi) \iff \Phi \in \mathsf{S}, \quad SFR(\Phi) \iff \Phi \notin \mathsf{S} \wedge \Phi \notin \mathsf{L}, \quad NFR(\Phi) \iff \Phi \in \mathsf{L},$$

$$AFS(\Phi) \iff \Phi \in \mathsf{C}, \quad SFS(\Phi) \iff \Phi \notin \mathsf{C} \wedge \Phi \notin \mathsf{K}, \quad NFS(\Phi) \iff \Phi \in \mathsf{K}.$$

The verdict-axis partition is the product $\{AFR, SFR, NFR\} \times \{AFS, SFS, NFS\}$, with nine cells.

Theorem 8.2 (Refinement of Peled–Havelund 9-Cell Partition)**.** *Let $\mathcal{V}$ be the verdict-axis partition of Definition 8.1 and let $\mathcal{M}$ be the membership-axis cells of table 1, each read exclusively as the set of properties with one complete membership tuple (s, c, ℓ, k) (as in Theorem 8.4). The following hold.*

(1) *$\mathcal{M}$ refines $\mathcal{V}\backslash\{SFR\cap SFS\}$: every cell of $\mathcal{M}$ other than the trivial singletons is contained in exactly one verdict-axis cell of $\mathcal{V}$; $\{\Sigma^\omega\}$ lies in $AFR\cap AFS$ and $NFR\cap AFS$, and $\{\varnothing\}$ in $AFR \cap AFS$ and $AFR \cap NFS$, since $\Sigma^\omega \in AFR \cap NFR$ and $\varnothing \in AFS \cap NFS$* [50, Sec. 3].

(2) *The Peled–Havelund cell $SFR \cap SFS$ coincides exactly with $\Sigma^\omega \setminus (\mathsf{S} \cup \mathsf{C} \cup \mathsf{L} \cup \mathsf{K})$.*

(3) *The four* corner *cells of $\mathcal{V}$ coincide, modulo the two trivial singletons $\{\Sigma^\omega, \varnothing\}$, with the four pair-intersections of $\mathcal{M}$ that cross axes: $AFR\cap AFS = \mathsf{S}\cap\mathsf{C}$, $AFR\cap NFS = \mathsf{S}\cap\mathsf{K}$, $NFR \cap AFS = \mathsf{C} \cap \mathsf{L}$, $NFR \cap NFS = \mathsf{L} \cap \mathsf{K}$.*

(4) *The four* edge *cells of $\mathcal{V}$ each coincide with a single non-empty cell of $\mathcal{M}$: $AFR\cap SFS = (\mathsf{S} \cap \overline{\mathsf{C}} \cap \overline{\mathsf{K}})$ is the atomic cell with tuple $(1, 0, 0, 0)$, the tuple $(1, 0, 1, 0)$ being empty (and analogously for the other three edges).*

Proof. Part (1) is by case analysis on $\mathcal{M}$. Each cell of $\mathcal{M}$ specifies a complete membership tuple $(s, c, \ell, k) \in \{0, 1\}^4$ with $s \vee c \vee \ell \vee k = 1$; the tuple uniquely determines exactly one verdict-axis coordinate per axis by Definition 8.1, except that $(s, \ell) = (1, 1)$ and $(c, k) = (1, 1)$, which occur only for Σ^ω and $\varnothing$ (Theorem 8.4), give two coordinates on one axis.

Part (2). $SFR \cap SFS$ is the set of properties with $\Phi \notin \mathsf{S}, \mathsf{L}, \mathsf{C}, \mathsf{K}$, which is exactly $\Sigma^\omega \setminus (\mathsf{S} \cup \mathsf{C} \cup \mathsf{L} \cup \mathsf{K})$ by the closure of these four classes under complement of membership. Nontriviality of $SFR \cap SFS$ is witnessed by $((p \vee \Box\Diamond p) \wedge q)$, a variant of the witness $((p \vee \Box\Diamond p) \wedge \bigcirc q)$ of [50, Sec. 3].

Part (3). Direct substitution into Definition 8.1.

Part (4). The cell $AFR \cap SFS$ requires $\Phi \in \mathsf{S} \cap \overline{\mathsf{C}} \cap \overline{\mathsf{K}}$, i.e. the tuple $(1, 0, \ell, 0)$, and does not constrain ℓ. The tuple $(1, 0, 1, 0)$ is empty: $\mathsf{S} \cap \mathsf{L} = \{\Sigma^\omega\}$ (table 1) and $\Sigma^\omega \in \mathsf{C}$. Hence $AFR \cap SFS$ is the atomic cell $(1, 0, 0, 0)$. The arguments for the other three edges are symmetric, via $\mathsf{C} \cap \mathsf{K} = \{\varnothing\}$ and $\varnothing \in \mathsf{S}$. □

Figure 4 visualises the correspondence: the 3×3 Peled–Havelund verdict-axis grid is annotated with the membership-axis cell(s) of Theorem 8.4 living inside each PH cell, making the 8-of-9 one-to-one structure and the unique 3-to-1 aggregation at $AFR\cap AFS$ (collecting the non-trivial clopen with the two trivial singletons) directly readable.

Remark 8.3 (Quaestio Outside the Lattice)**.** Theorem 8.2 (2) restates in membership-axis terms the observation of [50, Sec. 3]: the Peled–Havelund *quaestio* class is precisely the set of properties *outside* the union of all four primitive classes $\mathsf{S}\cup\mathsf{C}\cup\mathsf{L}\cup\mathsf{K}$. In membership-axis terms it occupies the "$\varnothing$" node of the subset lattice over $\{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$, which is not among our 15 cells (we index cells by *nonempty* subfamilies). The complete classification of ω-regular properties therefore decomposes as

$$\Sigma^\omega = \big(\mathsf{S} \cup \mathsf{C} \cup \mathsf{L} \cup \mathsf{K}\big) \;\sqcup\; \textit{Quaestio}.$$

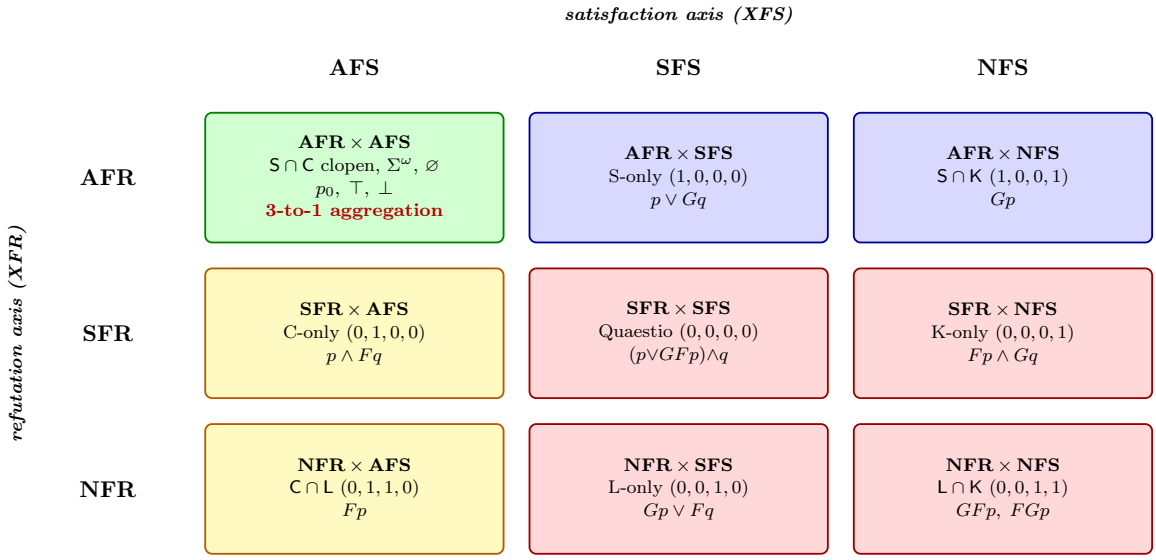


Figure 4: **Bijection between the Peled–Havelund 9-cell verdict-axis partition and the nine non-degenerate cells of the 11-cell membership-axis partition** (Theorem 8.2). Each cell of the 3×3 PH grid (axes XFR and XFS) is annotated with the membership tuple (s, c, ℓ, k) and canonical default expression of the CLASP cell(s) it contains. Eight of the nine PH cells are in 1-to-1 correspondence with a single CLASP cell. The remaining cell, $AFR \cap AFS$, aggregates three CLASP cells: the non-trivial clopen $\mathsf{S} \cap \mathsf{C}$ together with the two trivial singletons Σ^ω and $\varnothing$. Cell colours track those of figure 3: green for the clopen aggregate cell, blue for safety-monitorable PH cells, yellow/orange for cosafety-monitorable, red for the cells outside $\mathsf{S} \cup \mathsf{C}$ (Remark 5.8); of these only $NFR \cap NFS$ is non-monitorable throughout, the other three contain both monitorable and non-monitorable properties [50, Sec. 4.3] (Lemma 5.9).

8.1. **Totality: The Exhaustive 16-Cell Membership Partition.** The membership-axis lattice of table 1 is a *subset-monotone* indexing: each cell Γ is the intersection $\bigcap_{X \in \Gamma} X$. This indexing is not a partition of Σ^ω because different cells share elements (for instance Σ^ω belongs to all of $\mathsf{S} \cap \mathsf{C}$, $\mathsf{S} \cap \mathsf{L}$, $\mathsf{C} \cap \mathsf{L}$, and $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L}$). The natural partition–the one used by Peled and Havelund and referenced implicitly in section 9 for the worked-example tuples–is given by the *exclusive membership tuple* $(s, c, \ell, k) \in \{0, 1\}^4$ assigning each property exactly one of $2^4 = 16$ partition cells. This section establishes that this 16-cell partition is exhaustive (every $\Phi \subseteq \Sigma^\omega$ falls in exactly one cell) and identifies precisely which cells are non-empty. The conceptual bridge from the subset-monotone 15-cell view to the 16-cell exclusive partition is summarised in Remark 3.8.

Theorem 8.4 (Totality of the 16-Cell Exclusive Partition)**.** *For every property* $\Phi \subseteq \Sigma^\omega$*, define* $t(\Phi) := (s_\Phi, c_\Phi, \ell_\Phi, k_\Phi) \in \{0, 1\}^4$ *by*

$$s_\Phi = [\Phi \in \mathsf{S}],\ c_\Phi = [\Phi \in \mathsf{C}],\ \ell_\Phi = [\Phi \in \mathsf{L}],\ k_\Phi = [\Phi \in \mathsf{K}],$$

with $[\cdot]$ *the Iverson bracket. Then:*

(1) *t partitions $\mathcal{P}(\Sigma^\omega)$ into 16 pairwise disjoint cells $C_v := t^{-1}(v)$ for $v \in \{0,1\}^4$, with $\bigsqcup_{v\in\{0,1\}^4} C_v = \mathcal{P}(\Sigma^\omega)$.*

(2) *Exactly* eleven *of the 16 cells are non-empty:*

- *the four atomic single cells $C_{(1,0,0,0)}$, $C_{(0,1,0,0)}$, $C_{(0,0,1,0)}$, $C_{(0,0,0,1)}$;*
- *the four* cross-axis *pair cells $C_{(1,1,0,0)}$, $C_{(1,0,0,1)}$, $C_{(0,1,1,0)}$, $C_{(0,0,1,1)}$;*
- *the two singleton triple cells $C_{(1,1,1,0)} = \{\Sigma^\omega\}$ and $C_{(1,1,0,1)} = \{\varnothing\}$;*
- *the all-zeros cell $C_{(0,0,0,0)}$, which coincides with the Peled–Havelund* quaestio *class.*

(3) *The remaining* five *cells are provably empty: $C_{(1,0,1,0)}$, $C_{(0,1,0,1)}$, $C_{(1,0,1,1)}$, $C_{(0,1,1,1)}$, $C_{(1,1,1,1)}$.*

Proof. Part (1) is immediate: each Φ has a uniquely determined membership tuple by the four class definitions, and every tuple has a corresponding cell.

Part (2). Non-emptiness of each named cell is witnessed by an explicit LTL formula in section 9: $p \vee Gq$, $p \wedge Fq$, $Gp \vee Fq$, $Fp \wedge Gq$ for the four atomic cells respectively (table 7 for the first three; Example A.4 for the fourth); p, Gp, Fp, GFp for the four cross-axis pair cells (Examples 9.1, 9.3, A.1 and A.2); $\top$ and $\bot$ for the two singleton triple cells (Examples 9.7 and 9.8 and Corollary 8.12); and $(p \vee GFp) \wedge q$ for the quaestio cell (Example 9.14), a variant of the witness $((p \vee \square\lozenge p) \wedge \bigcirc q)$ of [50, Sec. 3].

Part (3). $C_{(1,0,1,0)} = \mathsf{S} \cap \overline{\mathsf{C}} \cap \mathsf{L} \cap \overline{\mathsf{K}} \subseteq \mathsf{S} \cap \mathsf{L} = \{\Sigma^\omega\}$ by Lemma 8.8; but $\Sigma^\omega \in \mathsf{C}$ (it is open) and $\Sigma^\omega \notin \mathsf{K}$ (it has full interior), so Σ^ω does not satisfy $\overline{\mathsf{C}}$, hence $C_{(1,0,1,0)} = \varnothing$. Symmetric argument shows $C_{(0,1,0,1)} = \varnothing$ via Lemma 8.9. The three remaining cells $C_{(1,0,1,1)}$, $C_{(0,1,1,1)}$, and $C_{(1,1,1,1)}$ are subcells of $\mathsf{S} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$ (Proposition 8.10 (3)), of $\mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$ (Proposition 8.10 (4)), and of $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$ (Proposition 8.10 (5)) respectively; all three are therefore empty. □

Corollary 8.5 (Quaestio Witnessed)**.** *The all-zeros cell $C_{(0,0,0,0)}$ of Theorem 8.4 is non-empty. Hence the union of the four primitive classes does not exhaust Σ^ω: there exist properties $\Phi \subseteq \Sigma^\omega$ that belong to none of $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$. A canonical witness is $\Phi_Q := [\![(p \vee GFp) \wedge q]\!]$, a variant of the witness $((p \vee \square\lozenge p) \wedge \bigcirc q)$ of* [50, Sec. 3].

Proof. Direct from Theorem 8.4 (2) and Example 9.14. □

Remark 8.6 (Relation to the Manna–Pnueli Hierarchy)**.** The Manna–Pnueli safety-progress hierarchy [48, 15] classifies ω-regular properties by Borel complexity in the Cantor topology:

$$\underbrace{\Pi^0_1}_{\text{Safety}}, \quad \underbrace{\Sigma^0_1}_{\text{Guarantee}} \quad \subseteq \quad \underbrace{\Delta^0_2}_{\text{Obligation}} \quad \subseteq \quad \underbrace{\Pi^0_2}_{\text{Recurrence/Response}} \quad , \quad \underbrace{\Sigma^0_2}_{\text{Persistence}} \quad \subseteq \quad \underbrace{\Delta^0_3}_{\text{Reactivity}} \quad .$$

Here Safety/Guarantee and Recurrence/Response / Persistence are *incomparable sibling* classes, not a linear chain: $\Pi^0_1 \not\subseteq \Sigma^0_1$ and $\Pi^0_2 \not\subseteq \Sigma^0_2$ (each pair is De Morgan dual, cf. Remark 9.6). The inclusions above are the edges of the Hasse diamond of figure 5: every Δ-level is the join of the two sibling classes immediately below it. The Peled–Havelund verdict-axis tripartition and the Manna–Pnueli hierarchy are *orthogonal* refinements of ω-regular properties, but the following inclusions hold at the boundary between them. We write MP_i for the i-th level of the hierarchy.

(1) $\mathrm{MP}_1 = \mathsf{S} = AFR$ exactly: safety properties are precisely the closed sets, i.e. the "always-finitely-refutable" class.

(2) $\mathrm{MP}_2 = \mathsf{C} = AFS$ exactly: guarantee properties are precisely the open sets, i.e. the "always-finitely-satisfiable" class.
(3) MP_3 (Obligation, Δ_2^0) decomposes into Boolean combinations of S and C and may populate any cell of the verdict-axis 3×3 partition except the strictly Π_2^0-or-higher cells—in particular obligation properties can sit in any of AFR, SFR, or even NFR on the refutation axis, and analogously on the satisfaction axis.
(4) MP_4 (Recurrence/Response, Π_2^0) is contained in $\overline{AFR} = SFR \cup NFR$: a strictly-$\Pi_2^0$ property cannot also be closed (Π_1^0), since a closed property is already safety (level MP_1), contradicting strict MP_4 membership.
(5) MP_5 (Persistence, Σ_2^0) is contained in $\overline{AFS} = SFS \cup NFS$ by the dual argument.
(6) MP_6 (Reactivity, Δ_3^0; every LTL-definable property is reactivity) can populate every one of the nine $XFR \times XFS$ cells, including the quaestio cell $C_{(0,0,0,0)} = SFR \cap SFS$.

Direct verification on the Peled–Havelund Fig. 1 witnesses [50, Fig. 1] places them at MP_1 (witness p, $\Box p$, $p \vee \Box q$), MP_2 ($\Diamond p$, $p \wedge \Diamond q$), MP_3 ($\Box p \vee \Diamond q$, $\Diamond p \wedge \Box q$), and MP_4 ($\Box\Diamond p$ and the quaestio witness $(p \vee \Box\Diamond p) \wedge q$). The quaestio cell admits no witness at MP_1 or MP_2, since $\mathrm{MP}_1 \cup \mathrm{MP}_2$ is contained in $\mathsf{S} \cup \mathsf{C}$ and therefore disjoint from the quaestio; it does admit strict witnesses from MP_3 upward. At MP_3, $(a \wedge Gc) \vee (\neg a \wedge Fb)$ is the union of a closed and an open set and lies in the quaestio cell (Lemma 5.9), as does $((p \vee q)\, U\, r) \vee Gp$ [7, Lemma 3.8]; the default witness $(p \vee \Box\Diamond p) \wedge q$ sits at MP_4.

This positioning agrees with the Falcone–Fernández–Mounier analysis [24], who show that every obligation (MP_3) property is monitorable [24, Thm. 2] (there for r-properties over finite and infinite executions; for ω-languages, every set in the Boolean algebra generated by the open and the closed sets is monitorable [19, Cor. 1]; so the MP_3 members of the quaestio cell above are monitorable, whereas $(p \vee \Box\Diamond p) \wedge q$ is not) and that monitorable and non-monitorable properties co-exist at levels MP_4, MP_5, and MP_6; the verdict-axis tripartition refines this observation by localizing exactly which monitoring verdicts are available at each level. The Manna–Pnueli hierarchy and the verdict-axis tripartition are therefore neither in inclusion nor in opposition: they are independent classification axes, intersected as a matter of verdict-availability $\times$ Borel-complexity rectangle.

Figure 5 renders the Manna–Pnueli Borel hierarchy as a Hasse diamond and annotates each level with the CLASP cells whose canonical default expression (table 2) realises a strict witness at that level. The figure makes the $\mathsf{L} \cap \mathsf{K}$ Recurrence/Persistence bipartition of Remark 9.6 visible: a single CLASP cell splits across the two sibling levels MP_4 (Π_2^0, GFp) and MP_5 (Σ_2^0, FGp).

8.2. **Composability Examples for Each Membership-Axis Cell.** To verify that the lattice cells are realizable, this subsection provides an explicit LTL composition for every nonempty cell of table 1, plus an LTL witness for the *quaestio* cell of Remark 8.3. Each example is annotated with the membership tuple (s, c, ℓ, k), the Peled–Havelund verdict-axis cell, and the LTL formula whose denotation realizes the cell. The atomic propositions p, q are taken distinct.

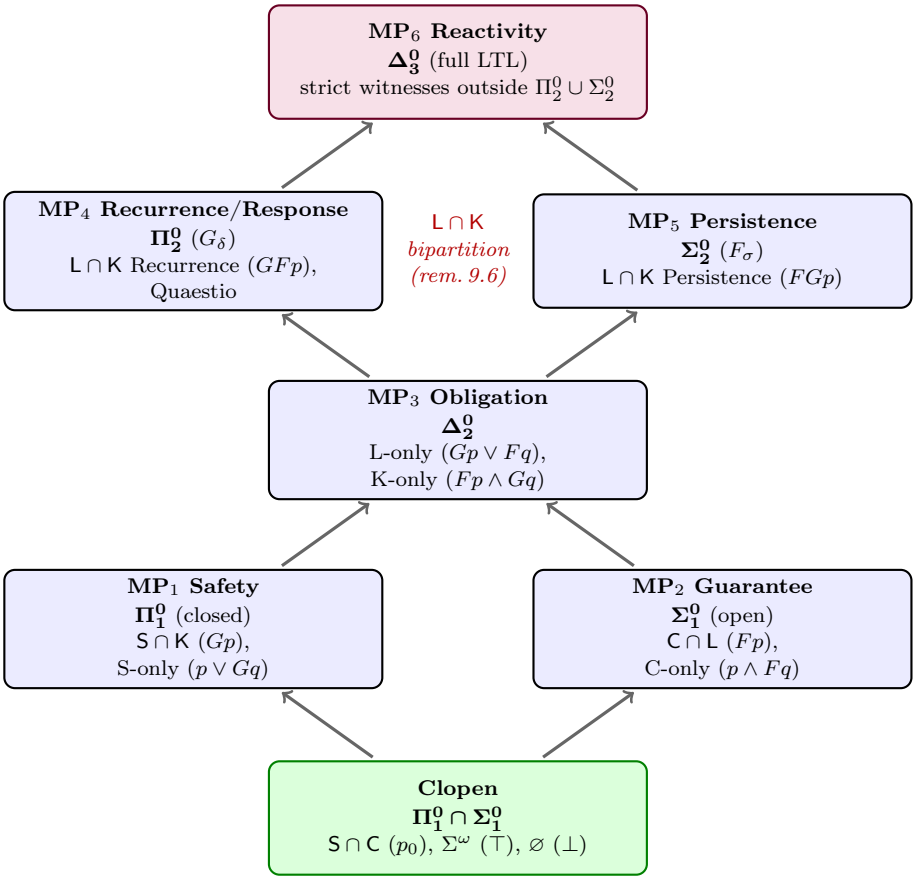


Figure 5: **Manna–Pnueli hierarchy with CLASP-cell witnesses.** The Borel tower of Remark 8.6 rendered as a Hasse diamond: arrows indicate strict inclusion of complexity classes. Each node is annotated with the CLASP cells whose canonical default expression (table 2) realises a *strict* witness at that level (i.e., that does not admit a witness at a lower level). The single CLASP cell L∩K splits across the two sibling levels MP$_4$ (Recurrence, $GFp \in \Pi_2^0$) and MP$_5$ (Persistence, $FGp \in \Sigma_2^0$), realising the bipartition of Remark 9.6. The Manna–Pnueli levels and the CLASP partition are independent classification axes (Remark 8.6); the figure depicts their intersection on the worked-example witnesses of section 9.

Exclusive cell Γ	(s, c, ℓ, k)	LTL witness	PH cell
{S}	$(1,0,0,0)$	$p \vee \mathsf{G}q$	$AFR \cap SFS$
{C}	$(0,1,0,0)$	$p \wedge \mathsf{F}q$	$SFR \cap AFS$
{L}	$(0,0,1,0)$	$\mathsf{G}p \vee \mathsf{F}q$	$NFR \cap SFS$
{K}	$(0,0,0,1)$	$\mathsf{F}p \wedge \mathsf{G}q$	$SFR \cap NFS$
{S, C}	$(1,1,0,0)$	p (length-1 decidable)	$AFR \cap AFS$
{S, K}	$(1,0,0,1)$	Gp	$AFR \cap NFS$
{C, L}	$(0,1,1,0)$	Fp	$NFR \cap AFS$
{L, K}	$(0,0,1,1)$	GFp, FGp	$NFR \cap NFS$
{S, C, L}	$(1,1,1,0)$	$\top$ (only Σ^ω)	$AFR \cap AFS$
{S, C, K}	$(1,1,0,1)$	$\bot$ (only $\varnothing$)	$AFR \cap AFS$
{S, L}	$(1,0,1,0)$	*provably empty*	—
{C, K}	$(0,1,0,1)$	*provably empty*	—
{S, L, K}	$(1,0,1,1)$	*provably empty*	—
{C, L, K}	$(0,1,1,1)$	*provably empty*	—
{S, C, L, K}	$(1,1,1,1)$	*provably empty*	—
Quaestio (outside)	$(0,0,0,0)$	$((p \vee \Box\Diamond p) \wedge q)$	$SFR \cap SFS$

Remark 8.7 (Composability versus Realizability)**.** The witnesses of table 6 are obtained constructively from the corresponding membership tuple (s, c, ℓ, k) by reading off prefix-quantifier obligations from Definitions 3.1 to 3.4: a witness must (i) admit invalidating prefixes for non-satisfying traces iff $s = 1$, (ii) admit confirming prefixes for satisfying traces iff $c = 1$, (iii) make every prefix extendable to satisfaction iff $\ell = 1$, and (iv) make every prefix extendable to violation iff $k = 1$. When a tuple admits no joint LTL realization (rows marked empty), the obstruction is topological: closed-and-dense forces $\Phi = \Sigma^\omega$, open-and-empty-interior forces $\Phi = \varnothing$, and the three remaining empties are consequences of the same identities. This composability methodology generalizes to any future layer of refinement: each new class added to the lattice introduces a new membership bit, and witnessing the resulting cells is a matter of independently satisfying or refuting the corresponding prefix-quantifier obligation.

8.3. **Logical Derivation of Triple-Intersection Witnesses.** The triples and the quadruple of table 1 were classified in the proof of Theorem 5.4 by topological arguments (closed-and-dense, open-and-empty-interior, and chained consequences). We now restate and prove the same five facts *directly from the prefix-quantifier definitions* Definitions 3.1 to 3.4, without invoking the topological characterization Theorem 4.6. The two derivations are isomorphic in content but logically independent: the prefix-quantifier derivation can be carried out before topology is introduced, which justifies the placement of the four primitive definitions as the syntactic skeleton of the framework.

The five claims are stated together as a single proposition whose five parts share auxiliary lemmas. We use the shorthand $\pi \sqsubseteq \sigma$ for "π is a finite prefix of σ" (Definition 2.1) and $\pi \sqsubseteq \tau$ inside a $\forall \tau \in \Sigma^\omega$ quantifier for "τ extends π".

Lemma 8.8 (Safety–Liveness Tension)**.** *If* $\Phi \in \mathsf{S} \cap \mathsf{L}$ *then* $\Phi = \Sigma^\omega$.

Proof. Assume $\Phi \in \mathsf{S} \cap \mathsf{L}$ and suppose for contradiction that there exists $\sigma \in \Sigma^\omega \setminus \Phi$. By Definition 3.1 applied to σ, there exists a prefix $\pi \sqsubseteq \sigma$ such that

$$\forall \tau \in \Sigma^\omega,\ \pi \sqsubseteq \tau \Rightarrow \tau \notin \Phi. \tag{8.1}$$

On the other hand, applying Definition 3.3 to this same prefix π, there exists $\tau^* \in \Sigma^\omega$ with $\pi \sqsubseteq \tau^* \wedge \tau^* \in \Phi$. This τ^* contradicts (8.1). Hence $\Sigma^\omega \setminus \Phi = \varnothing$, i.e., $\Phi = \Sigma^\omega$. □

Lemma 8.9 (Cosafety–Coliveness Tension)**.** *If* $\Phi \in \mathsf{C} \cap \mathsf{K}$ *then* $\Phi = \varnothing$.

Proof. Symmetric to Lemma 8.8. Suppose for contradiction $\sigma \in \Phi$. By Definition 3.2 applied to σ, there exists $\pi \sqsubseteq \sigma$ with

$$\forall \tau \in \Sigma^\omega,\ \pi \sqsubseteq \tau \Rightarrow \tau \in \Phi. \tag{8.2}$$

Applying Definition 3.4 to this same prefix π, there exists $\tau^* \in \Sigma^\omega$ with $\pi \sqsubseteq \tau^* \wedge \tau^* \notin \Phi$, contradicting (8.2). Hence $\Phi = \varnothing$. □

Proposition 8.10 (Triple and Quadruple Cells via Prefix Quantifiers)**.** *Let* $\Phi \subseteq \Sigma^\omega$.

(1) $\Phi \in \mathsf{S} \cap \mathsf{C} \cap \mathsf{L} \iff \Phi = \Sigma^\omega$.
(2) $\Phi \in \mathsf{S} \cap \mathsf{C} \cap \mathsf{K} \iff \Phi = \varnothing$.
(3) $\mathsf{S} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$.
(4) $\mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$.
(5) $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$.

Proof. **(1)** ($\Rightarrow$) If $\Phi \in \mathsf{S} \cap \mathsf{L}$ then $\Phi = \Sigma^\omega$ by Lemma 8.8, and $\Sigma^\omega \in \mathsf{C}$ is automatic. ($\Leftarrow$) Σ^ω satisfies each of Definitions 3.1 to 3.3 vacuously (Σ^ω has no non-members, every prefix is a confirming prefix, every prefix extends to a member).

(2) ($\Rightarrow$) If $\Phi \in \mathsf{C} \cap \mathsf{K}$ then $\Phi = \varnothing$ by Lemma 8.9, and $\varnothing \in \mathsf{S}$ is automatic. ($\Leftarrow$) $\varnothing$ satisfies each of Definitions 3.1, 3.2 and 3.4 vacuously.

(3) Suppose $\Phi \in \mathsf{S} \cap \mathsf{L} \cap \mathsf{K}$. By Lemma 8.8, $\Phi = \Sigma^\omega$. Pick any prefix $\pi \in \Sigma^*$. By Definition 3.4, there exists $\tau \in \Sigma^\omega$ with $\pi \sqsubseteq \tau$ and $\tau \notin \Phi = \Sigma^\omega$, which is impossible. Hence no such Φ exists.

(4) Symmetric. Suppose $\Phi \in \mathsf{C} \cap \mathsf{L} \cap \mathsf{K}$. By Lemma 8.9, $\Phi = \varnothing$. Pick any prefix π. By Definition 3.3, there exists $\tau \in \Sigma^\omega$ with $\pi \sqsubseteq \tau$ and $\tau \in \Phi = \varnothing$, which is impossible.

(5) $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L} \cap \mathsf{K} \subseteq \mathsf{S} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$ by part (3). □

Remark 8.11 (Dual Derivation: Topology Mirrors Logic)**.** The proofs of Proposition 8.10 use only the quantifier patterns of the four primitive class definitions and elementary contradiction; no topological notion appears. Translating the four key contradictions through Theorem 4.6 recovers the topological argument of Theorem 5.4:

$$\begin{aligned}
\mathsf{S} \cap \mathsf{L} = \{\Sigma^\omega\} &\equiv \text{closed} \cap \text{dense} = \text{universe},\\
\mathsf{C} \cap \mathsf{K} = \{\varnothing\} &\equiv \text{open} \cap \text{empty-interior} = \varnothing,\\
\mathsf{S} \cap \mathsf{L} \cap \mathsf{K} = \varnothing &\equiv \Sigma^\omega \text{ has nonempty interior},\\
\mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing &\equiv \varnothing \text{ is not dense}.
\end{aligned}$$

The two derivations agree pointwise; their independence matters methodologically because subsequent layers of the framework (those that extend $\{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$ by adding further class bits, or those that operate over trace structures where the Cantor topology is unavailable, e.g. finite-trace prefixes) can inherit the logical derivation without committing to the topological one.

Corollary 8.12 (Witness Existence for Triple Cells)**.** *The two nonempty triple cells of table 1 admit the following canonical LTL witnesses, verifiable entirely from Definitions 3.1 to 3.4:*

$$\begin{aligned}
[\![\top]\!] = \Sigma^\omega &\in \mathsf{S} \cap \mathsf{C} \cap \mathsf{L},\\
[\![\bot]\!] = \varnothing &\in \mathsf{S} \cap \mathsf{C} \cap \mathsf{K}.
\end{aligned}$$

The three empty triple/quadruple cells admit no LTL witness by Proposition 8.10 (3)–(5).

Proof. Membership of Σ^ω and $\varnothing$ in the asserted cells follows from the converse directions of Proposition 8.10 (1) and (2) respectively. Non-existence of LTL witnesses for the empty cells is immediate: an LTL formula φ denotes a property $[\![\varphi]\!] \subseteq \Sigma^\omega$ which would have to be an element of an empty class. □

8.4. **De Morgan Duality on the 16-Cell Partition.** The LTL De Morgan equivalences (Theorem 2.4) lift to a class-level involution on the membership-axis lattice. At the property level, set complement $\neg\colon \mathcal{P}(\Sigma^\omega) \to \mathcal{P}(\Sigma^\omega)$, $\Phi \mapsto \Sigma^\omega \setminus \Phi$, is an involution that exchanges the closed/open and dense/empty-interior topological characters in matched pairs.

Theorem 8.13 (Class-Level De Morgan Duality)**.** *The complement involution $\neg$ on $\mathcal{P}(\Sigma^\omega)$ satisfies:*

(1) $\neg(\mathsf{S}) = \mathsf{C}$ *and* $\neg(\mathsf{C}) = \mathsf{S}$*;*

(2) $\neg(\mathsf{L}) = \mathsf{K}$ *and* $\neg(\mathsf{K}) = \mathsf{L}$.

Consequently, for every cell index $\Gamma \subseteq \{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$ *of table 1,*

$$\neg\Big(\bigcap_{X\in\Gamma} X\Big) \;=\; \bigcap_{X\in\Gamma} \neg X \;=\; \bigcap_{X\in\overline{\Gamma}} X,$$

where $\overline{\Gamma}$ *is the index obtained from* Γ *by applying the dual swap* $\mathsf{S} \leftrightarrow \mathsf{C}$, $\mathsf{L} \leftrightarrow \mathsf{K}$.

Proof. (1) $\mathsf{S} \subseteq \neg\mathsf{C}$: if Φ is closed then $\neg\Phi$ is open, so $\neg\Phi \in \mathsf{C}$, i.e. $\Phi \in \neg\mathsf{C}$. Conversely, every open set has closed complement. Hence $\mathsf{S} = \neg\mathsf{C}$, equivalently $\neg\mathsf{S} = \mathsf{C}$. (2) $\mathsf{L} = \neg\mathsf{K}$: $\mathrm{cl}(\Phi) = \Sigma^\omega$ iff $\mathrm{int}(\neg\Phi) = \varnothing$, by the standard identity $\mathrm{int}(\neg\Phi) = \neg\,\mathrm{cl}(\Phi)$. The cell-level statement is then immediate from set-theoretic De Morgan $\neg(\bigcap X_i) = \bigcup \neg X_i$ combined with $\neg$ being an involution on $\mathcal{P}(\Sigma^\omega)$. □

Corollary 8.14 (16-Cell Duality Structure)**.** *The involution* $\neg$ *on* $\mathcal{P}(\Sigma^\omega)$ *permutes the 16 exclusive cells of Theorem 8.4 as follows:*

(1) *Four swap-pairs of non-empty cells:*

$$\{\mathsf{S}\} \;\overset{\neg}{\leftrightarrow}\; \{\mathsf{C}\}, \quad \{\mathsf{L}\} \;\overset{\neg}{\leftrightarrow}\; \{\mathsf{K}\},$$

$$\{\mathsf{S},\mathsf{K}\}_{excl} \;\overset{\neg}{\leftrightarrow}\; \{\mathsf{C},\mathsf{L}\}_{excl}, \quad \{\Sigma^\omega\} \;\overset{\neg}{\leftrightarrow}\; \{\varnothing\}.$$

(2) *Three self-dual non-empty cells:* $\{\mathsf{S},\mathsf{C}\}_{excl}$, $\{\mathsf{L},\mathsf{K}\}_{excl}$, *and the* quaestio *cell* $C_{(0,0,0,0)}$.

(3) *Two empty-cell swap-pairs and one self-dual empty cell:* $C_{(1,0,1,0)} \leftrightarrow C_{(0,1,0,1)}$; $C_{(1,0,1,1)} \leftrightarrow C_{(0,1,1,1)}$; $C_{(1,1,1,1)}$ *self.*

Equivalently: $\neg$ *acts on the 4-bit membership tuple* (s, c, ℓ, k) *by simultaneous transpositions* $s \leftrightarrow c$ *and* $\ell \leftrightarrow k$, *yielding* (c, s, k, ℓ).

Proof. Direct from Theorem 8.13: each membership bit $s_\Phi = [\Phi \in \mathsf{S}]$ transforms under $\Phi \mapsto \neg\Phi$ according to $s_{\neg\Phi} = c_\Phi$ and $\ell_{\neg\Phi} = k_\Phi$. The orbit structure of the resulting tuple transposition is the disjoint union of swap-pairs and fixed points enumerated in the statement. □

Remark 8.15 (LTL De Morgan Equivalences as Witness Pairs)**.** The LTL De Morgan laws Theorem 2.4 produce witness pairs for the four non-trivial swap-pairs of Corollary 8.14:

$$\begin{array}{lcl}
Gp \;(\in \mathsf{S}\cap\mathsf{K}) & \overset{\neg}{\leftrightarrow} & F\neg p \;(\in \mathsf{C}\cap\mathsf{L}) \\
Fp \;(\in \mathsf{C}\cap\mathsf{L}) & \overset{\neg}{\leftrightarrow} & G\neg p \;(\in \mathsf{S}\cap\mathsf{K}) \\
p \vee Gq \;(\in \{\mathsf{S}\}) & \overset{\neg}{\leftrightarrow} & \neg p \wedge F\neg q \;(\in \{\mathsf{C}\}) \\
\top \;(= \Sigma^\omega) & \overset{\neg}{\leftrightarrow} & \bot \;(= \varnothing)
\end{array}$$

For the three self-dual cells, the same witness up to atomic renaming serves both as an element and as its image: e.g. $GFp \overset{\neg}{\leftrightarrow} FG\neg p$ are both in $\{\mathsf{L},\mathsf{K}\}_{\mathrm{excl}}$; and the quaestio witness $(p \vee GFp) \wedge q$ is sent by $\neg$ to $(\neg p \wedge FG\neg p) \vee \neg q$, again in $C_{(0,0,0,0)}$.

8.5. **Counting Resolution and Equivalence to Peled–Havelund.** The paper uses two distinct cell-counts: the *subset-monotone* count of table 1 and the *exclusive partition* count of Theorem 8.4. We now reconcile them and locate the present work precisely relative to the Peled–Havelund classification.

Remark 8.16 (Reconciliation of 15-vs-16 and 11-vs-9 Counts)**.** There are two coherent ways to count cells:

(1) *Subset-monotone (V3 table 1).* Index by nonempty subsets $\Gamma \subseteq \{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$. This gives $2^4 - 1 = 15$ cells, of which 12 are non-empty (with 4 of them degenerate singletons $\{\Sigma^\omega\}$ or $\{\varnothing\}$) and 3 are provably empty. This indexing is not a partition.

(2) *Exclusive partition (Theorem 8.4).* Index by the membership tuple $(s, c, \ell, k) \in \{0, 1\}^4$. This gives $2^4 = 16$ cells partitioning $\mathcal{P}(\Sigma^\omega)$ exactly, of which 11 are non-empty (2 of them being the degenerate singletons $\{\Sigma^\omega\}$ and $\{\varnothing\}$) and 5 are provably empty.

The two counts are not in conflict: the subset-monotone indexing identifies 4 of its 12 non-empty cells with the two degenerate singletons because $\{\mathsf{S}, \mathsf{L}\} = \{\mathsf{S}, \mathsf{C}, \mathsf{L}\} = \{\Sigma^\omega\}$ and $\{\mathsf{C}, \mathsf{K}\} = \{\mathsf{S}, \mathsf{C}, \mathsf{K}\} = \{\varnothing\}$, whereas the exclusive partition counts each singleton as a single cell. Subtracting the singleton-valued rows from each scheme makes the accounting explicit: the subset-monotone scheme yields $12 - 4 = 8$ non-degenerate non-empty cells, while the exclusive partition yields $11 - 2 = 9$. The two schemes thus agree on *eight non-degenerate non-empty cells* – the four atomic single cells $\{\mathsf{S}\}, \{\mathsf{C}\}, \{\mathsf{L}\}, \{\mathsf{K}\}$ and the four pairs $\{\mathsf{S}, \mathsf{C}\}, \{\mathsf{S}, \mathsf{K}\}, \{\mathsf{C}, \mathsf{L}\}, \{\mathsf{L}, \mathsf{K}\}$ – and the exclusive partition contributes a *ninth*, the quaestio cell $C_{(0,0,0,0)}$, which is the empty-subfamily node and so has no subset-monotone index (Remark 8.3).

These nine non-degenerate non-empty cells coincide exactly with the nine $XFR \times XFS$ cells of Peled–Havelund [50, Fig. 1]**.** In particular:

- the four atomic single cells $\{\mathsf{S}\}, \{\mathsf{C}\}, \{\mathsf{L}\}, \{\mathsf{K}\}$ are $AFR \cap SFS$, $SFR \cap AFS$, $NFR \cap SFS$, $SFR \cap NFS$ respectively;
- the four cross-axis pair cells $\{\mathsf{S}, \mathsf{C}\}_{\text{excl}}, \{\mathsf{S}, \mathsf{K}\}_{\text{excl}}, \{\mathsf{C}, \mathsf{L}\}_{\text{excl}}, \{\mathsf{L}, \mathsf{K}\}_{\text{excl}}$ are $AFR \cap AFS$, $AFR \cap NFS$, $NFR \cap AFS$, $NFR \cap NFS$;
- the quaestio cell $C_{(0,0,0,0)}$ is $SFR \cap SFS$.

Remark 8.17 (Equivalence with Peled–Havelund 2019)**.** Remark 8.16 makes explicit that the membership-axis 16-cell partition of the present work is *equivalent* to the verdict-axis 9-cell partition of [50] once one factors out the two degenerate singletons $\{\Sigma^\omega\} = [\![\top]\!]$ and $\{\varnothing\} = [\![\bot]\!]$. Peled and Havelund's classification subsumes ours up to this trivial refinement. The contribution of the present work is therefore not a new classification of ω-regular properties. It is rather:

(1) an alternative *lens* on the same partition (membership-axis, in terms of the four primitive classes $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$ rather than the verdict-axis $XFR \times XFS$);

(2) an explicit bijection (Theorem 8.2) and totality theorem (Theorem 8.4) tying the two lenses together;

(3) two independent derivations of empty-cell results (topological in Theorem 5.4; prefix-quantifier logical in Proposition 8.10), neither of which is given in [50];

(4) a De Morgan duality structure on the 16-cell partition (Theorem 8.13 and Corollary 8.14) that follows from the LTL temporal dualities (Theorem 2.4);

(5) a literature-anchored worked-example catalog (table 8, section 9) with provenance for each cell.

The classification itself is, modulo the two trivial singletons, the one introduced by Peled and Havelund.

9. Worked Examples and Applied Patterns

This section illustrates each nonempty membership-axis cell with a real-world property pattern. Each example fixes a concrete operational scenario, names the LTL formula that realizes the cell, exhibits the membership tuple (s, c, ℓ, k), and traces the monitoring consequences. The format is uniform: *Pattern* (real-world specification), *Formula*, *Cell*, *Operational verdict*, *Source*.

Vocabulary convention for this section. All cell classifications $\mathsf{S}/\mathsf{C}/\mathsf{L}/\mathsf{K}$ below refer to *Layer-1 membership* on Σ^ω in the sense of Definitions 3.1 to 3.4 and 10.1. The *Stratified reading* blocks within individual examples use "Level-k" in the Manna–Pnueli sense of modal-nesting depth (compositional bracket order); the bridge from Level-k stratification to Layer-1/Layer-2 verdict-monitor semantics is established in Remark 9.13. These two stratifications are formally distinct: Layer-1/Layer-2 partitions the verdict-trace position in the monitor pipeline, while Level-k counts modal-operator nesting in the LTL formula.

We use p, q, r as atomic propositions throughout, and write a property through its LTL formalisation rather than via the underlying ω-language. The provenance column attributes the LTL witness to its earliest occurrence in the literature: PH-2019 Fig. 1 [50] provides the canonical witnesses for the four corner pair cells and the four atomic single cells of the membership-axis lattice; the two singleton pair cells $\{\mathsf{S}, \mathsf{L}\} = \{\Sigma^\omega\}$ and $\{\mathsf{C}, \mathsf{K}\} = \{\varnothing\}$ are the degenerate overlaps identified in [50, Sec. 3] (*true* is the only property in both AFR and NFR, *false* the only one in both AFS and NFS; for $\mathsf{S} \cap \mathsf{L}$ see already [2]); the present work lists them as separate cells of the 16-cell partition.

9.1. **Pair Cells: Four Corners and Two Trivial Singletons.**

Example 9.1 (Pair $\mathsf{S}\cap\mathsf{C}$: Initialisation Check)**.** *Pattern.* At startup, exactly one of `init_OK` or `init_FAIL` is reported in the first state. *Formula.* $\varphi = p$, where $p =$ `init_OK`. *Cell.* $(1, 1, 0, 0)$; clopen. *Operational verdict.* Decidable after observing one symbol: the first state's truth of p both confirms and refutes φ. *Source.* PH-2019 Fig. 1, $AFR \cap AFS$ (their witness $\bigcirc p$) [50].

Example 9.2 (Pair $\mathsf{S} \cap \mathsf{K}$: Sustained Mutual Exclusion — Morbidity class)**.** *Pattern.* "At every step, at most one process is in the critical section." *Formula.* $\varphi = G\,(\neg(p_1 \wedge p_2))$, where $p_i = \mathtt{P}_i$ in CS. *Cell.* $(1, 0, 0, 1)$; closed with empty interior; *Morbidity class. Verdict (PH-2019 $AFR \cap NFS$).* A single joint-CS observation refutes (the S face); sustained mutex is never finitely confirmable as permanent (the K face). Full operational verdict, stratified reading, and sources are given in appendix A.1 (Example A.1).

Example 9.3 (Pair $\mathsf{C} \cap \mathsf{L}$: Consensus Termination)**.** *Pattern.* "Eventually, every correct process decides a value." The consensus protocol terminates with a decision. *Formula.* $\varphi = F\,p$, where $p =$ `decided` is the predicate that every correct process has output a decision value. *Cell.* $(0, 1, 1, 0)$; open and dense. *Operational verdict.* In Peled–Havelund verdict-axis terms: *Always finitely satisfiable* (AFS, the C face) — every satisfying trace has a finite good-prefix at the first global decision event. *Never finitely refutable* (NFR, the L face) — no violating trace admits a finite bad-prefix, since consensus may still be reached in a later round. Confirmation is uniform on the satisfying side; refutation is structurally unavailable from any finite observation. *Stratified reading* (Remark 9.13). The predicate `decided` is a Level-1 evaluation on each global state. The outer F operator lifts the Level-1

verdict to the Level-2 question "will the Level-1 verdict eventually be positive?". The L classification reflects that this Level-2 question always admits an extension into satisfaction — the Level-1 verdict stream remains structurally open to future confirmation, regardless of the current observation. *Source.* PH-2019 Fig. 1, $NFR \cap AFS$ [50]; Kupferman–Vardi [38]; consensus termination as formulated in Paxos [43] and Raft [49].

Example 9.4 (Pair $\mathsf{L} \cap \mathsf{K}$: Eventually Perfect Failure Detector ($\Diamond P$))**.** *Pattern.* "Eventually, the failure detector becomes permanently correct." *Formula.* $\varphi = F\,G\,p$, $p =$ `detector_correct`. Same cell hosts $G\,F\,q$ ("`checkpoint` infinitely often") by polarity flip. *Cell.* $(0, 0, 1, 1)$; full boundary $\mathrm{bd}(\varphi) = \Sigma^\omega$ (Theorem 4.7). *Verdict (PH-2019 $NFR \cap NFS$).* Neither finitely refutable nor finitely satisfiable — the prototype of the "zero monitoring information" class of [50, Sec. 4.3] (the term is reused in [32]). Full operational verdict, stratified reading and sources in appendix A.2 (Example A.2).

Example 9.5 (Pair $\mathsf{L} \cap \mathsf{K}$ continued: Weak-Fairness Recurrence (the KL polarity))**.** *Pattern.* "A heartbeat is emitted infinitely often." *Formula.* $\psi = G\,F\,q$, $q =$ `heartbeat`. *Cell.* $(0, 0, 1, 1)$; same membership-axis cell as Example A.2, opposite Borel polarity ($\mathbf{\Pi}^0_2$ versus $\mathbf{\Sigma}^0_2$); Manna–Pnueli class 4 (Recurrence / Response). *Verdict.* Compositional bipartition residue: inner F contributes L, outer G contributes K; net $\mathsf{L} \cap \mathsf{K}$ at the KL polarity. Full operational verdict, stratified reading, sources in appendix A.3 (Example A.3).

Remark 9.6 (Bipartition of $\mathsf{L} \cap \mathsf{K}$: Persistence vs Recurrence)**.** The cell $\mathsf{L} \cap \mathsf{K}$ is *not monolithic*. Although the membership-axis tuple $(0, 0, 1, 1)$ assigns the same L and K bits, the Borel hierarchy of the Manna–Pnueli temporal hierarchy (Remark 8.6) separates the cell into two complementary polarities:

- $\mathbf{\Sigma}^0_2$ (Persistence, Manna–Pnueli class 5): $F\,G\,p$ formulae, exemplified by Example A.2 ($\Diamond P$ failure detector). In the compositional stratification of Remark 9.13, this is the LK *polarity*: outer F contributes L (eventual-reachable), inner G contributes K (sustained-morbidity). Reading: *eventually-permanent stabilisation.*
- $\mathbf{\Pi}^0_2$ (Recurrence/Response, Manna–Pnueli class 4): $G\,F\,q$ formulae, exemplified by Example A.3 (heartbeat / strong-fairness). This is the KL *polarity*: outer G contributes K (sustained-not-finitely-confirmable), inner F contributes L (locally-reachable). Reading: *permanently-recurring with disruption between recurrences.*

The two polarities are linked by De Morgan duality (Theorem 8.13): $\neg(F\,G\,p) = G\,F\,\neg p$ and $\neg(G\,F\,p) = F\,G\,\neg p$, so the negation of a Persistence claim on p is a Recurrence claim on $\neg p$, and vice versa — the two sub-cells are internally De Morgan-dual within $\mathsf{L} \cap \mathsf{K}$. Both polarities share the verdict-axis cell NFR∩NFS, but their operational signatures are structurally distinct: Persistence describes *eventually-permanent* stabilisation, Recurrence describes *permanently-recurring* events between which the system may transit through arbitrary intermediate states. The compositional stratification LK vs KL (outer-vs-inner role of L and K) is the natural refinement of the CLASP cell that captures this distinction; it is implicit in the Manna–Pnueli hierarchy classes 4 vs 5 and is foregrounded here as a compositional reading of the $\mathsf{L} \cap \mathsf{K}$ membership-axis cell.

Structural uniqueness of the LK/KL *bipartition.* The compositional bipartition is structurally unique to the $\mathsf{L} \cap \mathsf{K}$ cell within the CLASP framework. A systematic check confirms this:

- *Borel-level-1 cells* ($\mathsf{S} \cap \mathsf{C}$ Clopen, $\mathsf{S} \cap \mathsf{K}$, $\mathsf{C} \cap \mathsf{L}$): characterised by formulae of nesting depth at most one (p, $G\,p$, $F\,p$, or bounded variants). No $\mathbf{\Sigma}^0_2/\mathbf{\Pi}^0_2$ distinction applies because no nested-modal composition is required to inhabit the cell.
- *Atomic single cells* $\{\mathsf{S}\}$, $\{\mathsf{C}\}$, $\{\mathsf{L}\}$, $\{\mathsf{K}\}$: characterised by boolean combinations of Level-1 modal patterns ($p \vee G\,q$, $p \wedge F\,q$, $G\,p \vee F\,q$, $F\,p \wedge G\,q$). The modal operators here are juxtaposed by boolean connective, not nested. No outer-vs-inner ordering choice arises.
- *Triple intersections involving* L *and* K *simultaneously*. Each combination is empty by the membership constraints: $\mathsf{S} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$ (Safety plus Liveness forces Σ^ω, which is not codense); $\mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$ (open dense plus codense forces complement-dense contradiction); $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$ (clopen interior excludes codense empty interior). See Theorem 5.4. No bipartition arises because the cells themselves are vacuous.
- *Trivial singletons* $\mathsf{S} \cap \mathsf{L} = \{\Sigma^\omega\}$ and $\mathsf{C} \cap \mathsf{K} = \{\varnothing\}$: single-element cells with no structural choice.
- *Quaestio cell* (outside the membership-axis lattice): may contain Level-2 sub-formulae (the example $(p \vee G\,F\,p) \wedge q$ does) but these are boolean-combined with Level-0 atomics, collapsing the LK/KL polarity into the larger boolean structure.

The LK/KL polarity emerges precisely because $\mathsf{L} \cap \mathsf{K}$ *requires* two nested modal operators — one to supply L, one to supply K — and the two operators admit a non-trivial choice of outer-vs-inner ordering. No other cell in the present partition supports this choice non-trivially. Compositional refinement at higher Borel levels (Reactivity, $\mathbf{\Delta}^0_3$, Manna–Pnueli class 6) lies beyond the CLASP four-class framework and is not addressed here.

Example 9.7 (Pair $\mathsf{S} \cap \mathsf{L} = \{\Sigma^\omega\}$: Universal Tautology)**.** *Pattern.* "Anything is fine." *Formula.* $\varphi = \top$. *Cell.* $(1, 1, 1, 0)$ by the subset-monotone reading (Theorem 5.4; since $[\![\top]\!] = \Sigma^\omega$ is closed, open, and dense but has full interior so $\notin \mathsf{K}$); singleton $\{\Sigma^\omega\}$. *Operational verdict.* Trivially satisfied by every prefix. This cell is not enumerated by Peled–Havelund and is V3-specific.

Example 9.8 (Pair $\mathsf{C} \cap \mathsf{K} = \{\varnothing\}$: Universal Contradiction)**.** *Pattern.* "Nothing works." *Formula.* $\varphi = \bot$. *Cell.* $(1, 1, 0, 1)$; singleton $\{\varnothing\}$. *Operational verdict.* Trivially refuted by every prefix. V3-specific cell.

9.2. **Atomic Single Cells: One Class without the Others.** The four atomic single cells (s, c, ℓ, k) with exactly one bit set each have a canonical PH-2019 witness. Three of them (atomic $\{\mathsf{S}\}$, $\{\mathsf{C}\}$, $\{\mathsf{L}\}$) are summarised in table 7; the fourth, atomic $\{\mathsf{K}\}$ (Example A.4), receives full treatment as the Byzantine-consensus pattern because it serves as the canonical witness for the $\mathsf{L} \cap \mathsf{K}$ bipartition discussion of Remark 9.6.

Cell	Formula φ	Pattern	Operational verdict (PH-2019 verdict-axis)
$\{\mathsf{S}\} = (1,0,0,0)$	$p \vee G\,q$	Initialised safety: valid token, else permanent maintenance mode	$AFR \cap SFS$ [50]. Finite invalidating prefix on $\neg p \wedge \neg q$-extension; positive interior $[\,p\,]$; not dense (prefix $\langle \neg p \rangle$ requires Gq).
$\{\mathsf{C}\} = (0,1,0,0)$	$p \wedge F\,q$	Conjunctive cosafety: valid certificate at handshake AND eventual session-key establishment	$SFR \cap AFS$ [50]. Confirming prefix at first q-event conditional on p at position 0; has interior $[\,p, q\,]$.
$\{\mathsf{L}\} = (0,0,1,0)$	$G\,p \vee F\,q$	Self-stabilising recovery: legal configuration maintained or eventually re-entered (Dijkstra [21])	$NFR \cap SFS$ [50]. No bad-prefix (the Fq-branch may save any violation); good-prefix only on the Fq-branch.

Table 7: Three atomic single cells $\{\mathsf{S}\}$, $\{\mathsf{C}\}$, $\{\mathsf{L}\}$. Each formula combines a position-0 predicate with a Level-1-with-modal pattern; the atomic classification reflects which Level-1 branch gates the verdict (Remark 9.13). The fourth atomic cell $\{\mathsf{K}\}$ is treated in Example A.4.

Example 9.9 (Atomic $\{\mathsf{K}\}$: Byzantine Consensus Correctness)**.** *Pattern.* "Eventually consensus is reached, AND no Byzantine fault occurs throughout the execution." *Formula.* $\varphi = F\,p \wedge G\,q$, $p =$ `consensus_reached`, $q = \neg$`byzantine_fault`. *Cell.* $(0,0,0,1)$. *Verdict (PH-2019 $SFR \cap NFS$).* Conjunctive asymmetry: Gq confirmable refutation, Fp confirmable positive — net cell $\{\mathsf{K}\}$. Full operational verdict, stratified reading and sources in appendix A.4 (Example A.4).

Remark 9.10 (Two complementary readings of $\mathsf{S} \cap \mathsf{K}$)**.** The cell $\mathsf{S} \cap \mathsf{K}$ supports two complementary narrative readings of the same topological structure:

- *Morbidity reading (sustained positive quality).* The property asserts that the system carries a *positive quality* (operational health, mutual exclusion, linearizability, confidentiality, immune competence, calibration, integrity, fairness) *throughout.* Each step provides snapshot evidence; permanence is never finitely confirmable. Refutation is the observation of one breach. Classical instances include mutual exclusion preserved (Example A.1); linearizability [33]; sequential consistency [42]; noninterference; memory safety.
- *Pejorative-judgment reading (imputed negative quality).* Natural-language judgments such as "X is unreliable", "X is unfair", "X is incompetent", "X is hostile", "X is corrupt" carry the same topological structure when formalised as $\varphi = G\,p$ with p a negative behavioural predicate ("X malfunctions at this step", "X discriminates at this step", etc.). Each observation of p adds evidence; one observation of $\neg p$ refutes the permanence claim; no length of confirming evidence settles the judgment definitively. The K bit is here the topological warrant of *epistemic humility*: definitive pejorative judgments are structurally unsettleable in finite time.

The two readings are topologically identical and narratively dual: Morbidity is the positive-quality side of $\mathsf{S} \cap \mathsf{K}$, pejorative judgment is the negative-quality side, of the same class of properties *always finitely refutable* (AFR) on one violation but *never finitely satisfiable* (NFS) as permanent — the canonical AFR∩NFS verdict-axis cell of Peled–Havelund [50].

Remark 9.11 (The K bit within S for non-trivial properties)**.** For any closed property $P \subseteq \Sigma^\omega$ with $P \notin \{\Sigma^\omega, \varnothing\}$, $P \in \mathsf{K}$ implies $P \notin \mathsf{C}$: a non-empty open set has non-empty interior. Hence $\mathsf{S} \cap \mathsf{K} \subseteq \mathsf{S} \setminus \mathsf{C}$ for non-trivial properties. The inclusion is strict: $p \vee Gq$ is closed and not open, yet $\{p\}$ is a good prefix, so $p \vee Gq \notin \mathsf{K}$; it is the atomic $\{\mathsf{S}\}$ witness of table 7. The K bit is therefore *not* derivable from "S and not Clopen": within $\mathsf{S} \setminus \mathsf{C}$ it separates $\mathsf{S} \cap \mathsf{K}$ from $\{\mathsf{S}\}$.

The explicit K-tag in the four-bit tuple is retained for three reasons. First, K is *not* redundant, neither within S (as just shown) nor in other cells: in $\mathsf{L} \cap \mathsf{K}$ (densely extendable both into satisfaction and into violation), $\{\mathsf{K}\}$ atomic, and the triple intersections involving K, the bit carries information not derivable from $\{\mathsf{S}, \mathsf{C}, \mathsf{L}\}$. Second, dropping K collapses the 16-cell partition into an 8-cell one and breaks the symmetry of the four-family generator. Third, the bit is pedagogically explicit: it names the *no-finite-confirmation* role that the cell carries, which is not visible to the reader without the K tag.

Remark 9.12 (Framework-relative reading of the K class)**.** The diagnosis of a property as K-class is relative to the *trace-only observation framework* of runtime verification (Pnueli–Zaks, Peled–Havelund, Bauer–Leucker–Schallhart): from the prefix of an actual execution, no finite-length good-prefix exists, hence the property is never finitely confirmable. Two complementary epistemic positions soften this verdict without contradicting it:

- *Model checking* [38]. Given a finite-state model M of the system, K-class safety properties such as $G\,p$ are routinely *decidable* via exhaustive reachability analysis on M. Mutual exclusion (Example A.1) is the textbook example: not finitely confirmable from a trace, but confirmed by construction on any finite Kripke structure that satisfies the protocol. The confirmation comes from the *model*, not from the trace.
- *Refinement-based confirmation* (sketched at framework level in section 10.7; the formal convergence theorem is left to a forthcoming dedicated treatment). When the verifier has access to an external structural-zero guarantee — Miller-GRH bound on primality, AKS polynomial congruence, Pratt-tree certificate, parametric monitor adaptation, or simply the finiteness of a Kripke structure — the Hansel–Newton refinement tuple (D, A_0, U, E) converts the otherwise non-confirmable K-class property into a finite-time certification.

The K classification therefore answers the question "what does the trace alone tell us?", not "what can be known about the property in principle". Both model checking and refinement-based confirmation are instances of *external supporting structure* that lifts the K verdict in a unifying language in which runtime monitoring sits at one endpoint (no supporting structure, K never lifted) and model checking sits at the other (closed-world model as full supporting structure, K always lifted on finite models), with a spectrum of conditional and parametric instances in between.

Remark 9.13 (Stratified compositional reading of the CLASP classes)**.** The four-class partition admits a stratified reading aligned with the Manna–Pnueli temporal hierarchy [48] (see also Remark 8.6). The stratification is by *modal-nesting depth* (Level-k); it is formally

distinct from the *verdict-trace position* partition Layer-1/Layer-2 of Definition 10.1, although the two are bridged at the end of this remark.

Level 1 — direct trace evaluation. Safety (S) and Cosafety (C) classify properties evaluated by *current-step monitors* on individual trace states. In a compositional setting, the trace states are produced by another process or sub-routine; the Level-1 monitor reports a verdict on each such state as it is observed. A Safety monitor for $G\,p$ flags violation on the first $\neg p$-event; a Cosafety monitor for $F\,p$ flags confirmation on the first p-event; both are local-snapshot evaluators.

Level 2 — temporal modalities on Level-1 outputs. Liveness (L) and Coliveness (K) classify properties about the *long-run behaviour of the Level-1 monitor's verdict stream*:

- $F\,G\,p$ (Persistence, Manna–Pnueli class 5): the Level-1 monitor's verdict on p *eventually stabilises* on *satisfied* — a Liveness claim about the verdict stream of the p-monitor.
- $G\,F\,p$ (Recurrence, Manna–Pnueli class 4): the verdict on p *revisits satisfaction infinitely often*, with the structural warrant that subsequent observations may always disrupt the revisit — in $\mathsf{L} \cap \mathsf{K}$.
- $G\,F\,\neg p$: the verdict on p *never stabilises* — the Level-1 judgement is perpetually disrupted by subsequent intervention — in K.

The Manna–Pnueli classes 4–5 are thus precisely the *liveness/coliveness lifts* of classes 1–2: temporal modalities applied to the Level-1 verdict stream.

Operational reading. Operationally, L and K formalise the question "does the Level-1 verdict stabilise?" on the verdict stream produced by current-state monitors. In multi-process systems where sub-routines $R_1 \to R_2 \to R_3$ communicate, R_2 may run Level-1 monitors on states produced by R_1, while R_3 may run Level-2 monitors on the long-run behaviour of R_2's verdicts. The stratification matches the Layer-1 / Layer-2 disambiguation of section 10.1 and the universal attribution theorem of Theorem 10.23: every Level-1 monitor with $|V| \geq 2$ canonically supplies a verdict-stream over which Level-2 LTL formulae of the $\mathsf{L} \cap \mathsf{K}$ class can be naturally evaluated.

This stratification is the foundation of the refinement-confirmation framework of section 10.7, which operates structurally at Level 2: refinement systems consume Level-1 verdict streams as their input domain and certify pullbacks of $\mathsf{L} \cap \mathsf{K}$ verdict-stream properties via the (D, A_0, U, E) tuple.

9.3. **The Quaestio Cell: Outside the Membership-Axis Lattice.**

Example 9.14 (Quaestio: Neither in S, C, L, nor K)**.** *Pattern.* "At the initial state q holds, and the property $p \vee \Box\Diamond p$ also holds." *Formula.* $\varphi = (p \vee GFp) \wedge q$. *Cell.* $(0, 0, 0, 0)$, i.e. $\Sigma^\omega \setminus (\mathsf{S} \cup \mathsf{C} \cup \mathsf{L} \cup \mathsf{K})$ (Remark 8.3). *Operational verdict.* PH-2019 classify its next-time variant $((p \vee \Box\Diamond p) \wedge \bigcirc q)$ as *Quaestio* and give it as a non-monitorable example [50, Sec. 4.3]; the same holds for this form. The quaestio cell as a whole is mixed: [50, Sec. 4.3] also gives the monitorable member $((\Diamond r \vee \Box\Diamond p) \wedge \bigcirc q)$. Verdicts depend on q at position 0 and on infinitary p behavior simultaneously. *Source.* PH-2019 Fig. 1, $SFR \cap SFS$ [50]; see also Henzinger–Saraç [32, Ex. 8] for an applied non-monitorable property with similar structure.

9.4. **Empty Cells and Their Operational Meaning.** The three empty triple cells and the empty quadruple cell of table 1 admit no LTL witness by Proposition 8.10 (3)–(5). The corresponding operational intuition is direct:

- $\mathsf{S} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$: any closed and dense property is the whole universe Σ^ω, whose interior is $\Sigma^\omega \neq \varnothing$, so no such property belongs to K. In monitoring terms: a property that finitely refutes every wrong trace *and* keeps every prefix open to satisfaction *and* keeps every prefix open to violation cannot exist—the three obligations are jointly inconsistent except for the trivial Σ^ω which fails the last obligation.
- $\mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$: symmetric, via the empty property $\varnothing$.
- $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L} \cap \mathsf{K} = \varnothing$: corollary of the previous two; included for completeness as the apex of the membership-axis lattice.

These three obstructions are not artifacts of the language LTL: they are obstructions at the level of the four primitive prefix-quantifier definitions, and therefore persist under any expressive extension that retains those definitions.

9.5. **Applied-Pattern Summary.** Table 8 consolidates the worked examples of sections 9.1 to 9.3 with their cell, verdict character, and source attribution.

Cell	LTL pattern	Operational verdict	Source
$\{\mathsf{S}, \mathsf{C}\}$	p	1-prefix decidable	PH-2019 Fig. 1
$\{\mathsf{S}, \mathsf{K}\}$	Gp	refute only	PH-2019; classical
$\{\mathsf{C}, \mathsf{L}\}$	Fp	confirm only	PH-2019; KV-2001
$\{\mathsf{L}, \mathsf{K}\}$	GFp	inconclusive (zero info)	PH-2019; D–L 2014
$\{\mathsf{S}, \mathsf{L}\} = \{\Sigma^\omega\}$	$\top$	trivially confirmed	AS85; PH-2019 Sec. 3
$\{\mathsf{C}, \mathsf{K}\} = \{\varnothing\}$	$\bot$	trivially refuted	PH-2019 Sec. 3
$\{\mathsf{S}\}$	$p \vee Gq$	refute conditionally	PH-2019 Fig. 1
$\{\mathsf{C}\}$	$p \wedge Fq$	confirm conditionally	PH-2019 Fig. 1
$\{\mathsf{L}\}$	$Gp \vee Fq$	no negative verdict	PH-2019 Fig. 1
$\{\mathsf{K}\}$	$Fp \wedge Gq$	no positive verdict	PH-2019 Fig. 1
$\{\mathsf{S}, \mathsf{C}, \mathsf{L}\} = \{\Sigma^\omega\}$	$\top$	trivial	AS85; PH-2019 Sec. 3
$\{\mathsf{S}, \mathsf{C}, \mathsf{K}\} = \{\varnothing\}$	$\bot$	trivial	PH-2019 Sec. 3
$\{\mathsf{S}, \mathsf{L}, \mathsf{K}\}$	—	empty	V3 Prop 8.10(3)
$\{\mathsf{C}, \mathsf{L}, \mathsf{K}\}$	—	empty	V3 Prop 8.10(4)
$\{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$	—	empty	V3 Prop 8.10(5)
Quaestio (outside)	$(p \vee GFp) \wedge q$	non-monitorable Quaestio	PH-2019 Fig. 1

Table 8: Applied-pattern summary: each membership-axis cell with its canonical LTL pattern, operational verdict character, and earliest literature attribution. PH-2019 = [50]; KV-2001 = [38]; D–L 2014 = [19]; V3 = the present work.

10. Inherited monitor systems and the verdict-alphabet ladder

The verdict-monitor constructions of this section sit on a small pipeline diagram with four interacting parts:

$$\underbrace{\text{function}}_{\sigma\in\Sigma^\omega} \xrightarrow{\text{production}} \underbrace{\text{monitor } M}_{\text{algebra } A \text{ on alphabet } V} \xrightarrow{\text{production}} \underbrace{\text{trace } E_M(\sigma)}_{\in V^\omega} \xleftarrow[\text{evaluation}]{} \underbrace{\text{LTL property } \varphi}_{\text{on } V^\omega}.$$

Reading left-to-right is *production*: the function (input trace) is observed by the monitor M, whose internal algebra A acts on each finite prefix to emit a verdict in V; iterating produces the verdict trace $E_M(\sigma) \in V^\omega$. Reading right-to-left is *evaluation*: an LTL formula φ on V^ω is tested against the produced trace. The two readings classify properties on *different* carriers: the monitor classifies properties of the function via its native CLASP cell at Layer 1; the formula φ classifies properties of the verdict trace itself at Layer 2 (Definition 10.1). This duality is the central structural observation around which the present section is organised.

The four primary classes of the intersection algebra — Safety (S), Cosafety (C), Liveness (L), and Coliveness (K), together with their non-trivial mixed cell $\mathsf{L} \cap \mathsf{K}$ — carry the analysis throughout; the twelve non-empty sub-cells of the full partition are referenced only when finer structure is required, and higher intersections together with the Quaestio cell are mentioned only as boundary cases. A *parametric* closure of the diagram — in which the monitor's algebra A is adapted online in response to a target φ, making the central box itself a feedback-controlled object — opens a direction beyond the present scope; see section 10.7 for the present paper's introduction to that thread.

The 16-cell membership partition of section 8, together with the monitorability gradient of Remark 5.8, classifies properties by *which* verdict information a runtime monitor can extract from a prefix (Lemma 5.9; whether a verdict stays reachable after every prefix, i.e. monitorability, is not fixed by the cell). The pipeline diagram lets us turn the perspective around and ask: *which monitor systems incarnate each cell of the central box.* A monitor system is defined here as the pair (verdict alphabet, semantics) emitted at every observation step. Different alphabets carry different semantic resolution, and each resolution determines which cells of the intersection algebra a monitor of that system can decide.

This perspective is operational rather than topological. The topological view [19, 58] and the operational-guide reading of monitorability of Aceto, Achilleos, Francalanza, Ingólfsdóttir and Lehtinen [1] characterise *which* properties are monitorable; the present section characterises *with what alphabet* a monitor reports its verdict, and how that alphabet's cardinality determines the cells covered. Several lines of prior work meet here. First, the catalogue of multi-valued LTL semantics for runtime verification [6] proposes specific finite truth domains ($n = 2, 3, 4$) and compares them on syntactic maxims, without indexing the comparison by cardinality nor mapping it to the CLASP lattice. Second, the multi-valued model-checking line of Li, Droste and Lei [46] studies linear-time properties over multi-valued systems but focuses on model-checking complexity rather than ladder-by-cardinality structure. Third, the algebraic framework of Jakšić et al. [34] unifies qualitative and quantitative semantics by varying the underlying semiring on a fixed automaton, without enumerating verdict cardinalities or relating them to monitorability cells. Fourth, the robust-LTL line [59] fixes a richer alphabet (the five-valued lattice $\{0000, 0001, 0011, 0111, 1111\}$) for a single LTL syntax, again without ladder-by-cardinality treatment. The present section weaves the four lines into a single progression indexed by verdict-alphabet cardinality $n \in \{1, 2, 3, 4, \infty\}$ (degenerate baseline at n=1, section 11.1; qualitative rungs $n \in \{2, 3, 4\}$; quantitative limit

at $n{=}\infty$), with each rung mapped explicitly to the cells of the intersection algebra established in section 8.

A subtle but essential distinction precedes the progression.

10.1. **Single-step alphabet vs verdict-trace pattern.** The expression "a monitor with n-valued verdicts" admits two readings that the literature does not always disentangle [6, 7, 34]. At a single observation step the monitor reports exactly one element of a finite domain; over time the sequence of reports is itself a word, and that word can be examined for any property one cares to specify. Conflating the two readings generates the apparent paradox that even a binary monitor seems to "express" Liveness/Coliveness patterns in its output stream.

Definition 10.1 (Layer 1 and Layer 2 of a verdict monitor)**.** Fix a verdict alphabet $\mathbb{V}$. A *verdict monitor* $M : \Sigma^* \to \mathbb{V}$ assigns to every finite trace prefix a verdict in $\mathbb{V}$. Two structures live on top of M.

- *Layer 1* (per-step alphabet): the function M itself. Its codomain $\mathbb{V}$ encodes the *single-step semantic resolution* of the monitor. The cells of section 8 captured by M at Layer 1 are exactly the cells whose membership question is resolved by the semantics that $\mathbb{V}$ can express.
- *Layer 2* (verdict-trace as ω-word): the sequence $M(u_0)M(u_1)M(u_2)\cdots \in \mathbb{V}^\omega$ obtained by feeding M successive prefixes $u_n = a_0 \cdots a_n$ of an input trace σ. This ω-word over $\mathbb{V}$ is a new $\mathbb{V}$-trace and admits its own classification in the intersection algebra of $\mathbb{V}^\omega$.

The two layers are formally distinct: Layer 1 captures what the monitor can *say* at one step; Layer 2 captures the pattern its sayings *form* over time. The following remark fixes the distinction.

Remark 10.2 (The bit-stream pattern is a Layer 2 artefact)**.** Consider the binary monitor $M : \Sigma^* \to \{0,1\}$ that emits 1 exactly when the input prefix satisfies a bounded $\mathrm{LTL}^{\leq k}$ formula φ. At Layer 1, M decides a Clopen property of σ in the sense of Theorem 4.6: the set of accepting traces is both closed and open in the Cantor topology of Σ^ω. At Layer 2, the verdict-trace $M(u_0)M(u_1)\cdots \in \{0,1\}^\omega$ *itself* can exhibit any LTL pattern over the alphabet $\{0,1\}$: "eventually all 1s" ($\mathbf{FG}(b=1)$, in $\mathsf{L}\cap\mathsf{K}$ over $\mathbb{V}_2^\omega$), "alternating" ($\mathbf{GF}(b=1)\wedge\mathbf{GF}(b=0)$, also in $\mathsf{L}\cap\mathsf{K}$ over $\mathbb{V}_2^\omega$), and so on. The Layer 2 phenomenon does not promote M to a Liveness/Coliveness monitor of the input σ: M still decides only a Clopen property of σ at Layer 1. The Layer 2 pattern is a property of M's output considered as a fresh ω-word over $\mathbb{V}$, not a property of σ.

Remark 10.3 (The ladder is indexed by the verdict-alphabet cardinality)**.** The four *non-trivial* rungs of sections 10.2 to 10.5 are organised by a single parameter: the cardinality $|\mathbb{V}|$ of the verdict alphabet, taking the discrete values $|\mathbb{V}| \in \{2,3,4\}$ and the limit $|\mathbb{V}| \to \infty$. The degenerate $|\mathbb{V}|{=}1$ baseline (trivial monitors emitting a constant verdict) is treated separately in section 11.1 and does not appear in the coverage matrix because it resolves no new cell beyond the trivial singletons of Theorem 5.4. The Layer-1 effect of climbing the ladder is recorded in the coverage matrix table 9. The Layer-2 effect is *independent* of the Layer-1 effect: the verdict trace $\mathbf{v} \in \mathbb{V}^\omega$ is a fresh ω-word over $\mathbb{V}$ and admits its own classification in the intersection algebra of $\mathcal{P}(\mathbb{V}^\omega)$. The per-rung enumeration of which signal-kinematic modalities each verdict-alphabet cardinality can host is left to

future work. We confine the present treatment to a single canonical Layer-2 example at the focus rung $|\mathbb{V}| = 3$ (Example 10.4); the threshold worth noting here concerns the two schemas of the atomic witnesses: the disjunctive-L schema $\mathbf{G}\,p \vee \mathbf{F}\,q$ of table 7, instantiated on single values $v{=}a$, $v{=}b$, and the conjunctive-K schema $\mathbf{F}\,p \wedge \mathbf{G}\,q$ of Example A.4, instantiated on $v{=}a$ and $v{\neq}b$ (as in Example 10.4), need $a \neq b$ and a third value, hence $|\mathbb{V}| \geq 3$ (over $\{0,1\}$ they degenerate, e.g. $\mathbf{G}(b{=}1) \vee \mathbf{F}(b{=}0)$ is valid). Atomic $\{\mathsf{L}\}$ and $\{\mathsf{K}\}$ Layer-2 properties as such already exist over a binary alphabet: $\mathbf{FG}(b{=}1) \vee (b_0{=}0)$ is in $\{\mathsf{L}\}$ and its complement $\mathbf{GF}(b{=}0) \wedge (b_0{=}1)$ in $\{\mathsf{K}\}$, where b_0 is the first verdict.

Example 10.4 (Layer-2 L and K patterns at the focus rung $|\mathbb{V}| = 3$)**.** Read the verdict-trace $\mathbf{v} = v_0v_1v_2\cdots \in \mathbb{V}_3^\omega$ as a fresh ω-word over $\mathbb{V}_3 = \{-1, 0, +1\}$. Instantiating the schemas of table 7 (atomic $\{\mathsf{L}\}$ row) and Example A.4 on the verdict-value predicates $\{v = +1, v = 0, v = -1\}$ gives canonical Layer-2 witnesses for atomic $\{\mathsf{L}\}$ and atomic $\{\mathsf{K}\}$:

$$\mathsf{L}\text{-witness:}\quad \mathbf{G}(v{=}{+}1) \vee \mathbf{F}(v{=}{-}1) \quad (\text{atomic } \{\mathsf{L}\}),$$
$$\mathsf{K}\text{-witness:}\quad \mathbf{F}(v{=}{+}1) \wedge \mathbf{G}(v{\neq}{-}1) \quad (\text{atomic } \{\mathsf{K}\}).$$

For the K-witness $\Phi = \mathbf{F}(v = +1) \wedge \mathbf{G}(v \neq -1)$: $\Phi \notin \mathsf{S}$ because a bad trace that never reaches $v = +1$ and never reaches $v = -1$ (e.g. all $v = 0$) has no finite bad prefix; $\Phi \notin \mathsf{L}$ because any prefix containing a $v = -1$ symbol permanently refutes $\mathbf{G}(v \neq -1)$ and admits no satisfying extension; $\Phi \notin \mathsf{C}$ because the conjunct $\mathbf{G}(v \neq -1)$ is never finitely confirmable; $\Phi \in \mathsf{K}$ because every prefix can be extended by an infinite tail of $v = -1$, violating the $\mathbf{G}$ conjunct.

Layer-1 invariance. Neither witness promotes the Layer-1 cell of the monitor on the input: the Layer-1 cell is fixed by the rung's resolution capacity (Propositions 10.6, 10.10, 10.15 and 10.17). The Layer-2 CLASP classification is a fresh instance of the same intersection algebra applied to a different alphabet — the alphabet $\mathbb{V}^\omega$ of verdict traces.

Per-rung extension. Analogous Layer-2 witnesses on $\mathbb{V}_4^\omega$ and $\mathbb{V}_\infty^\omega$ follow the same schemas with p, q instantiated as distinct value-predicates of the larger alphabet.

The remainder of this section reads "cell of M" as Layer 1 cell, unless explicitly noted. Layer 2 returns in section 10.6 as the mechanism by which monitor *composition* promotes Layer 1 cells to higher cells of the intersection algebra.

10.2. **The binary monitor ($n = 2$).** The simplest verdict alphabet is $\mathbb{V}_2 := \{0, 1\}$, the Boolean lattice $\mathbb{B}_2$ of [6]. The alphabet has no value that signals indecision: once the monitor commits, its value is an absorbing fixed point, and values emitted before commitment are provisional.

Definition 10.5 (Binary verdict monitor)**.** A *binary verdict monitor* for a property $P \subseteq \Sigma^\omega$ is a map $M_2^P : \Sigma^* \to \{0, 1\}$ together with a set $\mathrm{Com} \subseteq \Sigma^*$ of *committing* prefixes such that (i) every $u \in \mathrm{Com}$ is a good or a bad prefix of P (Lemma 5.9) and $M_2^P(u') = 1$ (resp. 0) for every $u' \sqsupseteq u$ if u is good (resp. bad): the commitment is sound and irrevocable; and (ii) every $\sigma \in \Sigma^\omega$ has a prefix in Com. Values of M_2^P outside the extensions of committing prefixes are provisional and unconstrained.

Proposition 10.6 (Layer-1 cell of a binary monitor)**.** *A binary verdict monitor M_2^P exists for P at Layer 1 if and only if $P \in \mathsf{S} \cap \mathsf{C}$, i.e., P is clopen in the Cantor topology on Σ^ω.*

Proof. If M_2^P exists, every trace σ has a committing prefix, which is good or bad; a good prefix of σ forces $\sigma \in P$ and a bad one $\sigma \notin P$. Hence every $\sigma \in P$ has a good prefix and every $\sigma \notin P$ a bad one, i.e. $P \in \mathsf{C} \cap \mathsf{S}$ (Lemma 5.9 (2)). Conversely, for $P \in \mathsf{S} \cap \mathsf{C}$ let Com be the set of good or bad prefixes of P and let $M_2^P(u) = 1$ if u is good and 0 otherwise; by Lemma 5.9 (2) every trace has a prefix in Com, and extensions of good (bad) prefixes are good (bad). □

Why commitment is required. Without clause (i) the correspondence fails: $M(u) = 1$ iff p occurs in u converges on every trace to the membership value of $\mathbf{F}p$, which is open but not closed; its value 0 is never a committing verdict, since every prefix extends to a satisfaction.

Remark 10.7 (Verdict-role disambiguation at the n=2 rung)**.** The mapping "n=2 ⇒ clopen" of Proposition 10.6 presupposes the *symmetric-conclusive* configuration $\mathbb{V}_2 = \{\top, \bot\}$ in which both verdicts are absorbing. Following the verdict-axis of Peled–Havelund [50] (cf. section 8, Remark 3.6), a 2-valued monitor admits three inequivalent configurations:

- $\{\top, \bot\}$ both conclusive ⇒ AFR∩AFS = *clopen* $\mathsf{S} \cap \mathsf{C}$.
- $\{\top, \varepsilon\}$ positive-only conclusive with "no-information" second value ⇒ AFS∩NFR = C*-region (cosafety-only)*: a positive verdict is reachable in finite time, a negative verdict never is.
- $\{\bot, \varepsilon\}$ negative-only conclusive ⇒ AFR∩NFS = S*-region (safety-only)*: a negative verdict is reachable in finite time, a positive verdict never is.

Cardinality $|V| = 2$ alone is therefore not sufficient to fix the resolvable property class; the *verdict role* (which value, if any, is absorbing on satisfaction versus violation) matters and is what the Peled–Havelund verdict-axis tracks. The "ladder rung n=2 ⇒ clopen" identification in the remainder of this section refers exclusively to the symmetric-conclusive configuration; the two asymmetric configurations sit at the S-region and C-region rungs in the verdict-axis classification and, we conjecture, are operationally subsumed by the LTL_3 rung (section 10.3) when the second value ε is identified with the inconclusive ? of LTL_3.

The binary alphabet is therefore the operational incarnation of Clopen. It corresponds to the singleton cell $\mathsf{S} \cap \mathsf{C}$ of the intersection algebra and to the bounded-LTL fragment $\mathrm{LTL}^{\leq k}$ (Kupferman–Vardi safety closure with a finite bound).

Example 10.8 (Canonical $n = 2$ formula)**.** The bounded recurrence $\varphi \equiv (\mathbf{F}^{\leq 5} p) \wedge (\mathbf{G}^{\leq 5} \neg q)$ over $\Sigma = 2^{\{p,q\}}$ is clopen and admits a binary monitor that emits 1 iff the first five samples contain at least one p and no q. The verdict is decided no later than the fifth sample and is absorbing thereafter.

10.3. **The Kleene ternary monitor ($n = 3$).** Adding a third symbol *below* the two Boolean fixed points allows the alphabet to express "not yet decided" as a distinct, explicit value rather than an absence. The resulting alphabet is the three-valued lattice $\mathbb{V}_3 := \{-1, 0, +1\}$, with $-1 \sqsubseteq 0 \sqsubseteq +1$ in the information order, isomorphic to the truth domain $\mathbb{B}_3 = \{\bot, ?, \top\}$ of LTL_3 [6, 7]. The intended reading is: $+1$ = positive verdict, -1 = negative verdict, 0 = undetermined. Two semantic regimes are admissible.

Definition 10.9 (Strong and weak Kleene verdict regime)**.** Let $M_3^P : \Sigma^* \to \mathbb{V}_3$.

- *Strong (absorbing) regime*: once $M_3^P(u) = +1$ (resp. -1), every extension $u' \in u \cdot \Sigma^*$ satisfies $M_3^P(u') = +1$ (resp. -1). The values ± 1 are fixed points; 0 may transition to either. This is the verdict semantics of LTL_3 [6].
- *Weak (revocable) regime*: any value may transition to any other at the next step. ± 1 encode current-state evaluation, not commitment.

The strong regime is the one that the CLASP partition resolves naturally; the weak regime addresses current-state predicates and sits closer to the rLTL [59] reading. The proposition below characterises the cells covered by the strong regime.

Proposition 10.10 (Layer-1 cell footprint of strong Kleene)**.** *Let M_3^P be the LTL$_3$ monitor of P: $M_3^P(u) = +1$ if u is a good prefix of P, -1 if u is a bad prefix, and 0 otherwise* [7, Remark 3.2]*. It realises the strong regime of Definition 10.9 for every P, and:*

(1) (completeness) *every $\sigma \notin P$ receives -1 on some prefix iff $P \in \mathsf{S}$, and every $\sigma \in P$ receives $+1$ on some prefix iff $P \in \mathsf{C}$; so M_3^P is complete on at least one verdict side iff $P \in \mathsf{S} \cup \mathsf{C}$, and on both iff $P \in \mathsf{S} \cap \mathsf{C}$ (Clopen, the retract to $n = 2$);*

(2) (reachability) -1 *is reachable on Σ^* iff $P \notin \mathsf{L}$, and $+1$ iff $P \notin \mathsf{K}$; so only -1 is reachable iff $P \in \mathsf{K} \setminus \mathsf{L}$ (e.g. $\mathbf{G}p$), only $+1$ iff $P \in \mathsf{L} \setminus \mathsf{K}$ (e.g. $\mathbf{F}p$), both iff $P \notin \mathsf{L} \cup \mathsf{K}$, and neither iff $P \in \mathsf{L} \cap \mathsf{K}$;*

(3) (monitorability) *M_3^P has no ugly prefix, i.e. P is monitorable in the sense of Pnueli–Zaks* [52]*, for every $P \in \mathsf{S} \cup \mathsf{C}$ and for no $P \in \mathsf{L} \cap \mathsf{K}$; the inclusion $\mathsf{S} \cup \mathsf{C} \subsetneq \mathcal{M}(\Sigma^\omega)$ is strict* [7, Lemma 3.8]*, and on the L-only, K-only and quaestio cells monitorability is not determined by the cell (Lemma 5.9).*

Proof. M_3^P is well defined for every P, and it is absorbing because extensions of good (bad) prefixes are good (bad). (1) is Lemma 5.9 (2) read on the prefixes of σ; (2) is Lemma 5.9 (1), the four cases being the four values of (ℓ, k); (3) is Lemma 5.9 (3). □

Remark 10.11 (Connection to the monitorability gradient)**.** Proposition 10.10 gives the Layer-1 incarnation of the monitorability gradient of Remark 5.8: the gradient ranks cells by "how much" verdict information a prefix determines; the $n = 3$ alphabet captures *which* cells are fully or partially Layer-1 resolved. The two views are dual: gradient is property-side, ladder is monitor-side.

Example 10.12 (Cell-discriminating $n = 3$ formulae)**.** For $P_1 = [\![\mathbf{G}p]\!]$ (in $\mathsf{S} \setminus \mathsf{C}$, complete on the violation side by Proposition 10.10 (1); the formula itself sits in the membership cell $\mathsf{S} \cap \mathsf{K}$, Example A.1) the strong Kleene monitor never emits $+1$ in finite time and emits -1 exactly when $\neg p$ is seen. For $P_2 = [\![\mathbf{F}p]\!]$ (in $\mathsf{C} \setminus \mathsf{S}$; the formula sits in $\mathsf{C} \cap \mathsf{L}$, Example 9.3) it never emits -1 in finite time and emits $+1$ on first witness of p. For $P_3 = \left[\!\left[\mathbf{F}^{\leq k} p \wedge \mathbf{G}^{\leq k} \neg q\right]\!\right]$ (Clopen, $\mathsf{S} \cap \mathsf{C}$, Example 9.1) both verdicts are reachable. For $P_4 = [\![\mathbf{GF}p]\!]$ ($\mathsf{L} \cap \mathsf{K}$, Example A.2) the monitor emits 0 on every prefix: $\mathsf{L} \cap \mathsf{K}$ is exactly the cell on which M_3^P never emits a definitive verdict (Proposition 10.10 (2)), and it motivates the revocable verdicts of $n = 4$ in section 10.4.

Figure 6 consolidates the LTL_3 monitor structure cell-by-cell: each of the nine non-trivial cells of the 16-cell partition is rendered as a 3-state automaton $\{?, \top, \bot\}$ with the canonical default expression from table 2 as formula and the transitions labelled by their triggering observation pattern. The gallery makes visible which sinks are reachable per cell: the $\mathsf{L} \cap \mathsf{K}$ cell emits ? forever (no sink reachable, matching $n_{\text{eff}} = 1$ of Proposition 7.5); by

Proposition 10.10 (2) $\bot$ is reachable exactly when $\ell = 0$ and $\top$ exactly when $k = 0$, so $\mathsf{S} \cap \mathsf{K}$ and K-only reach only $\bot$, $\mathsf{C} \cap \mathsf{L}$ and L-only reach only $\top$, the S-only, C-only and quaestio cells reach both, and the clopen p_0 resolves at the first step.

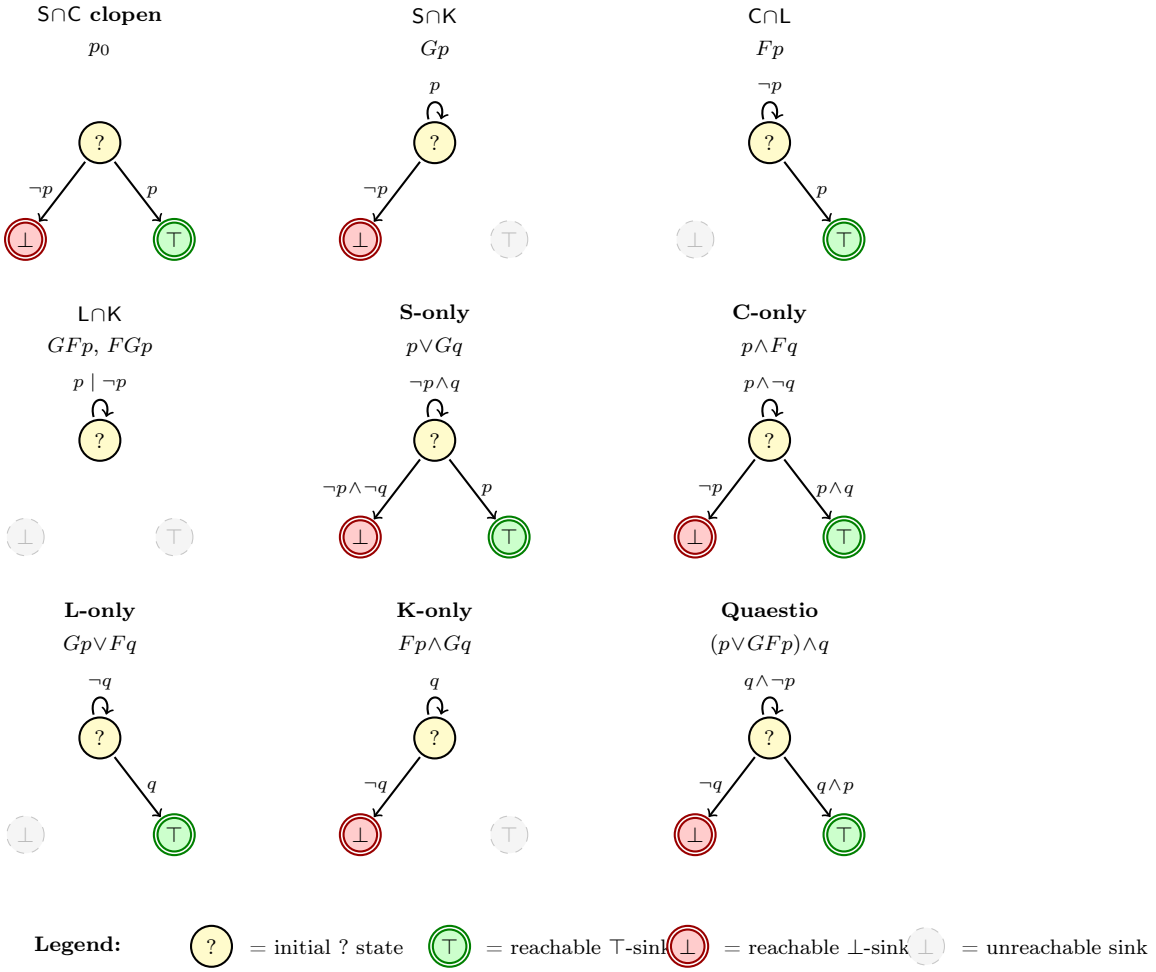


Figure 6: **LTL$_3$ verdict-monitor gallery.** Nine LTL$_3$ monitors, one per non-trivial cell of the 16-cell exclusive partition (Theorem 8.4). Each monitor has the canonical $\{?, \top, \bot\}$ state set; the initial state is ? and $\top$, $\bot$ are absorbing. Transition labels are the propositional triggers (representative; the precise labels depend on whether the position is $t = 0$ or $t \geq 1$ but the qualitative behaviour is preserved). Unreachable sinks are rendered in dashed grey: their absence is the diagnostic feature of the corresponding cell. The $\mathsf{L} \cap \mathsf{K}$ panel (no sink reachable) is the canonical witness of effective-$n = 1$ under LTL$_3$ (Proposition 7.5). Cell colours of the sinks track those of figure 3.

10.4. **The RV-LTL quaternary monitor ($n = 4$).** The Kleene ternary alphabet of section 10.3 has *one* positive value and *one* negative value, both interpreted as permanent

under the strong regime; the 0 symbol covers everything else. Properties such as $\mathbf{GF}p$ ($\mathsf{L}\cap\mathsf{K}$) therefore receive the verdict 0 on every finite prefix — the alphabet has no vocabulary for the intermediate situation in which the current prefix *looks satisfying* (or *looks violating*) but the verdict cannot be made permanent.

RV-LTL of [6] resolves this by *splitting* the positive and negative levels into permanent and current variants, yielding the four-valued alphabet $\mathbb{V}_4 := \{\top_s, \top_w, \bot_w, \bot_s\}$, ordered as a de Morgan lattice with $\bot_s \sqsubseteq \bot_w \sqsubseteq \top_w \sqsubseteq \top_s$. The subscripts read *strong* (absorbing, permanent) and *weak* (revocable, current).

Definition 10.13 (RV-LTL quaternary verdict monitor)**.** A *quaternary verdict monitor* for P is a map $M_4^P : \Sigma^* \to \mathbb{V}_4$ such that, for every $u \in \Sigma^*$,

$$M_4^P(u) = \begin{cases} \top_s & \text{if every } w \in \Sigma^\omega \text{ has } uw \in P, \\ \bot_s & \text{if every } w \in \Sigma^\omega \text{ has } uw \notin P, \\ \top_w & \text{else if the strong-LTL truth of } P \text{ on } u \text{ is } \top, \\ \bot_w & \text{otherwise.} \end{cases}$$

The values $\top_s, \bot_s$ are absorbing along Σ^*; $\top_w, \bot_w$ are revocable. The weak cases are defined for $u \neq \varepsilon$, since the finite-trace semantics is undefined on the empty word; for $P \notin \{\Sigma^\omega, \varnothing\}$ we set $M_4^P(\varepsilon) = \top_w$, as in the RV-LTL monitor of [6, Thm. 3]. Here P is presented by an LTL formula φ and the strong-LTL truth is the finite-trace (FLTL) semantics of φ, as in RV-LTL [6]. The values $\top_s, \bot_s$ depend only on the property P, whereas $\top_w, \bot_w$ depend on the formula: $\mathbf{GF}p$ and $\mathbf{GFX}p$ denote the same property, yet on the prefix $\{p\}$ the former yields $\top_w$ and the latter $\bot_w$, since a strong next beyond the prefix evaluates to false.

Proposition 10.14 (Differential expressiveness of $n = 4$ versus $n = 3$)**.** *At Layer 1, M_4^P strictly refines M_3^P* [6, Sec. 5.1]*: every Kleene verdict in $\mathbb{V}_3$ is recoverable from M_4^P by collapsing $\top_s \mapsto +1$, $\bot_s \mapsto -1$, and $\top_w, \bot_w \mapsto 0$. The converse does not hold: the distinction between $\top_w$ and $\bot_w$ cannot be recovered from M_3^P alone, because both project to 0.*

Proof. The projection $\pi : \mathbb{V}_4 \to \mathbb{V}_3$ defined above is well-defined: it sends the two absorbing values to ± 1 and both revocable values to 0. The composition $\pi \circ M_4^P$ satisfies the strong-Kleene fixed-point condition of Definition 10.9: ± 1 are reached exactly where $\top_s, \bot_s$ are reached, and remain absorbing by inheritance. Hence $\pi \circ M_4^P = M_3^P$.

For the converse, take $P = [\![\mathbf{GF}p]\!]$ presented by the formula $\mathbf{GF}p$. On any prefix u, $M_3^P(u) = 0$ because neither permanent verdict is determined, while on non-empty u, $M_4^P(u) = \top_w$ if the last letter of u contains p and $\bot_w$ otherwise. The map $0 \mapsto \{\top_w, \bot_w\}$ is therefore not single-valued, and the additional information cannot be recovered from M_3^P. □

Proposition 10.15 (Cells resolved by $n = 4$ beyond $n = 3$)**.** (1) *The absorbing verdicts of M_4^P are those of M_3^P: $M_4^P(u) = \top_s$ iff $M_3^P(u) = +1$, and $M_4^P(u) = \bot_s$ iff $M_3^P(u) = -1$. Hence M_4^P resolves no Layer-1 cell, in the definitive-verdict sense of Proposition 10.10 (1), beyond those resolved by M_3^P (cf. Proposition 10.17 (1)).*

(2) *M_4^P refines M_3^P exactly on the prefixes with $M_3^P(u) = 0$, where it emits a revocable $\top_w$ or $\bot_w$ that depends on the chosen formula (Definition 10.13). On an* ugly *prefix u (Lemma 5.9) M_3^P emits 0 on every extension of u, while M_4^P may still vary (Example 10.16). A property has an ugly prefix iff it is not monitorable: this holds for every property in the indifference cell $\mathsf{L}\cap\mathsf{K}$ (Example A.2), for none in $\mathsf{S}\cup\mathsf{C}$, and for*

some but not all properties of each of the cells $\{\mathsf{L}\}$, $\{\mathsf{K}\}$ *and quaestio. For instance* $\neg a \vee GFp \in \{\mathsf{L}\}$ *and* $a \wedge GFp \in \{\mathsf{K}\}$ *have ugly prefixes, whereas the atomic witnesses* $\mathbf{G}p \vee \mathbf{F}q$ *(table 7) and* $\mathbf{F}p \wedge \mathbf{G}q$ *(Example A.4) have none: from every prefix* u, $u\{q\}$ *is a good prefix of the former and* $u\{\}$ *(no proposition true) a bad prefix of the latter, so* $\top_s$ *(resp.* $\bot_s$*) remains reachable.*

Proof. (1) By Definition 10.13, $M_4^P(u) = \top_s$ iff every w has $uw \in P$, i.e. u is a good prefix, i.e. $M_3^P(u) = +1$; dually for $\bot_s$. The consequence is Proposition 10.17 (1). (2) The first sentence is (1) together with Definition 10.13; the characterisation of properties with an ugly prefix and the witnesses are Lemma 5.9 (3). □

Example 10.16 (Discriminating example for $n = 4$ vs. $n = 3$)**.** On $P = [\![\mathbf{GF}p]\!]$ and a trace $\sigma = p\neg p\, p\neg p\, p \cdots$, M_3^P emits 0 on every prefix; M_4^P alternates between $\top_w$ at positions where p is in the most recent window and $\bot_w$ elsewhere. The information "the current prefix is in the live pattern" is exposed by $n = 4$ and erased by $n = 3$.

10.5. **The robustness monitor ($n \to \infty$).** A natural continuation of the ladder takes the verdict alphabet to be unbounded. The standard incarnation is the Signal-Temporal-Logic *robustness* of Donzé–Maler and the algebraic-runtime framework of Jakšić et al. [34]. The alphabet is $\mathbb{V}_\infty := \mathbb{R} \cup \{\pm\infty\}$, ordered by the natural real order, with positive sign denoting satisfaction and absolute value denoting margin. Although the cardinality jumps from 4 to a continuum, the cells of the intersection algebra resolved at Layer 1 do *not* grow correspondingly.

Proposition 10.17 (Robustness adds no definitive verdict beyond $n = 3$)**.** *Let* $\rho^P : \Sigma^* \to \mathbb{V}_\infty$ *be the STL robustness monitor for* P *instantiated with the* ∞*-norm semiring as in* [34]*, and define the sign projection* $\operatorname{sgn} : \mathbb{V}_\infty \to \mathbb{V}_3$ *by* $\operatorname{sgn}(r) = +1$ *if* $r > 0$, -1 *if* $r < 0$, 0 *if* $r = 0$.

(1) *For every* P *and every verdict monitor* M*, of any alphabet: if* M *issues on a prefix* u *a sound irrevocable positive (negative) verdict, i.e. one that is correct for every trace extending* u*, then* u *is a good (bad) prefix of* P*, so* $M_3^P(u) = +1$ (-1)*. In particular, no Layer-1 cell of the intersection algebra that fails to be resolved by* M_3^P *becomes resolved by* ρ^P*.*

(2) $\operatorname{sgn} \circ \rho^P$ *is not* M_3^P *in general: the robustness of* [34] *evaluates the prefix as a finite trace, so* $\operatorname{sgn} \circ \rho^P$ *is a revocable (weak-regime) ternary map, and it differs from* M_3^P *already on* $P = \mathbf{G}p$ *and the prefix* pp*, which satisfies* $\mathbf{G}p$ *as a finite trace while* $M_3^P(pp) = 0$*; and, like the weak verdicts of Definition 10.13,* ρ^P *depends on the formula presenting* P*.*

Proof. (1) A verdict correct for every trace extending u is, by Lemma 5.9, exactly the statement that u is a good (bad) prefix, which is where M_3^P emits $+1$ (-1). (2) The robustness degree of [34, Sec. 3] is defined on finite traces: it is the distance of τ to the same-length traces violating the specification if τ satisfies it, and minus the distance to the satisfying ones otherwise, so that $\rho > 0$ implies satisfaction and $\rho < 0$ violation of the finite trace [34, Thm. 3]; the MinMax instantiation is the infinity-norm degree [34, Sec. 4.4]. The finite trace pp satisfies $\mathbf{G}p$ and, reading p as a $\{0,1\}$-valued variable, is at distance 1 from every violating trace of length 2, so $\operatorname{sgn} \rho^P(pp) = +1$; but pp is not a good prefix, since $pp\,\{\}^\omega$ (no proposition true after pp) violates $\mathbf{G}p$, so $M_3^P(pp) = 0$. □

Remark 10.18 (What $n \to \infty$ adds and what it does not)**.** Proposition 10.17 clarifies the role of robustness on the ladder: ρ^P contributes *quantitative*, revocable information, not a new Layer-1 cell. In particular ρ^P does not resolve $\mathsf{L} \cap \mathsf{K}$ at Layer 1 either: no prefix of a property in $\mathsf{L} \cap \mathsf{K}$ is good or bad, so no monitor issues a sound irrevocable verdict there (Proposition 10.17 (1)). The richer alphabet trades cardinality for resolution *inside* cells (margin-to-boundary for downstream noise tolerance, as exploited in CPS applications [54]), not *between* cells. The robust-LTL alphabet of [59] sits between $\mathbb{V}_4$ and $\mathbb{V}_\infty$ in this respect: it uses five truth values to grade degrees of violation rather than to enlarge the resolved cell set.

Table 9: Verdict-alphabet ladder coverage matrix: which cell of the intersection algebra each rung newly resolves, with a canonical LTL example and the verdict-side mechanism that witnesses the resolution. A rung *resolves* a cell when, for every property of the cell, its definitive verdicts are complete on at least one side (every violation, or every satisfaction, receives one; Proposition 10.10 (1)). Cells absent from this table fall into two categories: the singleton trivial cells $\mathsf{S} \cap \mathsf{L} = \{\Sigma^\omega\}$ and $\mathsf{C} \cap \mathsf{K} = \{\varnothing\}$ (resolved by the canonical monitor of every rung $n \geq 2$, including binary); and the cells outside $\mathsf{S} \cup \mathsf{C}$ — L-only, K-only, $\mathsf{L} \cap \mathsf{K}$ and Quaestio — which no rung resolves; of these only $\mathsf{L} \cap \mathsf{K}$ is non-monitorable throughout, the other three being mixed (Lemma 5.9).

Rung	**Newly resolved cell**	**Canonical LTL example**	**Witness mechanism**
$n = 2$ Boolean	$\mathsf{S} \cap \mathsf{C}$ (Clopen)	$\mathbf{F}^{\leq 5} p \wedge \mathbf{G}^{\leq 5} \neg q$	DFA with two absorbing sinks (Proposition 10.6)
$n = 3$ Kleene	$\mathsf{S} \setminus \mathsf{C}$	$\mathbf{G}p$ (in $\mathsf{S} \cap \mathsf{K}$, Example A.1)	-1 absorbing on every violation (Proposition 10.10)
$n = 3$ Kleene	$\mathsf{C} \setminus \mathsf{S}$	$\mathbf{F}p$ (in $\mathsf{C} \cap \mathsf{L}$, Example 9.3)	$+1$ absorbing on every satisfaction (Proposition 10.10)
$n = 4$ RV-LTL	(no new cell)	$\mathbf{GF}p$ (Example A.3)	$\top_s, \bot_s$ as $n{=}3$; revocable $\top_w/\bot_w$ on inconclusive prefixes (Proposition 10.15 and Example 10.16)
$n \to \infty$ STL	(no new cell)	any LTL with margin function	quantitative, revocable; no definitive verdict beyond $n{=}3$ (Proposition 10.17)

10.6. **Composition: from Layer 1 to Layer 2.** The progression $n = 2 \to 3 \to 4 \to \infty$ stays within Layer 1: each rung enlarges the per-step alphabet; the rungs $n = 2, 3$ resolve additional cells of the intersection algebra, while $n = 4$ refines inconclusive verdicts and $n \to \infty$ adds revocable quantitative information, neither resolving a further cell (Propositions 10.15 and 10.17). A second mechanism for resolving higher cells is qualitatively different: feed the Layer-2 verdict-trace of a monitor into a *second* monitor and let the composite decide a property of the second-layer ω-word. The composite acquires a Layer-1 cell that can be strictly higher than the base monitor's cell, even when the base alphabet is the simplest possible ($n = 2$). This mirrors the hierarchical organisation of monitors discussed for

distributed runtime verification [54] but read in the opposite direction: *bottom-up* class promotion rather than *top-down* decomposition.

Definition 10.19 (Monitor composition)**.** Let $M : \Sigma^* \to \mathbb{V}$ be a verdict monitor at Layer 1 on Σ, and let $N : \mathbb{V}^* \to \mathbb{V}'$ be a verdict monitor at Layer 1 on the alphabet $\mathbb{V}$. The *composite monitor* $N \circ M : \Sigma^* \to \mathbb{V}'$ is defined by

$$(N \circ M)(u) := N\big(M(u_0)M(u_1)\cdots M(u_{|u|-1})\big),$$

where $u_i := u\restriction_{i+1}$ is the length-$(i{+}1)$ prefix of u. The composite is itself a verdict monitor on Σ with verdict alphabet $\mathbb{V}'$.

Theorem 10.20 (Composition class promotion)**.** *Let $M : \Sigma^* \to \mathbb{V}_2$ be a verdict monitor with binary alphabet (Definition 10.1), for instance a binary verdict monitor of Definition 10.5 or the per-step indicator monitor of Remark 10.21, and let the ω-extension map $E_M : \Sigma^\omega \to \mathbb{V}_2^\omega$, $E_M(\sigma) := (M(\sigma\restriction_{n+1}))_{n\in\mathbb{N}}$, be continuous under the Cantor topology. Let $Q \in \mathsf{X}$ be any ω-regular property of $\mathbb{V}_2^\omega$, where X is any cell of the intersection algebra of $\mathbb{V}_2^\omega$, and let N be a verdict monitor for Q with alphabet $\mathbb{V}'$ of cardinality matching the resolution required by X. Then the composite $N \circ M$ is a verdict monitor for the property $P^Q := \{\sigma \in \Sigma^\omega \mid E_M(\sigma) \in Q\}$. For the open/closed cells, S and C of Q always lift to S and C of P^Q by continuity alone; for the cells involving L (dense) or K (empty interior) it suffices, in addition, that M be* prefix-surjective*: for every $u \in \Sigma^*$ the image $E_M([u])$ contains a basic cylinder of $\mathbb{V}_2^\omega$ (in particular, if E_M is an open map). If moreover E_M is an open surjection, the Layer-1 cell of $N \circ M$ in the intersection algebra of Σ^ω is exactly X. In particular, for every non-empty $\mathsf{X} \in \{\mathsf{S},\mathsf{C},\mathsf{L},\mathsf{K}\}^{\text{intersections}}$ there is a composition with a binary-alphabet base monitor whose Layer-1 cell is exactly X: the per-step indicator monitor (Remark 10.21), whose emission map E_M is an open surjection.*

Proof. The definition of P^Q pulls the property Q back through the ω-extension of M to a property of Σ^ω. The map E_M is continuous from the Cantor topology on Σ^ω to the Cantor topology on $\mathbb{V}_2^\omega$ (this is also clause (1) of Theorem 10.23). Continuous preimages preserve the open and closed properties by topological invariance, so S and C of Q lift to S and C of P^Q.

For L (dense) and K (empty interior), continuity alone is *not* sufficient: density of the preimage of every dense set is equivalent to prefix-surjectivity, which every open map satisfies, not to E_M having dense image (see Remark 10.21 for a continuous, dense-image map whose preimage of a dense set fails to be dense). We therefore invoke the open-map / prefix-surjectivity hypothesis. Assume that for every $u \in \Sigma^*$ the image $E_M([u])$ contains a basic cylinder $[w]$ of $\mathbb{V}_2^\omega$ (with $w \in \mathbb{V}_2^*$ depending on u).

- L*:* suppose Q is dense. Fix any cylinder $[u] \subseteq \Sigma^\omega$ and pick w with $[w] \subseteq E_M([u])$. By density of Q there is $\tau \in Q \cap [w] \subseteq Q \cap E_M([u])$, so some $\sigma \in [u]$ has $E_M(\sigma) = \tau \in Q$; hence $E_M^{-1}(Q)$ meets $[u]$. As $[u]$ was arbitrary, $E_M^{-1}(Q) = P^Q$ is dense, i.e. in L.
- K*:* suppose Q has empty interior, i.e. its complement Q^c is dense. Applying the same open-map argument to Q^c gives that $E_M^{-1}(Q^c) = (P^Q)^c$ is dense, so P^Q has empty interior, i.e. in K.

Hence under prefix-surjectivity each of the four memberships of Q transfers to P^Q. If E_M is moreover an open surjection, the converse transfers hold as well: by surjectivity $E_M(P^Q) = Q$ and $E_M((P^Q)^c) = Q^c$, so openness of P^Q (resp. of its complement) gives openness of Q (resp. Q^c); and if P^Q is dense then, by continuity and surjectivity, $\mathbb{V}_2^\omega =$

$E_M(\operatorname{cl} P^Q) \subseteq \operatorname{cl} E_M(P^Q) = \operatorname{cl} Q$, dually for the complement. Then P^Q has exactly the membership tuple of Q, i.e. it lies in the cell X of Q (Theorem 4.6). Without surjectivity the converse fails: with $M(u) = 1$ for $|u| = 1$ and $M(u) = 1$ iff p holds at the last letter of u otherwise, E_M is prefix-surjective and open onto [1], and $Q = (b_0{=}1) \vee \mathbf{GF}(b{=}1) \in \{\mathsf{L}\}$ has $P^Q = \Sigma^\omega$. The composite $N \circ M$ is a verdict monitor for P^Q by direct unfolding of Definition 10.19.

For the existence claim, the open-map base realising the L, K and $\mathsf{L} \cap \mathsf{K}$ cells is the *per-step indicator* monitor: fix a non-trivial predicate π on Σ and let M emit $v_t = 1$ iff the current symbol satisfies π, else $v_t = 0$. Its ω-extension E_M is a memory-0, letter-to-letter surjection onto $\{0,1\}^\omega$ (on non-empty prefixes, as in the proof of Theorem 10.23 (1)), which is open (prefix-surjective: $E_M([u]) = [\,\pi(u)\,]$ for the indicator string $\pi(u)$, and every basic cylinder of $\mathbb{V}_2^\omega$ arises as $E_M([u])$ for a one-symbol extension). Choosing Q in L (resp. K, $\mathsf{L} \cap \mathsf{K}$) then places $E_M^{-1}(Q) = P^Q$ in exactly that cell. The same per-step indicator base realises every other cell exactly, since its E_M is an open surjection. □

Remark 10.21 (Why prefix-surjectivity, not dense image, is the right hypothesis)**.** The property of E_M that lifts L and K is *prefix-surjectivity* (implied by openness), *not* dense image. Density of $E_M^{-1}(Q)$ for every dense Q is precisely prefix-surjectivity (every $E_M([u])$ has non-empty interior), which every open map satisfies; dense image is strictly weaker. A continuous, dense-image (indeed surjective) map can fail it: on $\{0,1\}^\omega$ let $E(\sigma) = 0^\omega$ if $\sigma_0 = 0$ and $E(\sigma) = \sigma_1\sigma_2\cdots$ if $\sigma_0 = 1$. Then E is continuous with full (hence dense) image, and $Q = \{\text{infinitely many 1s}\}$ is dense, yet $E^{-1}(Q) \subseteq [\sigma_0 = 1]$ is not dense (it misses the clopen set $[\sigma_0 = 0]$). The *per-step indicator* monitor of Theorem 10.20 — emitting 1 exactly when the current symbol satisfies a fixed non-trivial predicate — realises the open-surjection condition: its memory-0, letter-to-letter E_M is a surjection onto $\{0,1\}^\omega$ with $E_M([u])$ a basic cylinder for every u. By contrast, a uniformly-bounded clopen base monitor stabilises its verdict at some bounded depth, so its E_M has *finite* (hence nowhere-dense) image and is neither open nor prefix-surjective; such a base *cannot* lift L, K, or $\mathsf{L} \cap \mathsf{K}$. This is exactly why prefix-surjectivity is required and dense image alone does not suffice. Prefix-surjectivity is strictly weaker than openness: on $\{0,1\}^\omega$, $E(0\tau) = 0\tau$, $E(1^k0\tau) = 1^k00\tau$ and $E(1^\omega) = 1^\omega$ is prefix-surjective, but $E([1])$ is not open.

Remark 10.22 (Why composition is not a substitute for ladder rungs)**.** Theorem 10.20 might suggest that the Layer-1 ladder of sections 10.2 to 10.5 is redundant: pick the simplest base M and push complexity into N. Two structural facts prevent this reduction. First, the composite's input alphabet is Σ but the composite's working alphabet is $\mathbb{V}_2$: the composite cannot reflect any property of Σ^ω that is not factor-able through a clopen partition of Σ. Second, the resources needed by N scale with the Layer-2 cell complexity of the encoded Q, and the bound is tight by classical ω-language complexity [58]. The Layer-1 ladder distributes complexity into the per-step alphabet; the composition lifts it into the second-stage automaton. The two mechanisms are dual, not substitutable.

Theorem 10.20 establishes that a specific construction (binary base + secondary monitor) can promote the Layer-1 cell. A complementary universal statement holds: *every* verdict monitor, irrespective of its Layer-1 cell, gives rise to a canonical family of Persistence / Response properties over the input Σ^ω, simply by pulling back the canonical Persistence / Response properties on the verdict-trace, which are Liveness/Coliveness there, through the monitor's extension map; a pulled-back property is itself in $\mathsf{L} \cap \mathsf{K}$ exactly at the verdicts where the monitor is non-absorbing (Proposition 10.24).

Theorem 10.23 (Universal Liveness/Coliveness attribution to monitor output)**.** *Let $M : \Sigma^* \to \mathbb{V}$ be any Layer-1 verdict monitor with verdict alphabet $\mathbb{V}$ of cardinality $|\mathbb{V}| \geq 2$, and let $E_M : \Sigma^\omega \to \mathbb{V}^\omega$ be its ω-extension defined by $E_M(\sigma)_n := M(\sigma\restriction_{n+1})$ (non-empty prefixes, as in Definition 10.1). For every $v \in \mathbb{V}$ define the pullback properties*

$$\Phi_v^M := \big\{\sigma \in \Sigma^\omega \,\big|\, \exists n \in \mathbb{N},\ \forall m \geq n\colon M(\sigma\restriction_m) = v\big\},$$
$$\Psi_v^M := \big\{\sigma \in \Sigma^\omega \,\big|\, \forall n \in \mathbb{N},\ \exists m \geq n\colon M(\sigma\restriction_m) = v\big\}.$$

Then:

(1) *E_M is continuous with respect to the Cantor topologies on Σ^ω and $\mathbb{V}^\omega$;*
(2) *$\Phi_v^M \in \mathbf{\Sigma}_2^0(\Sigma^\omega)$ (Persistence-class) and $\Psi_v^M \in \mathbf{\Pi}_2^0(\Sigma^\omega)$ (Response-class);*
(3) *Φ_v^M and Ψ_v^M are pre-images under E_M of properties that lie in the $\mathsf{L} \cap \mathsf{K}$ cell of the intersection algebra of $\mathbb{V}^\omega$;*
(4) *if M is* non-degenerate at v *— meaning that for every prefix $u \in \Sigma^*$ both $\Phi_v^M \cap u{\cdot}\Sigma^\omega$ and its complement within the cylinder are non-empty — then $\Phi_v^M \in \mathsf{L}$ and $\Phi_v^M \in \mathsf{K}$ over Σ^ω (i.e. membership in the $\mathsf{L}\cap\mathsf{K}$ intersection; since $\mathsf{S}\cap\mathsf{L}\cap\mathsf{K} = \mathsf{C}\cap\mathsf{L}\cap\mathsf{K} = \varnothing$ (Theorem 5.4), this already excludes S and C and places Φ_v^M in the exclusive cell $(0,0,1,1)$). Dually for Ψ_v^M.*

Proof. (1) Pick $\sigma \in \Sigma^\omega$ and a basic open neighbourhood $U \subseteq \mathbb{V}^\omega$ of $E_M(\sigma)$ determined by its length-n prefix. The value $E_M(\sigma)\restriction_n = M(\sigma\restriction_1)\cdots M(\sigma\restriction_n)$ depends only on $\sigma\restriction_n$, so any $\sigma' \in \Sigma^\omega$ agreeing with σ on the first n samples maps into U. Hence $E_M^{-1}(U)$ contains a basic open neighbourhood of σ, and E_M is continuous.

(2) The property $\mathbf{FG}(x = v) = \bigcup_n \{x \in \mathbb{V}^\omega \mid \forall m \geq n,\, x_m = v\}$ is a countable union of closed cylinders, hence in $\mathbf{\Sigma}_2^0(\mathbb{V}^\omega)$. Symmetrically $\mathbf{GF}(x = v) \in \mathbf{\Pi}_2^0(\mathbb{V}^\omega)$. Borel classes are closed under continuous pre-images [12, §3.1] (see also [35]), so $\Phi_v^M = E_M^{-1}(\mathbf{FG}(x = v))$ inherits $\mathbf{\Sigma}_2^0$ and Ψ_v^M inherits $\mathbf{\Pi}_2^0$ on Σ^ω.

(3) Within $\mathbb{V}^\omega$ with $|\mathbb{V}| \geq 2$, every finite prefix $w \in \mathbb{V}^*$ admits the continuation v^ω (which makes $\mathbf{FG}(x = v)$ hold) and the continuation v'^ω for any $v' \neq v$ (which violates it). Therefore $\mathbf{FG}(x = v) \in \mathsf{L}$ (every prefix extends to satisfaction) and $\in \mathsf{K}$ (every prefix extends to violation), i.e. in $\mathsf{L} \cap \mathsf{K}$ over $\mathbb{V}^\omega$. The same holds for $\mathbf{GF}(x = v)$.

(4) Non-degeneracy says that no prefix-cylinder of Σ^ω is fully inside Φ_v^M nor fully outside. Equivalently, for every $u \in \Sigma^*$ there exist $\sigma_1, \sigma_2 \in u{\cdot}\Sigma^\omega$ with $\sigma_1 \in \Phi_v^M$ and $\sigma_2 \notin \Phi_v^M$. The first witness realises the Liveness condition (every prefix extends to satisfaction) and the second the Coliveness condition (every prefix extends to violation), placing Φ_v^M in $\mathsf{L}\cap\mathsf{K}$ over Σ^ω. The dual argument applies to Ψ_v^M. □

The non-degeneracy condition of Theorem 10.23(4) is the gate that controls when the universal attribution genuinely promotes a monitor's Layer-1 cell to the Layer-2 cell $\mathsf{L} \cap \mathsf{K}$. The following proposition sharpens (4) into an explicit dichotomy between absorbing and non-absorbing verdict regimes.

Proposition 10.24 (When the universal attribution promotes the cell)**.** *Let $M : \Sigma^* \to \mathbb{V}$ be a verdict monitor and $v \in \mathbb{V}$. The following are equivalent:*

(i) *Φ_v^M lies in $\mathsf{L} \cap \mathsf{K}$ of Σ^ω;*
(ii) *M is non-degenerate at v in the sense of Theorem 10.23(4);*
(iii) *M is* non-absorbing at v*: for every prefix $u \in \Sigma^*$ there exist both an extension whose verdict-trace eventually stabilises at v and an extension whose verdict-trace does not.*

In particular:

- *Strong-regime Kleene monitors (Definition 10.9) are absorbing at their fixed-point verdicts and* do not promote. *Concretely, for the LTL$_3$ monitor of P (Proposition 10.10): at the absorbing-*ACCEPT *verdict v, the pullback Φ_v^M is the open set of traces having a good prefix, which equals P (and so lies in its cell) exactly when $P \in \mathsf{C}$; at the absorbing-*REJECT *verdict v', $\Phi_{v'}^M$ is the set of traces having a bad prefix, which equals the complement of P (and so lies in the De Morgan-dual cell) exactly when $P \in \mathsf{S}$.*
- *Weak-regime Kleene monitors (Definition 10.9), RV-LTL revocable verdicts $\top_w, \bot_w$ (Definition 10.13), and Layer-2 composite monitors (Definition 10.19) are regimes in which (as is the per-step indicator of Theorem 10.20) non-absorbing verdicts occur, and they* promote *exactly when (iii) holds: for the RV-LTL monitor of $\mathbf{GF}p$ the verdict $\top_w$ stabilises exactly on $\mathbf{FG}p \in \mathsf{L} \cap \mathsf{K}$ (Example 10.16), whereas for a clopen P the RV-LTL monitor emits only $\top_s, \bot_s$ after finitely many steps, so $\Phi_{\top_w}^M = \varnothing$.*

Proof. The equivalence (i) $\Leftrightarrow$ (ii) is clause (4) of Theorem 10.23. For (ii) $\Leftrightarrow$ (iii): by construction, $\Phi_v^M \cap u \cdot \Sigma^\omega$ is exactly the set of σ extending u whose verdict-trace eventually stabilises at v, and its complement is the set of σ extending u whose verdict-trace does not. Non-emptiness of both sets for every prefix u is the verbatim definition of non-absorbing.

The particular cases follow. Strong-regime Kleene monitors have absorbing fixed-point verdicts v^*: once $M(u) = v^*$, all extensions $M(u \cdot a) = v^*$, so $\Phi_{v^*}^M \cap u \cdot \Sigma^\omega = u \cdot \Sigma^\omega$ or $\varnothing$ depending on whether v^* has been reached on u; hence $\Phi_{v^*}^M$ is open, and for LTL$_3$ it is the union of the cylinders of the good (resp. bad) prefixes, which is P (resp. its complement) exactly when $P \in \mathsf{C}$ (resp. $P \in \mathsf{S}$). Non-degeneracy fails at v^* on those prefixes that already determine the verdict. For the revocable regimes, (iii) is a property of the monitor, not of the regime: for $\mathbf{GF}p$ the finite-trace truth of the formula on u is that of p at the last letter of u, so $\top_w$ stabilises exactly on $\mathbf{FG}p$, which every prefix extends both into and out of; for a clopen P every trace reaches a good or bad prefix, after which only $\top_s, \bot_s$ are emitted. □

Remark 10.25 (Reading of Theorem 10.23)**.** The theorem says that *every* monitor automatically furnishes its consumer with a canonical family of Persistence and Response properties on the input, indexed by the verdict values $v \in \mathbb{V}$. Liveness and Coliveness attribution is not a privilege of specific monitor architectures: every monitor attributes pre-images of $\mathsf{L} \cap \mathsf{K}$ properties of its verdict trace, and these are $\mathsf{L} \cap \mathsf{K}$ on the input exactly at its non-absorbing verdicts. The Layer-2 cell $\mathsf{L} \cap \mathsf{K}$ is the canonical destination of this attribution, and the move from a Layer-1 cell of M to the Layer-2 $\mathsf{L} \cap \mathsf{K}$ cell is the universal version of the class-promotion phenomenon of Theorem 10.20. The non-degeneracy condition (4) is sharpened by Proposition 10.24: it excludes the absorbing verdicts (in particular those of strong-regime Kleene monitors), and admits exactly the verdicts at which the monitor is non-absorbing in the sense of Proposition 10.24 (iii), which occur only in the revocable regimes (e.g. weak Kleene, RV-LTL revocable verdicts, Layer-2 composites) and there depend on the monitored property.

Remark 10.26 (Boundary cases: what the hypotheses exclude)**.** The hypotheses of Theorem 10.23 and Theorem 10.20 are tight in the sense that removing them admits explicit counterexamples:

- $|\mathbb{V}| = 1$ *(degenerate alphabet).* With a single verdict value v, the verdict trace is constant v^ω regardless of the input; both Φ_v^M and Ψ_v^M equal Σ^ω, which sits in the trivial singleton cell $\{\Sigma^\omega\}$, not in $\mathsf{L} \cap \mathsf{K}$. The hypothesis $|\mathbb{V}| \geq 2$ in Theorem 10.23 excludes this case.

- *Constant monitor on* $|\mathbb{V}| \geq 2$. A monitor $M(\sigma\restriction_n) \equiv v_0$ trivialises $\Phi^M_{v_0} = \Sigma^\omega$ and $\Phi^M_v = \varnothing$ for $v \neq v_0$; the non-degeneracy hypothesis of clause (4) fails on every prefix, and the open-map / prefix-surjectivity hypothesis of Theorem 10.20 also fails (E_M has the single-point image $\{v_0^\omega\}$, whose preimage of any proper dense set is empty, hence not dense). Both theorems correctly refuse to promote the cell.
- *Absorbing-but-non-constant monitor.* Strong-regime LTL_3 monitors of pure Safety properties absorb at $\perp$ on the prefix where the property is first refuted; before that prefix, the verdict is ?. Non-degeneracy of clause (4) at $\perp$ fails on prefixes that already absorb; Proposition 10.24 characterises this failure as the absorbing regime.

These boundary cases are not pathological — they are common in classical monitor architectures. The theorems' hypotheses isolate exactly the regimes in which the conclusions go through: the non-absorbing regime for universal attribution (Theorem 10.23), and the open-map / prefix-surjective regime for class promotion (Theorem 10.20).

Example 10.27 (Promoting a binary per-step indicator to $\mathsf{L} \cap \mathsf{K}$)**.** Let M be the binary-alphabet monitor that reports, at each sample, whether p holds at the current sample (a one-bit indicator; its values are revocable, so it is not a committing monitor of Definition 10.5). This is precisely the per-step indicator monitor of Theorem 10.20: its bit stream is $(b_n)_{n\in\mathbb{N}} \in \{0,1\}^\omega$ where $b_n = \mathbf{1}[p \text{ holds at sample } n]$, and the emission map E_M is the memory-0 letter-to-letter surjection onto $\{0,1\}^\omega$, which is an open surjection (hence prefix-surjective), so the hypotheses of Theorem 10.20 hold. Let $Q = [\![\mathbf{GF}(b=1)]\!] \in \mathsf{L} \cap \mathsf{K}$ (a recurrence property on the bit stream). The composite $N \circ M$ is a verdict monitor for the trace property "p occurs infinitely often," which lies in $\mathsf{L} \cap \mathsf{K}$ over Σ^ω. The base monitor M is binary; the second-stage monitor N requires the $n = 4$ (or richer) alphabet to expose the current Liveness commitment of the bit stream, per Example 10.16.

10.7. **Refinement-based confirmation of verdict traces.**

Remark 10.28 (Scope of this subsection)**.** The contribution of this subsection (**O5** in the introduction) is a *framework-level connection point*, not a self-contained convergence theorem. Concretely: the existence and structural shape of the (D, A_0, U, E) refinement tuple are stated, and the two in-framework regularity sources (intrinsic / extrinsic) are identified, but the formal convergence theorem — bounding the number of verdict violations by the ratio of error-bound to strict-decrease constant — is left to future dedicated treatment. Within the present paper this subsection is therefore *conjectured at the framework level*; individual instances such as Miller–GRH primality certificates and the AKS deterministic primality test are established externally (Miller–GRH / AKS) and serve as witnesses of the framework on $\mathbf{\Pi}^0_1$ properties.

The constructions of sections 10.2 to 10.6 treat the monitor's verdict output as either a single-step value (Layer 1) or a verdict-trace viewed as ω-word (Layer 2). A third reading, complementary to both, is operationally important: at every finite step n, the monitor has produced a *finite-length verdict trace* $M(\sigma\restriction_1)M(\sigma\restriction_2)\cdots M(\sigma\restriction_n) \in V^n$. This finite trace is itself a concrete data object on which arbitrary computable functions can be defined, including refinement systems with Lyapunov-style error metrics that exploit the (V, A) structure of Remark 10.30 from a constructive angle. Refinement-based confirmation is the systematic study of such functions when their convergence on finite verdict traces implies

finite-time confirmation of properties that, on the infinite trace, would be non-monitorable Layer-2 properties, such as those of the cell $\mathsf{L} \cap \mathsf{K}$.

The essential pipeline composes a verdict-producing monitor M with a refinement system (D, A_0, U, E) whose domains D_k are finite-prefix verdict traces of length k, whose update U extends the trace by one observation (possibly invoking structural-knowledge computation about M), and whose error metric E measures the distance between the current finite trace and a target Layer-2 property such as $\mathbf{FG}(b = v)$. When E converges to zero in finite time *and* a structural-zero guarantee precludes future non-zero recurrence, the Layer-2 property is confirmed on the infinite trace from a finite observation — precisely the class of evidence that Theorem 10.23 and Proposition 10.24 identify as absent from absorbing verdict monitors alone. The refinement system supplies what the monitor cannot: external supporting structure that promotes a non-monitorable Layer-2 property to finite-time decidability.

Two distinct sources of regularity make refinement-based confirmation operationally feasible within the verdict-monitor framework treated by the present paper:

- *Intrinsic regularity of* (V, A): the algebraic structure of the verdict alphabet and its operations is itself the source of refinement-exploitable patterns. Number-theoretic instances are the clearest. The Miller primality test under the Generalised Riemann Hypothesis bounds the witness search at $2(\log n)^2$ bases and converts the natively $\mathbf{\Pi}^0_1$ predicate "n is prime" into a finite-time $\mathsf{S}\cap\mathsf{C}$ decision. The AKS deterministic certificate achieves the same lift without the GRH assumption. Both exploit intrinsic ring structure of $\mathbb{Z}$; no external bias is required.
- *Extrinsic regularity from a function behind the monitor*: when the input to M is itself the output of another process or function F, the verdict trace inherits statistical or structural bias from F. The refinement system can then exploit this induced bias as a Lyapunov source even when (V, A) alone would not suffice for finite-time confirmation. Examples include monitors of stochastic processes (where F is the process distribution), monitors of physically constrained signals (where F is the physical law), and monitors of agents with declared internal goals (where F is the agent's intention encoded in the trace).

A third source — regularity *produced by the refinement system itself* when the monitor is parametric and its parameters can be adapted online against a target Layer-2 property — is available in monitor frameworks that go beyond the verdict-only view of the present paper. This case sits outside the present scope and is left to future dedicated treatment.

This framing should be distinguished from two adjacent lines of work. *Assumption-based monitorability* (Henzinger–Saraç [32]; Cimatti et al. [16]) restricts the set of admissible infinite extensions of a prefix by an environment assumption and asks whether the residual property becomes monitorable under that restriction; the verdict semantics is unchanged and the lift comes from *shrinking the trace space*. The (D, A_0, U, E) refinement tuple instead leaves the trace space intact and adds an *external structural witness* — a convergent computation on the finite verdict trace, exploiting (V, A) regularity or a function behind the monitor — that licenses a finite-time positive verdict for a property which remains non-monitorable in the unrestricted setting. Where an assumption is itself an exploitable regularity, the two coincide: refinement-based confirmation is then the constructive, certificate-carrying instance of assumption-based confirmation, and we make no claim of generality beyond that overlap. Equally, the construction is *not* a truncated-path semantics: Eisner et al.'s strong/weak views [23] fix a finite horizon and reinterpret the formula on the truncated

word, whereas the refinement tuple confirms the *infinite-trace* property from finite evidence via a structural-zero guarantee, with no horizon truncation of the property itself.

The systematic development of refinement-based confirmation — including the formal definition of the Hansel–Newton liveness certificate, the convergence theorem that bounds the number of verdict violations by the ratio of error-bound to strict-decrease constant, and a worked catalogue of intrinsic and extrinsic regularity sources — is the subject of forthcoming dedicated treatment. The present subsection establishes the connection point only: every verdict monitor in the ladder of sections 10.2 to 10.5 is a potential input stage for a refinement-based confirmation system, and the choice of monitor architecture determines which (V, A) regularities the downstream refinement can exploit.

Table 10: Summary of the four primary CLASP classes against the verdict-monitor framework of this section: which ladder rung resolves the class at Layer 1, what refinement-confirmation promotion is available within the verdict-only framework of section 10.7, and a canonical LTL example. The table consolidates the analysis of sections 10.2 to 10.5 together with the composition (section 10.6) and refinement-confirmation (section 10.7) machinery developed earlier in the section.

Primary class	**Ladder rung resolving Layer 1**	**Refinement-confirmation promotion**	**Canonical LTL example**
Safety S	$n = 3$ Kleene strong, sub-cell $\mathsf{S} \setminus \mathsf{C}$ (Proposition 10.10)	intrinsic (V, A): Miller-GRH lifts $\mathbf{\Pi}_1^0$ to $\mathsf{S} \cap \mathsf{C}$	$\mathbf{G}p$ (in $\mathsf{S} \cap \mathsf{K}$, Example A.1)
Cosafety C	$n = 3$ Kleene strong, sub-cell $\mathsf{C} \setminus \mathsf{S}$ (Proposition 10.10)	intrinsic (V, A), symmetric to Safety	$\mathbf{F}p$ (in $\mathsf{C} \cap \mathsf{L}$, Example 9.3)
Liveness L	$n = 3$ for $\mathsf{C} \cap \mathsf{L}$; no rung for $\{\mathsf{L}\}$ (definitive $+1$ on some satisfactions only, Proposition 10.10; revocable $n = 4$ verdicts on inconclusive prefixes, Proposition 10.15)	extrinsic F-bias: function behind the monitor input induces statistical or structural bias exploitable as a Lyapunov source	$\mathbf{G}p \vee \mathbf{F}q$ (table 7)
$\mathsf{S} \cap \mathsf{C}$, $\mathsf{L} \cap \mathsf{K}$ (mixed)	Clopen $\mathsf{S} \cap \mathsf{C}$ at $n = 2$ (Proposition 10.6); $\mathsf{L} \cap \mathsf{K}$ at no rung, only revocable $n = 4$ verdicts (Proposition 10.15)	Layer-2 composition (Theorem 10.20) lifts a base monitor to higher cells via Definition 10.19	$\mathbf{F}^{\leq k}p \wedge \mathbf{G}^{\leq k}\neg q$; $\mathbf{GF}p$ (Example A.3)

10.8. **Beyond verdict monitors: event-generator algebras.** The four rungs $n = 2, 3, 4, \infty$ exhaust the *verdict-based* reading: a monitor is a function $\Sigma^* \to \mathbb{V}$ that assigns a truth-domain element to every observed prefix. A structurally different organisation is the *event-generator* reading: a monitor whose output is not a verdict on

a fixed property, but an *event* drawn from a small alphabet such as $\{\mathsf{V}, \mathsf{I}, \varepsilon\}$ (validate / invalidate / no-event), *parameterised* by configuration data and *generated* from a continuous or discrete signal.

This shift is more than a renaming. In the verdict-based reading the property is fixed and the monitor's job is to evaluate it; in the event-generator reading the property is implicit in the configuration parameters, and the monitor's job is to emit a discrete event-stream that downstream layers consume. The alphabet $\{\mathsf{V}, \mathsf{I}, \varepsilon\}$ is isomorphic as a set to the strong Kleene alphabet $\{+1, 0, -1\}$ of section 10.3, so the cardinality coincides; but the algebraic structure is different. Verdict-based monitors form a function-space $\Sigma^* \to \mathbb{V}$ ordered by truth-domain refinement; event-generator monitors form a parametric family $(\theta, \pi) \mapsto E_{\theta,\pi} : \Sigma^* \to \{\mathsf{V}, \mathsf{I}, \varepsilon\}$ ordered by parameter specialisation. The two organisations meet at the alphabet but diverge in their primitive operations.

A complete development of the event-generator face, including the case of two *dual primitives* (crossing-based and touch-based) for the same continuous trace, is a natural further direction we identify but do not pursue here. We close the present section at the boundary where verdict monitors stop being the natural framework.

Remark 10.29 (Scope of the present section)**.** The verdict-alphabet ladder of sections 10.2 to 10.5 together with the composition mechanism of section 10.6 cover the *verdict-based* incarnation of monitor systems against the CLASP intersection algebra. The event-generator incarnation, where two primitive algebras can produce the same event stream from the same signal under different parameterisations, opens a parallel direction that we identify as a face complementary to the present one but technically distinct; the relationship between the two faces is left to future dedicated treatment.

Remark 10.30 (The (V, A) parameter pair of this section)**.** The constructions of this section are parameterised by a pair (V, A) where V is a verdict alphabet and A is an algebra of operations over V (and over the underlying input alphabet Σ). In the ladder of sections 10.2 to 10.5, A is held fixed (the standard LTL operator semantics interpreted in the appropriate truth domain $\mathbb{B}_n$), while V ranges over $\{0, 1\}$, $\{-1, 0, +1\}$, $\{\top_s, \top_w, \bot_w, \bot_s\}$, and $\mathbb{R} \cup \{\pm\infty\}$. The CLASP cells reachable at each rung depend on V's cardinality and on A's expressiveness: varying V at fixed A traces the verdict-cardinality axis, while varying A at fixed V traces a complementary axis along which the alphabet stays the same but the algebra that defines operations on it changes. The complementary axis is a natural further direction we identify but do not pursue here.

11. The trivial rung and meta-classification of CLASP classes

The verdict-alphabet ladder of section 10 starts at n=2 because the binary Boolean monitor is the smallest cardinality at which a non-trivial Layer-1 discrimination can occur. Two adjacent questions remain. First, what happens below n=2 — the *degenerate* rungs where the monitor's output stream carries at most one distinct value across its lifetime? Second, the four CLASP classes are themselves objects in $\mathcal{P}(\mathcal{P}(\Sigma^\omega))$; can they be classified by the same prefix-quantifier discipline, applied one level up? Both questions admit clean answers that we develop in this closing section.

11.1. **The trivial rung n=1 and the empty rung n=0.** A verdict monitor whose Layer-1 output stream $\mathbf{v} = v_0 v_1 v_2 \cdots \in \mathbb{V}^\omega$ uses at most one distinct value $v \in \mathbb{V}$ across its lifetime is called *trivial*: every tick emits the same value v, regardless of input. Reading $\mathbb{V}_3 = \{\top, \bot, ?\}$ as in section 10.3, the three trivial monitors and their Layer-1 cells are tabulated in table 11.

Output stream	Reading	Property classified
$\top^\omega$	"property holds at every tick, by axiom"	$\{\Sigma^\omega\}$ (trivial top, $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L}$)
$\bot^\omega$	"property fails at every tick, by axiom"	$\{\varnothing\}$ (trivial bottom, $\mathsf{S} \cap \mathsf{C} \cap \mathsf{K}$)
$?^\omega$	"no finite verdict ever achievable"	$\mathsf{L} \cap \mathsf{K}$ (non-monitorable)

Table 11: The three canonical trivial-monitor configurations at n=1 and the CLASP cells they characterise (the LTL_3 output is constantly v exactly on the listed cell, Proposition 7.5). The first two correspond to the two trivial singletons of Theorem 5.4 (cells $\{\mathsf{S}, \mathsf{C}, \mathsf{L}\}$ and $\{\mathsf{S}, \mathsf{C}, \mathsf{K}\}$); the third corresponds to the $\mathsf{L} \cap \mathsf{K}$ cell, the canonical non-monitorable class.

A formally lower rung n=0 corresponds to the *empty* stream (no verdict ever emitted, not even ?): an operational impossibility for any synchronous-tick monitor, included here only for notational completeness. We collapse n=0 and n=1:$?^\omega$ into a single "non-monitorable" rung.

The three trivial-rung cells of table 11 admit a particularly clean graphical realization in the threshold-monitor framework of [10]. The model used there is the *discontinuous-threshold* construction: by the structural degeneracy of the n=1 rung the signal is constant at the unique threshold value ($s \equiv \theta$, the only value-position available when no other thresholds exist), so the signal and the threshold are in permanent geometric overlap by construction. What distinguishes the three configurations is therefore not a signal-vs-threshold offset but the *binary detection state* of the monitor: an on/off resource interrupted across time. Each $\mathsf{OFF} \to \mathsf{ON}$ re-activation fires exactly one event of the type fixed by the configuration, because the geometric overlap is already in place at the moment detection returns on.

V-mode pulsed re-activation.: Detection cycles $\mathsf{OFF} \to \mathsf{ON}$ per tick and fires a Validates on each rising edge, producing the V^ω verdict stream of cell $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L}$.

I-mode pulsed re-activation.: Symmetric, producing the I^ω stream of cell $\mathsf{S} \cap \mathsf{C} \cap \mathsf{K}$.

Detection permanently OFF.: No rising edge ever occurs, so no event is fired and the verdict stream is the empty (silent) stream realising the non-monitorable cell $\mathsf{L} \cap \mathsf{K}$.

The absence of an ε value at this rung follows structurally: the cause-paradox of the wall-induced touch construction in [10] requires *wall mobility* to generate the strict-vs-direction-resolved ambiguity that yields ε; removing mobility (as the trivial rung does by collapsing to a single available value-position) removes the ambiguity. See the n=1 discontinuous-threshold figure (the degenerate-overlap model with three detection-state configurations) in [10] for the visualisation of the three configurations stacked side by side.

11.2. **The split sub-ladder at n=2.** A monitor whose Layer-1 output uses exactly two distinct values can manifest in three canonical patterns, depending on which pair of $\{\top, \bot, \varepsilon\}$ appears. Each pattern admits a Layer-1 cell identification:

- $\{\top, \varepsilon\}$**.:** Monitor emits ε until a good-prefix is observed, then $\top$. Region resolved: the cosafety region (open, good-prefix detectable; witness $\mathbf{F}p$): it confirms every satisfaction in finite time iff $P \in \mathsf{C}$ (Proposition 10.10 (1)), and it discards no available verdict iff no bad prefix exists, i.e. $P \in \mathsf{L}$ (Lemma 5.9 (1)); both hold iff $P \in \mathsf{C} \cap \mathsf{L}$. On $\{\mathsf{C}\}$, e.g. $p \wedge \mathbf{F}q$, it is still complete on satisfactions but silent on the finitely detectable violations.
- $\{\bot, \varepsilon\}$**.:** Monitor emits ε until a bad-prefix is observed, then $\bot$. Region resolved: the safety region (closed, bad-prefix detectable; witness $\mathbf{G}p$): dually, it refutes every violation in finite time iff $P \in \mathsf{S}$, and discards no available verdict iff $P \in \mathsf{K}$; both hold iff $P \in \mathsf{S} \cap \mathsf{K}$; on $\{\mathsf{S}\}$, e.g. $p \vee \mathbf{G}q$, it is silent on the good prefix $\{p\}$.
- $\{\top, \bot\}$**.:** Monitor never emits ε: its values are provisional until a committing prefix and irrevocable afterwards, and every trace reaches a committing prefix (Definition 10.5). Property classified: clopen $\mathsf{S} \cap \mathsf{C}$ (Proposition 10.6).

The three patterns are pairwise distinguishable on a single signal model: the central observation is that pure-S and pure-C monitors are *absorbing* (one of the two verdicts is final), while the clopen monitor may *alternate* (the two values may alternate across ticks before commitment). The latter is the binary-Boolean monitor of Definition 10.5; the former two are the strong-Kleene LTL_3 monitors of Definition 10.9 restricted to the $\{\top, ?\}$ or $\{\bot, ?\}$ sub-alphabet (where $? = \varepsilon$ in our notation).

11.3. **Extended verdict-alphabet ladder.** Combining sections 10, 11.1 and 11.2 yields the full extended ladder:

n	Canonical alphabet structure	Cells resolvable (at Layer 1)
1	single constant $v \in \{\top, \bot, \varepsilon\}$	trivial singletons; $?^\omega$ characterises $\mathsf{L} \cap \mathsf{K}$ (table 11)
2	$\{v, v'\} \subset \{\top, \bot, \varepsilon\}$ with $v \neq v'$	safety S, cosafety C, clopen $\mathsf{S} \cap \mathsf{C}$ (section 11.2)
3	full LTL_3 $\{\top, \bot, ?\}$	$\mathsf{S} \cup \mathsf{C}$ in one monitor (Proposition 10.10)
4	RV-LTL $\{\top_s, \top_w, \bot_w, \bot_s\}$	no new cell; revocable verdicts (Proposition 10.15)
∞	STL robustness $\mathbb{R} \cup \{\pm\infty\}$	no new cell; quantitative (Proposition 10.17)

Table 12: The extended verdict-alphabet ladder including the trivial rung n=1 and the split sub-ladder at n=2. The resolved families are not strictly increasing along the ladder: in the sense of table 9, n=3 resolves the union of the families of the two asymmetric n=2 configurations, and definitive verdicts saturate at n=3 (Proposition 10.17 (1)); the n=4 and n=∞ rungs add revocable and quantitative information only.

11.4. **Bootstrap of CLASP at the meta-level.** The classes $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$ are introduced in section 3 as *Level-1 CLASP classes*: each is a subset of $\mathcal{P}(\Sigma^\omega)$ defined by a prefix-quantifier predicate on a single property P. Each class is itself an object of one level up:

$$\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K} \ \in \ \mathcal{P}(\mathcal{P}(\Sigma^\omega)).$$

A natural further question follows. The CLASP axioms (closed, open, dense, empty-interior) are stated in terms of a finite-prefix evidence discipline over an underlying domain. That discipline is not specific to Σ^ω: it applies, with the same axiom shapes, to subsets of any

second-countable Polish space (for the Effros Borel space of its closed subsets see Kechris [35, §12.C]; for the Borel hierarchy, §§22–23). Applying CLASP *at the meta-level* amounts to asking, for each Level-1 class X, in which Level-2 CLASP cell does X itself reside?

We formalise the meta-level predicates on a property P (recovered via prefix-membership queries on its cylinders) by spelling out the natural prefix-quantifier shapes:

$$\begin{aligned} P \in \mathsf{L} &\iff \forall \pi \in \Sigma^* \, \exists \tau \sqsupseteq \pi : \tau \in P, \\ P \in \mathsf{K} &\iff \forall \pi \in \Sigma^* \, \exists \tau \sqsupseteq \pi : \tau \notin P, \\ P \in \mathsf{S} &\iff \forall \sigma \notin P \, \exists \pi \sqsubseteq \sigma : [\pi] \cap P = \varnothing, \\ P \in \mathsf{C} &\iff \forall \sigma \in P \, \exists \pi \sqsubseteq \sigma : [\pi] \subseteq P. \end{aligned}$$

All four are Π_2 at the meta-level, but their quantifier structures split into two pairs: the *limit-classes* L, K quantify $\forall_\pi \exists_\tau$ over prefixes and extensions; the *fixed-witness classes* S, C quantify $\forall_\sigma \exists_\pi$ over traces and prefixes. This structural split drives the meta-classification.

Conjecture 11.1 (Meta-classification of the four CLASP classes)**.** *Read each Level-1 class $\mathsf{X} \in \{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$ as a subset of $\mathcal{P}(\Sigma^\omega)$. Then:*

(1) L *and* K *each sit in* $\mathsf{S} \cap \mathsf{K}$ *(Morbidity) at the meta-level: their meta-class is always finitely refutable (one prefix witness suffices to refute the meta-claim "P is liveness" resp. "P is coliveness") but never finitely satisfiable.*

(2) S *and* C *each sit in* $\mathsf{L} \cap \mathsf{K}$ *(Mixed) at the meta-level: neither finitely refutable nor finitely satisfiable in general.*

This is stated as a conjecture at the framework level*: the prefix-quantifier reformulations of each meta-class are made explicit, but the full descriptive-set-theoretic proofs (verifying the CLASP-cell assignment with rigour matching the underlying Borel-rank computations) require additional care beyond the scope of the present paper.*

Supporting argument (evidence for the conjecture). (1) The meta-claim "$P \in \mathsf{L}$" is the Π_2 statement $\forall \pi \, \exists \tau \sqsupseteq \pi : \tau \in P$. A *single* prefix π^* with $[\pi^*] \cap P = \varnothing$ falsifies the universal quantifier and so refutes the meta-claim: the meta-class L admits an explicit bad-prefix $\pi^* \in \Sigma^*$ for every $P \notin \mathsf{L}$, hence $\mathsf{L} \in \mathsf{S}$ at the meta-level. The meta-class admits no finite good-prefix (confirming $P \in \mathsf{L}$ requires verifying all π), hence $\mathsf{L} \in \mathsf{K}$ at the meta-level. Combining, $\mathsf{L} \in \mathsf{S} \cap \mathsf{K}$. The argument for K is symmetric: a single good prefix $[\pi^*] \subseteq P$ refutes "$P \in \mathsf{K}$". (2) The meta-claim "$P \in \mathsf{S}$" is $\forall \sigma \notin P \, \exists \pi \sqsubseteq \sigma : [\pi] \cap P = \varnothing$. A finite refutation would require exhibiting a specific $\sigma \notin P$ such that every prefix of σ admits a continuation in P — itself a Π_1 condition on the infinite trace σ that no finite prefix-cylinder of P resolves. A finite confirmation would require verifying the universal over σ, an explicit infinity of cases. Hence $\mathsf{S} \in \mathsf{L} \cap \mathsf{K}$ at the meta-level. The argument for C is symmetric. □

Level-1 class	Quantifier shape (Π_2)	Meta-class (Level-2)
L (Liveness)	$\forall_\pi \exists_\tau$ over prefixes	$\mathsf{S} \cap \mathsf{K}$ (Morbidity)
K (Coliveness)	$\forall_\pi \exists_\tau$ over prefixes	$\mathsf{S} \cap \mathsf{K}$ (Morbidity)
S (Safety)	$\forall_\sigma \exists_\pi$ over traces	$\mathsf{L} \cap \mathsf{K}$ (Mixed)
C (Cosafety)	$\forall_\sigma \exists_\pi$ over traces	$\mathsf{L} \cap \mathsf{K}$ (Mixed)

Table 13: The conjectured meta-classification of the four CLASP classes, applying CLASP recursively at one level up (Conjecture 11.1). Conjecturally, the *limit-classes* L, K map to Morbidity at the meta-level and the *fixed-witness classes* S, C map to Mixed; if so, none of the four meta-classes is in C (Cosafety) at the meta-level and no Level-1 class can be finitely confirmed for an arbitrary candidate property.

Remark 11.2 (Relation to Camerlo–Dagnino at the CLASP level)**.** Camerlo and Dagnino [12, Prop. 3.1] show that, for a countable second-countable space X, the family $\mathcal{M}(X) \subseteq \mathcal{P}(X) \cong 2^X$ of monitorable subsets is $\mathbf{\Pi}^0_3$; the bound does not apply directly to the uncountable Σ^ω. In our notation $\mathcal{M}(\Sigma^\omega) = \{\Phi : \mathrm{int}(\mathrm{bd}(\Phi)) = \varnothing\}$ (Diekert–Leucker [19]; Camerlo–Dagnino [12, Prop. 2.2(4), Cor. 2.5]). Hence $\mathsf{S} \cup \mathsf{C} \subsetneq \mathcal{M}(\Sigma^\omega)$ ([7, Lemma 3.8]) and $(\mathsf{L} \cap \mathsf{K}) \cap \mathcal{M}(\Sigma^\omega) = \varnothing$ (by Theorem 4.7, $\mathrm{bd}(\Phi) = \Sigma^\omega$ on $\mathsf{L} \cap \mathsf{K}$); the L-only, K-only, and quaestio cells are *mixed* (Lemma 5.9 (3)), so $\mathcal{M}(\Sigma^\omega)$ is not a union of cells of the 16-cell partition. The meta-classification of Conjecture 11.1 would place the four Level-1 classes at two CLASP meta-cells, each shared by one pair; if the conjecture holds, none of the four meta-classes admits a finite positive verdict.

Remark 11.3 (Why no meta-class is in Cosafety)**.** The conjectured table 13 has no Cosafety entry at the meta-level. This would reflect a structural asymmetry: the prefix-quantifier predicate "P is in class X" is universal at its outermost layer, for every $\mathsf{X} \in \{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$. We conjecture, as part of Conjecture 11.1, that no outermost-existential reformulation exists at the property-level for "P is closed" or "P is dense" that would yield a finite positive witness for the meta-claim. Operationally, if this holds, *no monitor at any finite verdict alphabet can confirm membership in a CLASP class in finite time.* Refutation is available for the limit-classes (L, K, one prefix witness: a bad, resp. good, prefix of P refutes the meta-claim), while its absence for the fixed-witness classes is part of the conjecture.

12. Conclusion

The four class families Safety (S), Cosafety (C), Liveness (L), and Coliveness (K) generate a 15-cell intersection algebra over Σ^ω (Theorems 5.4 and 5.5): 12 nonempty and 3 provably empty, anchored in the Cantor-topological characterization of ω-regular properties (Theorem 4.6). Each nonempty cell admits a constructive LTL witness (Lemma 5.3), the boundary-operator refinements (Theorem 4.7) expose the interior/closure structure that governs cell membership, and the resulting monitorability gradient (Remark 5.8) connects cell membership to runtime-verification verdict types.

The present results sit within three structural boundaries. First, the entire framework operates within ω-regular properties over finite alphabets: all topological arguments rely on the metric structure of the Cantor space Σ^ω, and the algebraic classification is formulated for the Boolean algebra of ω-regular languages; the relationship to properties outside

the ω-regular class is not addressed. Second, the 15-intersection enumeration is horizontal: it classifies properties by the lattice of class-family membership, not by syntactic or automata-theoretic complexity. Third, the prefix definitions and topological correspondences are classical (Alpern–Schneider, Manna–Pnueli, Kupferman–Vardi). The four-class skeleton and the nine-cell verdict-axis enumeration are due to Peled–Havelund [50, 28]; novelty claims of the present work are restricted to the exhaustive 15-cell *membership-axis* enumeration (including the four triple cells and the quadruple), the constructive LTL witness table for triples and the quadruple (table 6), the boundary-operator characterization, and the refinement bijection between the membership and verdict axes (Theorem 8.2).

Beyond the 15-cell classification, the verdict-alphabet ladder of section 10 adds an operational complement to the topological-and-membership analysis. The cells of the intersection algebra are mapped onto a sequence of verdict-monitor systems parameterised by alphabet cardinality $n \in \{1, 2, 3, 4, \infty\}$ (table 9; n=1 trivial baseline, section 11.1), with the definitive-verdict rungs newly resolving a specific portion of the lattice: $n = 2$ resolves the Clopen cell $\mathsf{S} \cap \mathsf{C}$, and $n = 3$ the safety and cosafety regions $\mathsf{S} \setminus \mathsf{C}$ and $\mathsf{C} \setminus \mathsf{S}$ (one verdict side each), where definitive verdicts saturate; $n = 4$ adds revocable verdicts on inconclusive prefixes, informative in particular on the non-monitorable properties, which fill $\mathsf{L} \cap \mathsf{K}$ and occur in the mixed cells $\{\mathsf{L}\}$, $\{\mathsf{K}\}$ and quaestio (Lemma 5.9), and $n \to \infty$ revocable quantitative information. Universal attribution (Theorem 10.23) shows that every verdict monitor with $|V| \geq 2$ canonically furnishes persistence and response pullback properties on the input via the verdict-stream extension map, which lie in $\mathsf{L} \cap \mathsf{K}$ exactly at its non-absorbing verdicts (Proposition 10.24); class-promotion via composition (Theorem 10.20) lifts a base monitor's Layer-1 cell to higher cells reachable through the verdict-trace lattice. Refinement-based confirmation (section 10.7) frames the use of finite verdict-trace observations as inputs to refinement systems whose convergence certifies properties that would otherwise be refutation-only monitorable. A summary against the four primary CLASP classes is given in table 10.

The pair (V, A) of verdict alphabet and operating algebra identified in Remark 10.30 is the structural invariant under which the ladder is read: V varies along the cardinality axis at fixed LTL semantics for A. The complementary axis (at fixed V, varying A) and the systematic refinement-confirmation theory sketched in section 10.7 are both natural further directions we identify but do not pursue here. Beyond these, the 15-cell intersection algebra provides the classification substrate for monitoring frameworks where physical observability constraints determine which temporal fragment governs each specification pattern. Future work extends the algebra with operational semantics relative to generator models, automata-theoretic characterizations, and bridge theorems connecting the topological classification to monitoring architectures.

Appendix A. Extended worked examples

This appendix collects four extended worked examples whose formal blocks (Pattern / Formula / Cell / Operational verdict / Stratified reading / Source) are too long to sit inline in section 8 without fragmenting the narrative. The examples are referenced from the main text by their original labels (Example A.1, Example A.2, Example A.3, Example A.4); the short stubs that remain in the main text record the cell, formula, and PH-2019 verdict-axis position; the full operational discussion, stratified reading, and provenance live here.

A.1. **Sustained Mutual Exclusion ($\mathsf{S} \cap \mathsf{K}$).**

Example A.1 (Pair $\mathsf{S} \cap \mathsf{K}$: Sustained Mutual Exclusion — Morbidity class)**.** *Pattern.* "At every step, at most one process is in the critical section." The protocol *maintains* the mutual-exclusion invariant throughout execution. *Formula.* $\varphi = G\,(\neg(p_1 \wedge p_2))$, where $p_i = \mathtt{P}_i$ in CS. *Cell.* $(1, 0, 0, 1)$; closed with empty interior; *Morbidity class. Operational verdict.* The verdict-axis reading of Peled–Havelund decomposes the cell into its two finitary faces: *Always finitely refutable* (AFR, the S face): every violating trace admits a finite bad-prefix — a single observed state in which both p_1 and p_2 hold simultaneously refutes the mutual-exclusion claim from that prefix onward. Refutation is uniform on the violating side. *Never finitely satisfiable* (NFS, the K face, *morbidity*): no satisfying trace has a finite good-prefix. Every prefix in which the protocol has so far preserved mutual exclusion admits an extension into joint critical-section occupancy; by definition of K, sustained mutex is *never finitely confirmable as permanent.* The cell sits in the *Morbidity class*: observable as a snapshot quality at every step (membership in the legal-state set is a clopen single-step predicate), yet *never finitely satisfiable as permanent* – the K component encodes that no length of conforming verdict trace settles the maintenance claim. *Stratified reading* (Remark 9.13). The predicate $\neg(p_1 \wedge p_2)$ is a Level-1 evaluation of the joint-CS condition at each individual step. The outer G operator lifts the Level-1 verdict to the question "has the Level-1 condition been maintained at *every* step so far?" — a Safety claim on the verdict stream. The K classification is the stratified observation that the running maintained-status claim is structurally unstable as a permanent verdict: every subsequent observation may disrupt it, and the Level-2 question of *permanent maintenance* is never finitely settled. *Source.* PH-2019 Fig. 1, $AFR \cap NFS$ [50]; canonical mutual-exclusion problem of Dijkstra [20] and Lamport [40]; classical safety formulation [41, 2, 3].

A.2. **Eventually Perfect Failure Detector ($\mathsf{L} \cap \mathsf{K}$).**

Example A.2 (Pair $\mathsf{L} \cap \mathsf{K}$: Eventually Perfect Failure Detector ($\Diamond P$))**.** *Pattern.* "Eventually, the failure detector becomes permanently correct: from some point on, it suspects all and only the actually crashed processes." *Formula.* $\varphi = F\,G\,p$, where $p = \texttt{detector_correct}$ encodes the perfect-output condition. Equivalently, the recurrence-class instance "`checkpoint` infinitely often" is given by $G\,F\,q$ with $q = \texttt{checkpoint}$, in the same cell. *Cell.* $(0, 0, 1, 1)$; full boundary $\mathrm{bd}(\varphi) = \Sigma^\omega$ (Theorem 4.7; boundary-dense nontriviality, Remark 5.7). *Operational verdict.* In Peled–Havelund verdict-axis terms: *Never finitely refutable* (NFR) and *never finitely satisfiable* (NFS) — neither side admits a finite witness. Every prefix extends both to a satisfying and to a violating trace: from any observation, the failure detector may either stabilise correctly forever (satisfying $FG\,p$) or flip into permanent incorrectness (violating $FG\,p$). This is the prototype of the "zero monitoring information" class of [50, Sec. 4.3] (the term is reused in [32]), characterised via the Diekert–Leucker boundary-condition [19]. *Stratified reading* (Remark 9.13). This example is the *prototype of Level-2 properties*: the predicate `detector_correct` (or `heartbeat`) is the Level-1 evaluation on each state, and the outer FG (resp. GF) operator asks the Level-2 question of *long-run stability* (eventual permanence, resp. recurrence with structural disruption) of the Level-1 verdict stream. The $\mathsf{L} \cap \mathsf{K}$ classification names the structural openness of the Level-2 question in both directions: stabilisation is never finitely confirmed (NFS), perpetual destabilisation is never finitely refuted (NFR). This cell is precisely the Manna–Pnueli classes 4–5 viewed as

liveness/coliveness lifts of class 1 predicates. *Source.* PH-2019 Fig. 1, $NFR \cap NFS$ [50]; $\Diamond P$ failure detector of Chandra–Toueg [14] (the JACM canonical $\mathsf{L} \cap \mathsf{K}$ specification of distributed systems); Persistence class of Manna–Pnueli [48].

A.3. **Weak-Fairness Recurrence ($\mathsf{L} \cap \mathsf{K}$, KL polarity).**

Example A.3 (Pair $\mathsf{L} \cap \mathsf{K}$ continued: Weak-Fairness Recurrence (the KL polarity))**.** *Pattern.* "A heartbeat is emitted infinitely often throughout the execution" — equivalently, in fair-scheduling terms, "every continuously-enabled process is scheduled infinitely often". *Formula.* $\psi = G\,F\,q$, where $q = \texttt{heartbeat}$ (or $q = \texttt{scheduled}_i$ for each $i \in \Pi$ in the fairness reading). *Cell.* $(0, 0, 1, 1)$; same membership-axis cell as Example A.2, opposite Borel polarity ($\mathbf{\Pi}^0_2$ versus $\mathbf{\Sigma}^0_2$); Manna–Pnueli class 4 (Recurrence / Response). Boundary-dense membership per Remark 5.7. *Operational verdict.* As in Example A.2: *Never finitely refutable* (NFR) and *never finitely satisfiable* (NFS). However, the operational signature differs in the compositional structure: outer G requires an F-witness at every prefix, which is itself never finitely refutable (a $\neg q$-run can always be broken later); inner F provides local-event confirmability of single heartbeats, but the outer G blocks finite-time settlement of the global recurrence claim. *Stratified reading* (Remark 9.13). Right-to-left semantic parsing using the Manna–Pnueli modal-nesting levels (distinct from the Layer-1/Layer-2 of Definition 10.1; see Remark 9.13): Level-1 predicate q (snapshot heartbeat-event detection). Level-2 modal F on Level-1: "q eventually" — Cosafety (C), Liveness (L). Level-3 modal G on Level-2: "the eventual-q claim holds at every prefix forward" — Safety (S), Coliveness (K). Compositional residue: inner F contributes L, outer G contributes K. The C of inner F is blocked by outer G's never-finite-confirmation; the S of outer G is blocked by inner F's never-finite-refutation. Net: $\mathsf{L} \cap \mathsf{K}$ at Borel polarity $\mathbf{\Pi}^0_2$, the KL *polarity* (outer-K, inner-L) of the bipartition of $\mathsf{L} \cap \mathsf{K}$. *Source.* PH-2019 Fig. 1, $NFR \cap NFS$ [50]; weak-fairness (justice) condition of Manna–Pnueli [48] (a recurrence property; strong fairness is a reactivity property there); heartbeat pattern canonical in distributed-system failure detection [14].

A.4. **Byzantine Consensus Correctness (K-only).**

Example A.4 (Atomic $\{\mathsf{K}\}$: Byzantine Consensus Correctness)**.** *Pattern.* "Eventually consensus is reached, AND no Byzantine fault occurs throughout the execution." *Formula.* $\varphi = F\,p \wedge G\,q$, where $p = \texttt{consensus_reached}$ and $q = \neg\texttt{byzantine_fault}$. *Cell.* $(0, 0, 0, 1)$. *Operational verdict.* In Peled–Havelund verdict-axis terms: *Never finitely satisfiable* (NFS) — no satisfying trace has a finite good-prefix, since the Gq conjunct cannot be confirmed in finite time even when Fp has been witnessed. *Sometimes finitely refutable* (SFR) — a $\neg q$-event at any step refutes the Gq conjunct and yields a finite bad-prefix; conversely, traces in which $\neg p$ continues forever satisfy the violation only asymptotically without a finite bad-prefix. The asymmetry is structural to the conjunctive shape: refutation is available exactly on Gq-violation, never on Fp-violation. *Stratified reading* (Remark 9.13). The formula is a conjunction of $F\,p$ (Level-1 cosafety, finitely confirmable on the first p-event) and $G\,q$ (Level-1 safety, never finitely confirmable as permanent). The atomic $\{\mathsf{K}\}$ classification names the structural blocker: the confirmable Level-1 sub-witness $F\,p$ is gated by the non-confirmable Level-1 maintenance claim $G\,q$, so the overall judgment is never finitely settled positively. Refutation, conversely, is available only via $G\,q$-violation, not via $F\,p$-violation (which has no finite witness at all). *Source.* PH-2019 Fig. 1, $SFR \cap NFS$ [50];

a simplification of the Byzantine agreement setting of Lamport–Shostak–Pease [44] and Castro–Liskov [13], whose protocols guarantee agreement while at most a bounded fraction of participants is faulty; the formula here instead asserts that no fault occurs.

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Appendix B. The monitor and trace axis taxonomy

This appendix catalogues the 23-axis design space referenced in the body: the eleven monitor axes (D1–D11) and twelve trace axes (T1–T12), each with this paper's adopted default. None of the individual axes is consumed by the main results; only the cross-axis dependency stated there is load-bearing.

B.1. **Monitor axes (D1–D11).** The monitor axes were introduced informally in section 10; we summarise them here as table 14.

ID	Axis	Values	Default of this paper
D1	Verdict-alphabet n	$\{1, 2, 3, 4, \infty\}$	varied along ladder
D2	Horizon discipline	$\{\omega, N, \text{variable}, \text{piecewise}\}$	ω (section 6)
D3	Trace domain	$\{\Sigma^\omega, \Sigma^N, \Sigma^*, \Sigma^* \subset \Sigma^\omega\}$	Σ^ω
D4	Observation alphabet	$\{2^{AP}$, $\mathbb{R}^k$, abstract events$\}$	2^{AP}
D5	Monitor state space	$\{1, 2, 3, 4, \text{finite}, \infty\}$	matches n
D6	Determinism	$\{\text{det}, \text{non-det}, \text{alternating}, \text{prob}\}$	deterministic
D7	Acceptance condition	$\{\text{sink}, \text{Büchi}, \text{co-Büchi}, \text{parity}\}$	sink-reachability
D8	Sink absorbance	$\{\text{abs}, \text{non-abs}, \text{refinable}, \text{resettable}\}$	absorbing (with section 10.7 as refinable extension)
D9	Output emission mode	$\{\text{sync}, \text{async}, \text{event}, \text{continuous}\}$	synchronous per-tick
D10	Composition layer	$\{1, 2, \text{hierarchical}\}$	1 (Layer 1), 2 (via section 10.6)
D11	Verdict revisability	$\{\text{final}, \text{current}, \text{refinable}, \text{resettable}\}$	final (with current/refinable as ladder rungs)

Table 14: The eleven monitor axes. Values listed are the distinct *types* of monitor configuration; combinatorial products of these values cover all monitor frameworks treated in section 10.

B.2. **Trace axes (T1–T12).** The trace itself is parameterised by twelve axes (table 15). Most of these are fixed silently by the conventional adoption of Σ^ω as the trace domain; the present table makes the defaults explicit and identifies the alternative regimes that future work can address.

ID	Axis	Values	Default of this paper
T1	Time domain	$\{\mathbb{N}_{<N}, \mathbb{N}, \mathbb{Z}, \mathbb{R}_{\geq 0}, \text{ordinals}, \text{poset}, \text{tree}\}$	$\mathbb{N}$
T2	Intrinsic length	$\{N, \Sigma^*, \omega, \text{transfinite}, \text{mixed}\}$	ω
T3	Per-tick alphabet	$\{2^{AP}$, k-symbol, $\mathbb{R}^k$, record, V_{k-1}, heterogeneous$\}$	2^{AP}
T4	Observation density	$\{\text{total}, \text{sparse}, \text{continuous}, \text{event-driven}\}$	total
T5	Trace determinism	$\{\text{det}, \text{non-det}, \text{prob}, \text{quantitative}\}$	deterministic
T6	Layer	$\{1, 2, 3, \ldots, k, \omega\}$	1 (raw); 2 via sections 10.6 and 11
T7	Producer / origin	$\{\text{synthetic}, \text{empirical}, \text{adversarial}, \text{derived}, \text{spec}\}$	unspecified (with derived at $T_6 \geq 2$)
T8	Observability	$\{\text{total}, \text{partial}, \text{noisy}, \text{hidden-state}\}$	total
T9	Stationarity	$\{\text{stationary}, \text{non-stationary}, \text{time-varying}\}$	stationary
T10	Multi-dimensionality	$\{\text{single}, \text{bundle}, \text{multi-agent}\}$	single
T11	Boundary	$\{\text{init-defined}, \text{end-defined}, \text{lasso}, \text{open}\}$	open
T12	Identifiability	$\{\text{anonymous}, \text{labelled}, \text{indexed}\}$	anonymous

Table 15: The twelve trace axes. Eight of the twelve were completely implicit in the paper before this section; two were partially treated (T6 by sections 10.6 and 11, T7 by the event-generator algebras of section 10.8); two were explicitly declared (T2 in section 2, T3 in section 2).

B.3. **The defaults of this paper.** Table 16 consolidates the slice of the 23-axis space that this paper inhabits. The classification results of sections 5, 6, 8 and 11 are statements about properties on this slice. Other settings (e.g. robust signal monitoring, multi-agent governance, continuous-time STL) inhabit different slices; the variable framework above is the basis for stating which results transfer and which require re-derivation.

Axis	Value adopted by this paper	Where varied (if any)	Where alternative cited
T1	$\mathbb{N}$ discrete	–	STL [47] for $\mathbb{R}_{\geq 0}$
T2	ω	–	LTLf [18] for Σ^*
T3	2^{AP}	meta-layers via Proposition 7.1	robust [22] for $\mathbb{R}^k$
T4	total per-tick	–	event-driven [47] for sparse
T5	deterministic trace	–	–
T6	Layer 1	Layer 2 in section 10.6	–
T7	unspecified raw	derived at $T_6 \geq 2$	event-generator [45]
T8	total observability	–	–
T9	stationary	–	–
T10	single	–	–
T11	open-ended	boundary marker in section 11 via n=1 rung	–
T12	anonymous	–	–
D1	varied $n \in \{1, 2, 3, 4, \infty\}$	section 10	–
D2	ω	finite-horizon in section 6	BMC [8], LTLf [18]
D3	Σ^ω	–	–
D4	2^{AP}	–	–
D5	matches n	along ladder	–
D6	deterministic	–	–
D7	sink-reachability	–	Büchi [60]
D8	absorbing	refinable in section 10.7	–
D9	synchronous	–	event-generator [45]
D10	Layer 1; Layer 2 via $N \circ M$	section 10.6	–
D11	final	current at n=4 in section 10.4, refinable in section 10.7	–

Table 16: The slice of the 23-axis design space inhabited by this paper. Eight trace axes (T1, T4, T5, T8, T9, T10, T12) were completely implicit before section 7 and are now declared. Monitor axes are varied or fixed as indicated; other settings connect to specific slices of this design space and may use this table to position their contributions relative to the present classification.

Trace axes (T)

ID	Axis	**Default** / alternatives
T1	Time domain	$\mathbb{N}$ / $\mathbb{R}_{\geq 0}$, $\mathbb{Z}$, poset, tree
T2	Length	ω / Σ^*, N, transfinite
T3	Σ per-tick	2^{AP} / $\mathbb{R}^k$, V_{k-1}
T4	Density	total / sparse, event-driven
T5	Determinism	det / non-det, prob
T6	Layer	1 / 2 (section 11), ω
T7	Origin	unspec / empirical, adversarial
T8	Observability	total / partial, noisy
T9	Stationarity	stat / non-stat
T10	Multi-dim	single / bundle, MAS
T11	Boundary	open / init, end, lasso
T12	Identifiability	anon / labelled, indexed

Monitor axes (D)

ID	Axis	**Default** / alternatives
D1	Verdict-n	varied $\{1, 2, 3, 4, \infty\}$
D2	Horizon	ω / N, piecewise
D3	Trace domain	Σ^ω / Σ^N, Σ^*
D4	Obs alphabet	2^{AP} / $\mathbb{R}^k$, events
D5	States	matches n
D6	Determinism	det / non-det
D7	Acceptance	sink / Büchi, parity
D8	Sink absorbance	abs / refinable
D9	Emission	sync / async, event
D10	Composition	L1; L2 via $N \circ M$
D11	Revisability	final / current, refinable
constraint: $\mathrm{T3}_{k+1} = \mathrm{D1}_k$ *(Proposition 7.1)*		

Figure 7: **Variable-space slice.** Left: the 12 trace axes; right: the 11 monitor axes. For each axis the value before the slash is the default adopted by this paper; values after are alternatives explored in related work or left to future work. The unique mandatory cross-axis dependency (bottom row of the monitor column, Proposition 7.1) ties the per-tick alphabet of a meta-trace at layer $k{+}1$ to the verdict alphabet of the monitor at layer k. Visualises table 16 as a side-by-side trace-vs-monitor view.

Table 17: **Rosetta stone of the three "layer" notions.** A single property carries one coordinate on each of three independent axes: *Axis A* (syntactic depth, operator nesting), *Axis B* (architectural layer, monitor composition depth), and *Axis C* (topological level, Borel / Manna–Pnueli). The four representative properties pin the axes against each other. Axis A counts operator nesting only; Axis B counts monitor cascades only; Axis C is not determined by the CLASP cell: GFp and $GFGp \equiv FGp$ share $\mathsf{L} \cap \mathsf{K}$ but sit at Π^0_2 and Σ^0_2. The three coordinates are correlated (e.g. GFp rises on A, B, and C) but *not* one-to-one: $GFGp$ is deeper than GFp on A and B without rising on C, illustrating the correlation-without-coincidence.

Property	**A: Syntactic** op. nesting	**B: Architectural** monitor depth	**C: Topological** Borel / MP	**CLASP cell**
p_1 (clopen base)	depth=0	L_1	$\Sigma^0_1 \cap \Pi^0_1$ (MP class 0)	$\mathsf{S} \cap \mathsf{C}$
Gp (safety)	depth=1	L_1	Π^0_1 (MP class 1)	$\mathsf{S} \cap \mathsf{K}$
GFp (recurrence)	depth=2	L_2	Π^0_2 (MP class 4)	$\mathsf{L} \cap \mathsf{K}$
$L_3{+}$ pattern (e.g. $GFGp$)	depth ≥ 3	L_3	Σ^0_2 (MP class 5; $GFGp \equiv FGp$)	$\mathsf{L} \cap \mathsf{K}$

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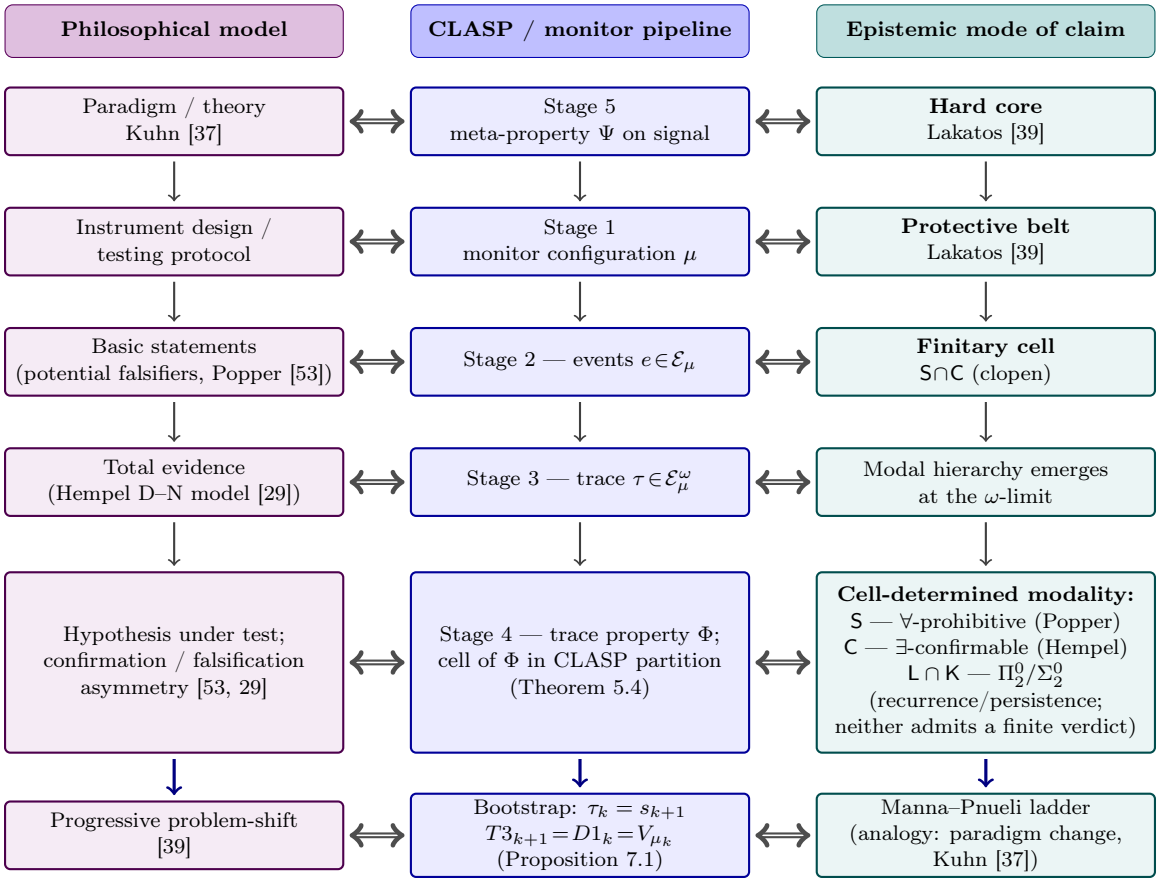


Figure 8: The CLASP / monitor pipeline as an operationalisation of the Popper–Hempel–Lakatos–Kuhn inquiry cycle. *Left column:* stages of empirical inquiry in classical philosophy of science, with the canonical attributions. *Middle column:* the six-stage CLASP / monitor pipeline of table 5. *Right column:* the epistemic mode of trace-property claims at each stage, keyed to the eleven CLASP cells of Theorem 5.4. Vertical arrows trace the flow of inquiry within each track; horizontal double arrows mark the identifications between the tracks. At the hypothesis row, the CLASP partition resolves the Popperian asymmetry of confirmation [53]: S cells host claims that are universally prohibitive (every counter-instance refutes the claim at a finite prefix), C cells host claims that are existentially confirmable (every instance confirms the claim at a finite prefix), L excludes finite refutation and K finite confirmation, while $\mathsf{L} \cap \mathsf{K}$, with witnesses $GFp \in \Pi^0_2$ and $FGp \in \Sigma^0_2$, admits neither finite verdict. The bottom row is the *bootstrap*: when a Stage 4 property falls in an infinitary cell, treating τ_k as a new signal s_{k+1} at meta-layer $k+1$ is read as a Lakatosian progressive problemshift [39]; the abandonment of a whole programme, which Lakatos ties to degenerating problemshifts and a better rival, is the analogue of Kuhn's paradigm change [37]; the cross-axis constraint $T3_{k+1} = D1_k = V_{\mu_k}$ ties the alphabets across layers. Read as a dictionary, the figure shows that the CLASP partition is a classification of *epistemic modalities* of empirical claims: which cell a property inhabits determines what kind of evidence can decisively support or refute it.

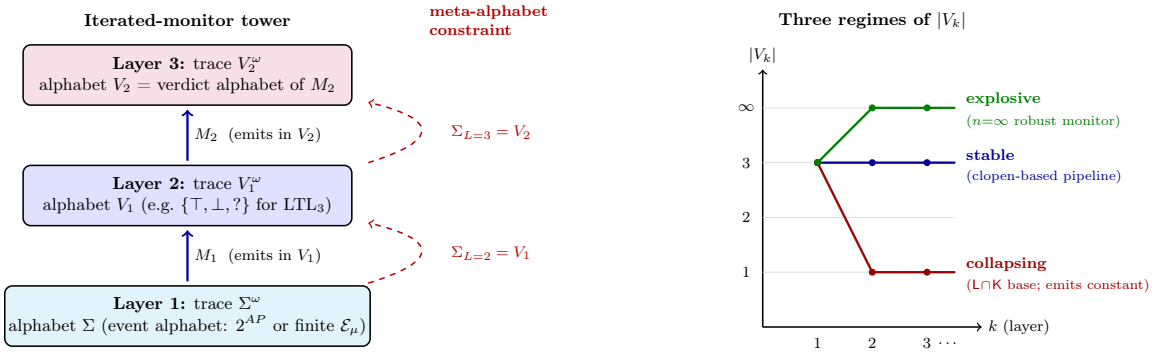


Figure 9: **Layer hierarchy and meta-classification regimes.** *Left panel*: the iterated-monitor tower. Layer 1 is the raw observation trace over Σ (any finite event alphabet: the textbook propositional-LTL convention is 2^{AP}; concrete monitors may use a smaller finite $\mathcal{E}_\mu$ defined by the configuration μ). Monitor M_1 running on Layer 1 emits a verdict trace over alphabet V_1 – which by Proposition 7.1 *is* the Layer 2 trace. A Layer-2 monitor M_2 in turn produces a Layer 3 trace over V_2, and so on. The values in V_2 are informally "verdicts on verdicts" (a Layer-2 monitor consumes a verdict-trace from M_1 and emits its own verdicts). The red dashed arcs visualise the cross-axis constraint $\Sigma_{L=k+1} = V_k$: the per-tick alphabet of the Layer-$(k{+}1)$ trace coincides with the verdict alphabet of the Layer-k monitor. *Right panel*: three regimes for the evolution of $|V_k|$ along the tower, each triggered by a different combination of (cell of base property) and (monitor architecture). *Stable*: $|V_k|$ stays constant – the CLASP partition reproduces at every meta-layer (as conjectured in Conjecture 11.1). **Example.** LTL$_3$ chain on the $\mathsf{S} \cap \mathsf{K}$ safety property Gp: $V_1 = \{\top, \bot, ?\}$ (effective cardinality 2: only $\bot$/? ever appear under a finite bad-prefix trigger); M_2 then watches the V_1-trace with another LTL$_3$ monitor, again with $V_2 = \{\top, \bot, ?\}$; $M_3, \ldots$ idem. The base cell $(\mathsf{S} \cap \mathsf{K})$ and the verdict cardinality are both preserved across the bootstrap. *Collapsing*: the *effective* verdict cardinality drops to 1 at layer 2 when the underlying property sits in $\mathsf{L} \cap \mathsf{K}$ or in a trivial singleton (Proposition 7.5). **Example.** LTL$_3$ monitor on the recurrence $GFp \in \mathsf{L} \cap \mathsf{K}$: no finite prefix yields $\top$ or $\bot$, so M_1 emits $?^\omega$ – nominally $|V_1|{=}3$ but effectively $|V_1|_{\text{eff}}{=}1$. At layer 2, M_2 sees a constant trace and has nothing to classify; the bootstrap collapses. *Explosive*: the alphabet grows beyond any finite cardinality while remaining *within LTL syntax*, when a *quantitative* LTL semantics is selected (section 10.5, $n = \infty$); verdicts carry distance / counting information rather than discrete tags. **Example (LTL-internal).** The counting-semantics monitor of Bartocci–Bloem–Ničković et al. [4] on the standard LTL formula $\varphi = G\,F\,p$ over an event trace: the verdict at position i is the pair of natural numbers $(d_\top(i), d_\bot(i)) \in \mathbb{N}^2$ recording the *number of steps* to the nearest witness of satisfaction and the nearest witness of violation, respectively. The verdict alphabet is $V_1 = \mathbb{N}^2$ (or its quotient to $\mathbb{N} \cup \{\infty\}$), a countably infinite lattice; the layer-2 monitor M_2 on this quantitative verdict trace inherits the same codomain. Saturation note: like $n{=}4$ RV-LTL [6], counting semantics also informs on non-monitorable properties such as $G\,F\,p$ [4]; the gain is quantitative refinement (distance-to-witness), not a new Layer-1 cell.